

Provenance map: the draft’s sentence, beside Lean’s verdict on it

Quantum Gadgets for quantum reduction: a new approach for investigating the class BQ-NP

(Manish et al., draft)

508 ledger rows; 124 carry a machine-checked witness (106 live, 10 comment, 7 figure, 1 support). Generated by `tools/gen_provenance.py`; do not hand-edit.

1 Pivotal claims

Eight findings the rest of the draft leans on. Verdict, tier, status and every declaration are read from the ledger row named; the prose beside them is authored, and generation fails if the two disagree (`tools/gen_provenance.py:pivotal`). Two of the eight assert nothing external; two cite a source not frozen here; four turn on a source that is, quoted where the quote is literal.

G-94 Definition: the Quantum Boolean Function (main.tex:249)

FALSE lean / HOLDS main.tex:249

```
witness: lean/Qgadget/Verbatim/Definition_QuantumBooleanFunction.lean#isQuantumBooleanFunction_iff_mem_unitaryGroup;  
lean/Qgadget/Verbatim/Definition_QuantumBooleanFunction.lean#phaseGate_isQuantumBooleanFunction; lean/Qgadget/Verbatim/Definition_QuantumBooleanFunction.lean#phaseGate_sq_ne_one
```

Every unitary operator satisfies the draft’s clause $f^\dagger f = I$, so the definition names the whole unitary group rather than a subclass of it. Montanaro and Osborne square f instead: $f^2 = 1$ (`qboolean_arxiv_resubmit.tex:327-328`). The phase gate $S = \text{diag}(1, i)$ is unitary and meets the draft’s clause; it fails the source’s, since $S^2 = Z \neq I$. `isQuantumBooleanFunction_iff_mem_unitaryGroup` identifies the draft’s predicate with membership in the unitary group at every n ; `phaseGate_isQuantumBooleanFunction` and `phaseGate_sq_ne_one` carry the witness. The source is frozen, and its own wording is quoted below.

G-308 Lemma: operator norm equals spectral norm for a normal matrix (main.tex:1940)

TRUE lean / HOLDS main.tex:1940

```
witness: lean/Qgadget/Findings/Section_15_OperatorNormEigenvalue.lean#topRQ_isGreatest_eigenvalue; lean/Qgadget/Findings/Section_15_OperatorNormEigenvalue.lean#l2_opNorm_eq_sqrt_topRQ
```

The displayed chain equates the operator norm with the square root of the top eigenvalue of $N^\dagger N$, and nothing in it uses $NN^\dagger = N^\dagger N$. `topRQ_isGreatest_eigenvalue` shows the top Rayleigh quotient is itself an eigenvalue and dominates every other one, for an arbitrary square matrix; `l2_opNorm_eq_sqrt_topRQ` closes the chain. Neither declaration assumes normality: the draft’s own hypothesis is stronger than its proof needs. The claim is the draft’s own; no external source is at issue.

G-239 Theorem: the bounded-edge XY problem is in BQP (main.tex:1254)

TRUE PROOF INVALID lean / HOLDS main.tex:1254

```
witness: lean/Qgadget/Support/TensorGroundEnergy.lean#groundEnergy_kron_add; lean/Qgadget/Verbatim/Theorem_XYZHam_LogEdges_inBQP.lean#groundEnergy_edgeSplit;  
lean/Qgadget/Verbatim/Theorem_XYZHam_LogEdges_inBQP.lean#card_le_poly_of_edge_bound
```

The theorem’s conclusion holds, by more than the draft’s own route: $\lambda_{\min}(H_W \otimes I + I \otimes H_R) = \lambda_{\min}(H_W) + \lambda_{\min}(H_R)$ for arbitrary Hermitian H_W, H_R , together with a polynomial bound on H_W ’s dimension from the edge count, puts the problem in P outright. The draft’s proof does not reach this: its guiding state $|E\rangle_{\text{edge}} \otimes |S\rangle_{\text{singletons}}$ can be orthogonal to the ground space (G-240) and is not of the form the draft’s own definition requires (G-241). The eigenvalue bound is Sobczyk’s, ICALP 2025 – cited, not frozen here.

G-281 Appendix XI: the strict-gadget quantifier (main.tex:1670)

FALSE cited / OPEN main.tex:1670

Trevisan, Sorkin, Sudan and Williamson’s strict-gadget condition existentially quantifies the auxiliary assignment b ; the draft’s main.tex:1670 quantifies it universally. The two are not equivalent – the draft’s reading demands every b reach $\alpha - 1$, where the source asks only that some b does – and the source’s own introductory 7-gadget is strict only under its own, weaker reading. Their 2000 SIAM J. Comput. paper is not frozen in this repository; the finding rests on a reading recorded in data/claims.yaml, not on a re-checked quotation here.

G-322 Appendix XV: the block-matrix inverse lemma (main.tex:2066)

FALSE lean / HOLDS main.tex:2066

witness: lean/Qgadget/Verbatim/Lemma_BlockMatrixInverse.lean#blockMatrix_inv; lean/Qgadget/Verbatim/Lemma_BlockMatrixInverse.lean#printed_top_right_ne_true

The printed top-right block of the inverse is $-AB(D - CA^{-1}B)^{-1}$; the correct block needs A^{-1} in place of A . At the matrix with rows $(2, 1)$ and $(0, 1)$, the printed formula gives -2 against a true $-1/2$. `blockMatrix_inv` computes the genuine inverse from its adjugate; `printed_top_right_ne_true` confirms the disagreement. The source is the same frozen paper G-319’s quote below comes from, read here at a different line (2locall-lanl-resub7.tex:1061), where it already prints A^{-1} : only the draft’s transcription drops it.

G-340 Appendix XV: the third resolvent term’s scalar (main.tex:2179)

FALSE lean / HOLDS main.tex:2179

witness: lean/Qgadget/Support/ZXGadgetOperators.lean#term3_op_eq

The third term of $\Sigma_-(z)$ is printed with coefficient $B^2/(1 - \delta)$; the resolvent on L_+ gives $B^2/(z - \delta)$, and the two agree only at $z = 1$. `term3_op_eq` proves the operator part exactly as printed, general in the coupling B , which isolates the scalar as the sole defect: at $z = 0$, $\delta = 2$ the printed value is -1 against the true $-1/2$. The source is the same frozen paper G-319’s quote below comes from; this row reads its own appendix, not a line the quote covers.

G-319 Appendix XV: Theorem 3’s conclusion, restricted to the wrong space (main.tex:2045)

FALSE prose / OPEN main.tex:2045

Kempe, Kitaev and Regev’s Theorem 3 concludes about $\tilde{H}|_{\tilde{L}_-}$, $\tilde{H}$ ’s own low space; the draft writes $\tilde{H}|_{L_-}$ at main.tex:2045 (and again at :2030, :2037, :2137), H ’s low space, discarding the coupling V_{+-} . On $\mathbb{C}^2$, at $H = \text{diag}(0, 1)$, $\lambda_* = 1/2$, $\Delta = 1$, every hypothesis holds; the draft’s compression, [0], has eigenvalue $7/50$ away from the true value, against $\varepsilon = 1/50$. Not machine-checked: the arithmetic is exact-rational (tools/probes/appendix_xv_perturbation_algebra.py), read against the source directly at 2locall-lanl-resub7.tex, whose tilde this map’s own extract loses.

G-511 Figure: the HHL circuit’s clock-register label (pics/HHL-circuit.png)

FALSE lean / HOLDS pics/HHL-circuit.png

witness: lean/Qgadget/Findings/Section_Figures_HHLCircuitOffGrid.lean#clockPurity_off_eq; lean/Qgadget/Findings/Section_Figures_HHLCircuitOffGrid.lean#clockPurity_on_eq

The image's label claims the middle register returns to $|0\rangle^{\otimes m}$; that holds only when phase estimation on the drawn m clock qubits resolves every eigenvalue of A exactly, a hypothesis the image omits. At $m = 1$, $A = \text{diag}(\pi, \pi/2)$, $|b\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ – off the grid – the reduced state after the inverse block has purity $7/9$, not 1: on the grid, at $A = \pi I$, the same computation gives purity 1. The shortfall from 1 is the entanglement the label denies; the claim is the draft's own.

2 Cited sources

Only three works are frozen at `papers/cited/`: `cade2022complexity`, `kempe2004complexity` and `weggemans2023guidable`. Two entries below read a source frozen elsewhere under `papers/` instead – the Montanaro-Osborne and Kempe-Kitaev-Regev LaTeX sources – because both are quoted by a pivotal claim above and both are already read directly, not through a `papers/cited` extract, by `tools/check_cited_sources.py`. Every quotation below is re-read from its file at generation time; a phrase not literally present there fails generation.

Montanaro and Osborne, "Quantum boolean functions" (arXiv:0810.2435)

`papers/montanaro2008-quantum-boolean-functions/source/qboolean_arxiv_resubmit.tex` supports: G-94

“unitary operator f on n qubits such that $f^2 = I$ ”

G-94’s own condition, $f^\dagger f = I$, is satisfied by every unitary; the source squares f instead.

Kempe, Kitaev and Regev, "The complexity of the local Hamiltonian problem" (quant-ph/0406180), Theorem 3, LaTeX source

`papers/kempe2004-complexity-local-hamiltonian/source/2local-lanl-resub7.tex` supports: G-319, G-322, G-340

“Then each eigenvalue λ_j of $H|_{\text{call}_j}$ is”

The source’s own `\pert{...}` macro marks both the eigenvalue and the space with the perturbed-Hamiltonian tilde; G-319’s draft sentence carries neither. Read here rather than through the `papers/cited` extract below, whose OCR loses exactly this tilde.

Kempe, Kitaev and Regev, "The complexity of the local Hamiltonian problem" (quant-ph/0406180), arXiv extract

`papers/cited/kempe2004complexity.txt` supports: G-4, G-56, G-88

“Theorem 1 The 2-LOCAL HAMILTONIAN problem is QMA-complete.”

The 2-local completeness result Appendix XV’s gadget apparatus is built on.

Cade, Folkertsma and Weggemans, "Complexity of the guided local Hamiltonian problem" (arXiv:2207.10097)

`papers/cited/cade2022complexity.txt` supports: G-4, G-5

“GLHLE(k, c, ζ, δ) is BQP-hard for $k \geq 2$, $0 \leq c \leq 0(\text{poly}(n))$ ”

Cited for the BQP side of the draft’s guided-local-Hamiltonian apparatus. None of the eight pivotal claims turns on it.

Weggemans, Folkertsma and Cade, "Guidable local Hamiltonian problems" (arXiv:2302.11578)

`papers/cited/weggemans2023guidable.txt` supports: G-6

“ $\|\Pi_{\text{gs}} u\|_2 \geq \zeta$.”

Cited for the QCMA side of the same apparatus. Likewise outside the eight.

3 Live claims

Rows citing a line of the draft’s own uncommented text, in the draft’s line order.

THE DRAFT SAYS	LEAN’S VERDICT
G-43 main.tex:72 Author footnote: a mailto link whose target is manish@accelequant.in and whose displayed text is manish@accelequant.com.	FALSE measured / – method: tools/check_paper_integrity.py evidence: tools/paper.toml
G-94 Definition (unlabelled, attributed ‘\cite{montanaro2008quantum}’): "A Quantum Boolean Function of ‘n’ qubits is a unitary operator ‘f’ on ‘n’ qubits such that ‘f[†] = I’." main.tex:249 Definition [montanaro2008quantum]: "A Quantum Boolean Function of n qubits is a unitary operator f on n qubits such that f[†] = I."	FALSE lean / L2 lean/Qgadget/Verbatim/Definition_QuantumBooleanFunction.lean#isQuantumBooleanFunction_iff_mem_unitaryGroup theorem isQuantumBooleanFunction_iff_mem_unitaryGroup {n : ℕ} (f : Matrix (Fin n) (Fin n) ℂ) : IsQuantumBooleanFunction f ↔ f ∈ Matrix.unitaryGroup (Fin n) ℂ := by THE DRAFT’S CONDITION IS THE WHOLE UNITARY GROUP. ‘ff = I’ is exactly membership in ‘Matrix.unitaryGroup’, for every ‘n’ and every ‘f’ — not a further restriction on unitaries, since ‘ff = I’ (a ONE-SIDED identity, for a SQUARE matrix) already forces the other side ‘f ff = I’ too (‘Matrix.mem_unitaryGroup_iff’, via [...]) lean/Qgadget/Verbatim/Definition_QuantumBooleanFunction.lean#phaseGate_isQuantumBooleanFunction theorem phaseGate_isQuantumBooleanFunction : IsQuantumBooleanFunction phaseGate := by ‘phaseGate’ is unitary, hence a quantum Boolean function under the draft’s ‘ff = I’. REAL: entrywise, ‘phaseGate[†]phaseGate = diag(1, conj(i)·i) = diag(1, 1) = I’ genuinely uses ‘conj(i)·i = 1’ (equivalently ‘(-i)·i = 1’), which fails for a non-unit-modulus diagonal entry — replace ‘i’ by ‘2i’ and the product is ‘4’, not ‘1’. lean/Qgadget/Verbatim/Definition_QuantumBooleanFunction.lean#phaseGate_sq_ne_one theorem phaseGate_sq_ne_one : phaseGate * phaseGate ≠ 1 := by ‘phaseGate² ≠ I’: ‘phaseGate’ is NOT a quantum Boolean function under the source’s own ‘f² = I’. REAL: ‘phaseGate² = diag(1, i²) = diag(1, -1)’, and its ‘(1,1)’ entry ‘-1’ genuinely differs from ‘1’s ‘(1,1)’ entry ‘1’ in ‘ℂ’ — this is exactly the computation that would fail were the source’s involutive condition equivalent to the [...]

THE DRAFT SAYS (cont.)

G-95 the unlabelled ‘example’ immediately after Definition 249, the first worked instance of a Quantum Boolean Function.

main.tex:252

Example: for θ in $\mathbb{R}$, $f(\theta) = [[\cos \theta, \sin \theta], [\sin \theta, \cos \theta]]$ is a quantum boolean function.

G-113 two inline remarks inside the QLSA proof of §7.2 ("An Efficient Quantum Algorithm for QLSA"), Steps III and IV. Neither is a numbered environment: both are prose sentences inside the algorithm’s ‘Proof:’ block, so they are witnessed here rather than in Verbatim/.

main.tex:338

Remark: "Due to the assumption on the bound on the norm of the matrix A , $1/\kappa \leq 1/(\lambda_{\min} A)$, the above quantum state is well defined."

G-114 two inline remarks inside the QLSA proof of §7.2 ("An Efficient Quantum Algorithm for QLSA"), Steps III and IV. Neither is a numbered environment: both are prose sentences inside the algorithm’s ‘Proof:’ block, so they are witnessed here rather than in Verbatim/.

main.tex:341

Step IV: "Apply the inverse quantum Fourier Transform to the first two registers to obtain $\sum_j \beta_j |0^n\rangle |\psi_j\rangle (\dots)$ ".

LEAN’S VERDICT (cont.)

FALSE lean / L2

lean/Qgadget/Verbatim/Example_QuantumBooleanFunction_Theta.lean#thetaExample_not_QBF_at_pi_div_four

theorem thetaExample_not_QBF_at_pi_div_four :

¬ IsQuantumBooleanFunction (thetaExample (Real.pi / 4)) := by

FINDING (G-95): the draft’s example at ‘main.tex:252’-‘main.tex:261’ is not a Quantum Boolean Function. At ‘ $\theta = \pi/4$ ’ its determinant vanishes (‘thetaExample_det_pi_div_four’), so by ‘isQuantumBooleanFunction_det_ne_zero’ it cannot satisfy ‘ $\det f = I$ ’.

lean/Qgadget/Verbatim/Example_QuantumBooleanFunction_Theta.lean#thetaExampleCorrected_isQBF

theorem thetaExampleCorrected_isQBF ($\theta : \mathbb{R}$) :

IsQuantumBooleanFunction (thetaExampleCorrected θ) := by

FINDING: the source’s own matrix IS a Quantum Boolean Function, for every real ‘ θ ’. Combines self-adjointness with the involution just proved: ‘ $\det f = \det f^* = I$ ’.

lean/Qgadget/Verbatim/Example_QuantumBooleanFunction_Theta.lean#thetaExampleCorrected_sq

theorem thetaExampleCorrected_sq ($\theta : \mathbb{R}$) :

thetaExampleCorrected θ * thetaExampleCorrected θ = 1 := by

The corrected matrix squares to the identity, for every ‘ θ ’. This is the source’s own definition of a Quantum Boolean Function (‘ $\det f = I$ ’), and it holds identically — no special value of ‘ θ ’ is used.

TRUE lean / L2-PARTIAL

lean/Qgadget/Findings/Section_7_QLSA_Remarks.lean#qlsa_step3_interval

theorem qlsa_step3_interval ($\lambda \kappa : \mathbb{R}$) ($h\kappa : 1 \leq \kappa$)

($h\lambda_{\min} : 1 / \kappa \leq \lambda$) ($h\lambda_{\max} : \lambda \leq 1$) :

$1 / \kappa \leq 1 / (\lambda * \kappa) \wedge 1 / (\lambda * \kappa) \leq 1$:= by

The TRUE half of ‘main.tex:338’. For ‘ $\lambda \in [1/\kappa, 1]$ ’ with ‘ $\kappa \geq 1$ ’, ‘ $1/(\lambda\kappa) \in [1/\kappa, 1]$ ’.

FALSE lean / L2-PARTIAL

lean/Qgadget/Findings/Section_7_QLSA_Remarks.lean#hadamard_ket1_has_nonzero_ket1_amplitude

theorem hadamard_ket1_has_nonzero_ket1_amplitude :

qftBoolean 1 (fun _ => 1) (fun _ => 1) ≠ 0 := by

The ‘ $n = 1$ ’ instance of ‘main.tex:341’'s claim. The one-qubit transform ‘qftBoolean 1’, applied to the register holding ‘ $|1\rangle$ ’ (the constant function ‘ $1 : \text{Fin } 1 \rightarrow \mathbb{Z}/2\mathbb{Z}$ ’), has a NONZERO ‘ $|1\rangle$ ’-amplitude in its output — so the claimed uncomputation to ‘ $|0\rangle$ ’ fails at the smallest register size the claim covers.

THE DRAFT SAYS (cont.)

G-118 §7.3 "Quantum vs Classical Constraint" — the expanded-Pauli display for the clause Hamiltonian $H_i = |001\rangle\langle 001|$ ('main.tex:386'-'main.tex:390'), and the discretised-amplitude remark immediately after it ('main.tex:392'-'main.tex:400').

main.tex:389

" $H_i = |001\rangle\langle 001| = I_{1I_2I_3} + I_{1Z_2I_3} - I_{1I_2I_3} - I_{1Z_2Z_3} + Z_{1I_2I_3} + Z_{1Z_2I_3} - Z_{1I_2Z_3} - Z_{1Z_2Z_3}$."

G-119 §7.3 "Quantum vs Classical Constraint" — the expanded-Pauli display for the clause Hamiltonian $H_i = |001\rangle\langle 001|$ ('main.tex:386'-'main.tex:390'), and the discretised-amplitude remark immediately after it ('main.tex:392'-'main.tex:400').

main.tex:389

The expansion of $|001\rangle\langle 001|$ contains the term " $- I_{1I_2I_3}$ " and no " $- I_{1I_2Z_3}$ " term.

LEAN'S VERDICT (cont.)

FALSE lean / L2

lean/Qgadget/Findings/Section_7_QuantumVsClassicalConstraint.lean#eightTermsCoeffOne_eq_eight_smul_proj001

theorem eightTermsCoeffOne_eq_eight_smul_proj001 :
 eightTermsCoeffOne = fun x => 8 * proj001 x := by

FINDING (G-118): with the signs corrected, the printed coefficient '1' gives ' $8 \cdot |001\rangle\langle 001|$ ', not ' $|001\rangle\langle 001|$ '. REAL, and it is a genuine 8-case computation: at every ' $x \neq \text{ket001}$ ' at least one factor ' $(1 \pm zVal \cdot)$ ' in the underlying product ' $(1+Z_1)(1+Z_2)(1-Z_3)$ ' vanishes, and at ' $x = \text{ket001}$ ' all three factors are '2'; get any one sign [...]

FALSE lean / L2

lean/Qgadget/Findings/Section_7_QuantumVsClassicalConstraint.lean#printedOperator_trace_zero

theorem printedOperator_trace_zero :
 ($\sum x : \text{Bool} \times \text{Bool} \times \text{Bool}, \text{printedOperator } x$) = 0 := by

The printed operator has trace '0'. REAL, an 8-case sum: it is not that each individual basis state contributes '0' ('printedOperator_at_001' below shows ' ket001 ' alone contributes '6') — the eight contributions cancel only in total, which is the genuine content of "traceless" for an operator that is not the zero operator.

lean/Qgadget/Findings/Section_7_QuantumVsClassicalConstraint.lean#printedOperator_at_001

theorem printedOperator_at_001 : printedOperator ket001 = 6 := by

The printed operator's eigenvalue on ' $|001\rangle$ ' is '6'. REAL: it is the sum of six ' ± 1 ' products, each a genuine sign computation on ' $zVal \text{ false} = 1$ ', ' $zVal \text{ true} = -1$ ' — not a definitional unfolding, since the six terms do not all carry the same sign.

lean/Qgadget/Findings/Section_7_QuantumVsClassicalConstraint.lean#proj001_trace_one

theorem proj001_trace_one : ($\sum x : \text{Bool} \times \text{Bool} \times \text{Bool}, \text{proj001 } x$) = 1 := by

' $|001\rangle\langle 001|$'s trace is '1' — the projector's own defining property, recorded for the contrast with ' $\text{printedOperator_trace_zero}$ '.

lean/Qgadget/Findings/Section_7_QuantumVsClassicalConstraint.lean#printedOperator_ne_smul_proj001

theorem printedOperator_ne_smul_proj001 (c : $\mathbb{R}$) :
 printedOperator $\neq$ fun x => c * proj001 x := by

FINDING (G-119, the closing sentence): no overall factor repairs the printed operator into ' $|001\rangle\langle 001|$ '. ' $\text{printedOperator_ne_proj001}$ ' is the ' $c = 1$ ' case; this is the general statement the row's closing sentence makes.

THE DRAFT SAYS (cont.)

G-120 §7.3 "Quantum vs Classical Constraint" — the expanded-Pauli display for the clause Hamiltonian $H_i = |001\rangle\langle 001|$ ('main.tex:386'-'main.tex:390'), and the discretised-amplitude remark immediately after it ('main.tex:392'-'main.tex:400').

main.tex:396

"But the space is discretized. For example, one can work with α_i in $\{-1, 1\}$ while still able to do universal quantum computation."

LEAN'S VERDICT (cont.)

FALSE lean / L2

lean/Qgadget/Findings/Section_7_QuantumVsClassicalConstraint.lean#normSq_psiOf

theorem normSq_psiOf (α : Fin 8 $\rightarrow$ $\mathbb{R}$) ($h\alpha$: IsPMOne α) :

$\|\text{psiOf } \alpha\|^2 = 8$:= by

Every discretised vector of 'main.tex:398' has squared norm '8'. REAL, and it could have failed: it uses ' $h\alpha$ ' pointwise (' $\alpha_i^2 = 1$ ' from either branch of the disjunction) inside a sum of 8 non-constant terms — an ' α ' outside ' $\{-1, 1\}$ ' (say ' $\alpha_i = 1/2$ ' for one ' i ') gives a smaller sum, and 'decide'/'rfl' see nothing here since ' α ' is an [...]

lean/Qgadget/Findings/Section_7_QuantumVsClassicalConstraint.lean#overlap_sq_normalized_psiOf

theorem overlap_sq_normalized_psiOf (α : Fin 8 $\rightarrow$ $\mathbb{R}$) ($h\alpha$: IsPMOne α) (j : Fin 8) :

$\|(\text{inner } (k := \mathbb{C}) (\text{EuclideanSpace.single } j (1 : \mathbb{C}))$

FINDING (G-120): after normalisation, the squared overlap with every basis state is ' $1/8$ '. The vector 'main.tex:398' discretises resembles no basis state — ' $|000\rangle$ ', the starting state of every circuit, included.

THE DRAFT SAYS (cont.)

G-147 the unlabelled Lemma of §8.1 "Local Quantum Phase Estimation": > If we apply ‘V’ to the state $|\psi\rangle$ and measure the answer register in the > computational basis, the probability of obtaining outcome ‘1’ is equal > to $1 - (1/m)\langle\eta|H|\eta\rangle$. The draft’s proof (‘main.tex:539’-‘main.tex:573’) derives this from the action of ‘W_j’ on ‘ $|\eta\rangle$ ’ expanded in the eigenbasis of each ‘H_j’

main.tex:532

Lemma: "If we apply V to the state $|\psi\rangle$ and measure the answer register in the computational basis, the probability of obtaining outcome 1 is equal to $1 - (1/m)\langle\eta|H|\eta\rangle$."

LEAN’S VERDICT (cont.)

TRUE lean / L2

```
lean/Qgadget/Verbatim/Lemma_LocalVerifierAcceptanceProbability.lean#branchAcceptProb_eq
theorem branchAcceptProb_eq (Hj : Matrix d d ℂ) (hHj : Hj.IsHermitian)
  (η : EuclideanSpace ℂ d) (hη : ||η|| = 1) :
  branchAcceptProb Hj hHj η = 1 - Qgadget.Borrowed.energy Hj η := by
  ‘main.tex:552’-‘main.tex:567’: the per-branch acceptance probability is ‘ $1 - \langle\eta|H_j|\eta\rangle$ ’, for every
  unit ‘η’. REAL, and it could have failed: it needs ‘η’ a UNIT vector (Parseval, ‘Orthonormal-
  Basis.sum_sq_norm_inner_right’) — an unnormalised ‘η’ would leave ‘ $\sum_s |\alpha_s|^2 \neq 1$ ’ and the
  identity would gain that factor instead of vanishing.

lean/Qgadget/Verbatim/Lemma_LocalVerifierAcceptanceProbability.lean#averagedAcceptProb_eq
theorem averagedAcceptProb_eq {m : ℕ} (hm : m ≠ 0) (H : Fin m → Matrix d d ℂ)
  (hH : ∀ j, (H j).IsHermitian) (η : EuclideanSpace ℂ d) (hη : ||η|| = 1) :
  averagedAcceptProb H hH η = 1 - (1 / (m : ℝ)) * Qgadget.Borrowed.energy (∑ j, H j) η := by
  ‘main.tex:539’-‘main.tex:546’ and ‘eqn: local-to-global_Ham’ (‘main.tex:544’-‘main.tex:545’),
  together: the averaged acceptance probability is ‘ $1 - (1/m)\langle\eta|\sum_j H_j|\eta\rangle$ ’. REAL: the linearity
  step (‘henergy_sum’ below) genuinely uses that ‘energy’ is built from a LINEAR map (‘Ma-
  trix.toEuclideanLin’) and a sesquilinear inner product — [...]
```

```
lean/Qgadget/Verbatim/Lemma_LocalVerifierAcceptanceProbability.lean#averagedAcceptProb_eq_
acceptProb
theorem averagedAcceptProb_eq_acceptProb (n m : ℕ) (hm : m ≠ 0)
  (H : Fin m → Matrix (Qgadget.Borrowed.Qubits n) (Qgadget.Borrowed.Qubits n) ℂ)
  (hH : ∀ j, (H j).IsHermitian) (η : EuclideanSpace ℂ (Qgadget.Borrowed.Qubits n))
  (hη : ||η|| = 1) :
  averagedAcceptProb H hH η = Qgadget.Core.acceptProb H η := by
```

FINDING (G-147): the paper’s Lemma holds — the eigenbasis-level construction its own proof performs agrees exactly with ‘Core.acceptProb’, ‘Qgadget.Core.’s transcription of the Lemma’s closed-form conclusion, over the actual register ‘Qgadget.Borrowed.Qubits n’ the local verifier acts on.

THE DRAFT SAYS (cont.)

G-163 §8.2 "Difference between Ground State and Guiding State" — the Case I/II fidelity-versus-energy analysis. The proof of Lemma [exp-fidelity] ('main.tex:612'-'main.tex:651') writes the decomposition ' $|\psi\rangle = \alpha_0|\psi_0\rangle + \alpha_1|\psi_0^\perp\rangle$ ' ('main.tex:628'), the resulting expansion of ' $\langle\psi|H|\psi\rangle$ ' ('main.tex:632'), the amplitude bound on ' $|\psi_0^\perp\rangle$ ' the fidelity

main.tex:632

$$\langle\psi|H|\psi\rangle = |\alpha_1|^2 \langle\psi_0|H|\psi_0\rangle + |\alpha_2|^2 \langle\psi_0^\perp|H|\psi_0^\perp\rangle.$$

LEAN'S VERDICT (cont.)

FALSE lean / L2

lean/Qgadget/Findings/Section_8_GroundVsGuidingState.lean#energy_hCrossTerm_ketPlus

theorem energy_hCrossTerm_ketPlus : Qgadget.Borrowed.energy hCrossTerm ketPlus = 1 := by

The left side of 'main.tex:632' at the witness: ' $\langle\psi|H|\psi\rangle = 1$ '. REAL, and it could have failed: 'H' is not diagonal in this basis, so the value genuinely depends on the off-diagonal entry of 'pauliX' — replacing it by the zero matrix (leaving 'H = I/2') would give '1/2', not '1'.

lean/Qgadget/Findings/Section_8_GroundVsGuidingState.lean#energy_hCrossTerm_ket0

theorem energy_hCrossTerm_ket0 : Qgadget.Borrowed.energy hCrossTerm ket0 = 1 / 2 := by

' $\langle\psi_0|H|\psi_0\rangle = 1/2$ ' at the witness.

lean/Qgadget/Findings/Section_8_GroundVsGuidingState.lean#energy_hCrossTerm_ket1

theorem energy_hCrossTerm_ket1 : Qgadget.Borrowed.energy hCrossTerm ket1 = 1 / 2 := by

' $\langle\psi_0^\perp|H|\psi_0^\perp\rangle = 1/2$ ' at the witness.

lean/Qgadget/Findings/Section_8_GroundVsGuidingState.lean#overlap_ket0_ketPlus_sq

theorem overlap_ket0_ketPlus_sq : ||(inner (k := ℂ) ket0 ketPlus : ℂ)|| ^ 2 = 1 / 2 := by

' $|\alpha_1|^2 = |\langle\psi_0|\psi\rangle|^2 = 1/2$ '.

lean/Qgadget/Findings/Section_8_GroundVsGuidingState.lean#overlap_ket1_ketPlus_sq

theorem overlap_ket1_ketPlus_sq : ||(inner (k := ℂ) ket1 ketPlus : ℂ)|| ^ 2 = 1 / 2 := by

' $|\alpha_2|^2 = |\langle\psi_0^\perp|\psi\rangle|^2 = 1/2$ '.

lean/Qgadget/Findings/Section_8_GroundVsGuidingState.lean#guided_vs_ground_display_false

theorem guided_vs_ground_display_false :

Qgadget.Borrowed.energy hCrossTerm ketPlus ≠

|| (inner (k := ℂ) ket0 ketPlus : ℂ) || ^ 2 * Qgadget.Borrowed.energy hCrossTerm ket0 +

FINDING (G-163): 'main.tex:632's displayed identity fails at a witness meeting every hypothesis of the lemma it appears in the proof of — the left side is '1', the printed right side is '1/2'. The missing cross term '2 Re($\bar{\alpha}_1\alpha_0\langle\psi_0|H|\psi_0^\perp\rangle$)' (the 'corrected_statement') is exactly this discrepancy, and it vanishes only when ' $|\psi_0\rangle$ ' is an [...]

THE DRAFT SAYS (cont.)

G-164 §8.2 "Difference between Ground State and Guiding State" — the Case I/II fidelity-versus-energy analysis. The proof of Lemma [exp-fidelity] ('main.tex:612'- 'main.tex:651') writes the decomposition ' $|\psi\rangle = \alpha_0|\psi_0\rangle + \alpha_1|\psi_0^\perp\rangle$ ' ('main.tex:628'), the resulting expansion of ' $\langle\psi|H|\psi\rangle$ ' ('main.tex:632'), the amplitude bound on ' $|\psi_0^\perp\rangle$ ' the fidelity

main.tex:636

" $|\langle\psi|\psi_0\rangle| \geq 1 - 1/2^n$ implies $|\langle\psi|\psi_0^\perp\rangle| \leq 1/2^n$."

G-166 §8.2 "Difference between Ground State and Guiding State" — the Case I/II fidelity-versus-energy analysis. The proof of Lemma [exp-fidelity] ('main.tex:612'- 'main.tex:651') writes the decomposition ' $|\psi\rangle = \alpha_0|\psi_0\rangle + \alpha_1|\psi_0^\perp\rangle$ ' ('main.tex:628'), the resulting expansion of ' $\langle\psi|H|\psi\rangle$ ' ('main.tex:632'), the amplitude bound on ' $|\psi_0^\perp\rangle$ ' the fidelity

main.tex:644

YES instance: $\langle\psi|H|\psi\rangle = |\alpha_1|^2 \langle\psi_0|H|\psi_0\rangle + |\alpha_0|^2 \langle\psi_0^\perp|H|\psi_0^\perp\rangle \leq (1 - 1/2^n) a + 1/2^n$.

LEAN'S VERDICT (cont.)

FALSE lean / L2

lean/Qgadget/Findings/Section_8_GroundVsGuidingState.lean#overlap_ket0_psiG164

theorem overlap_ket0_psiG164 : $\|(inner (k := \mathbb{C}) \text{ket0 psiG164} : \mathbb{C})\| = 1 / 2 := by$

The hypothesis, met with equality: ' $|\langle\psi|\psi_0\rangle| = 1/2 = 1 - 1/2^1$ '.

lean/Qgadget/Findings/Section_8_GroundVsGuidingState.lean#tail_norm_psiG164

theorem tail_norm_psiG164 :

$\|Qgadget.Borrowed.Analysis.tail \text{ket0 psiG164}\| = \text{Real.sqrt } 3 / 2 := by$

The exact orthogonal amplitude, 'main.tex:637's own value: ' $\sqrt{3}/2$ '. REAL: 'Real.sqrt_div' and 'Real.sqrt_sq' turn the squared-norm fact into the exact value the statement names, not merely a bound — ' $\sqrt{(3/4)} = \sqrt{3}/\sqrt{4} = \sqrt{3}/2$ ' uses that '4' is a perfect square, which '3' is not, so the two square roots do not simplify the same way.

lean/Qgadget/Findings/Section_8_GroundVsGuidingState.lean#half_lt_tail_norm_psiG164

theorem half_lt_tail_norm_psiG164 :

$(1 : \mathbb{R}) / 2 < \|Qgadget.Borrowed.Analysis.tail \text{ket0 psiG164}\| := by$

The refutation, in the paper's own amplitude language: the orthogonal AMPLITUDE ' $\|\text{tail ket0 psiG164}\| = \sqrt{3}/2$ ' is strictly more than ' $1/2^n = 1/2$ ', where 'main.tex:636' concludes it is at most ' $1/2^n$ '.

FALSE lean / L2

lean/Qgadget/Findings/Section_8_GroundVsGuidingState.lean#energy_projKet1_ket0

theorem energy_projKet1_ket0 : $Qgadget.Borrowed.energy \text{projKet1 ket0} = 0 := by$

The ground energy is ' $0 = a$ ', as the lemma's YES hypothesis requires.

lean/Qgadget/Findings/Section_8_GroundVsGuidingState.lean#energy_projKet1_psiG164

theorem energy_projKet1_psiG164 : $Qgadget.Borrowed.energy \text{projKet1 psiG164} = 3 / 4 := by$

The witness's energy is ' $3/4$ '. REAL: the value depends on BOTH components of 'psiG164', since 'projKet1' is diagonal with a nonzero entry precisely where the ' $\sqrt{3}/2$ '-amplitude sits.

lean/Qgadget/Findings/Section_8_GroundVsGuidingState.lean#yes_bound_fails

theorem yes_bound_fails : $\neg Qgadget.Borrowed.energy \text{projKet1 psiG164} \leq (1 - 1 / 2^n) * 0 + 1 / 2^n := by$

FINDING (G-166): the YES bound of 'main.tex:644' fails at a witness meeting every one of its hypotheses — the ground state 'ket0's overlap with 'psiG164' is ' $1 - 1/2^1$ ' and its ground energy is ' $a = 0$ ', yet the energy of 'psiG164' is ' $3/4$ ', strictly above the claimed bound ' $1/2$ '.

THE DRAFT SAYS (cont.)

G-167 §8.2 "Difference between Ground State and Guiding State" — the Case I/II fidelity-versus-energy analysis. The proof of Lemma [exp-fidelity] ('main.tex:612'-'main.tex:651') writes the decomposition ' $|\psi\rangle = \alpha_0|\psi_0\rangle + \alpha_1|\psi_0^\perp\rangle$ ' ('main.tex:628'), the resulting expansion of ' $\langle\psi|H|\psi\rangle$ ' ('main.tex:632'), the amplitude bound on ' $|\psi_0^\perp\rangle$ ' the fidelity

main.tex:648

NO instance: $\langle\psi|H|\psi\rangle = |\alpha_1|^2 \langle\psi_0|H|\psi_0\rangle + |\alpha_2|^2 \langle\psi_0^\perp|H|\psi_0^\perp\rangle \geq (1 - 1/2^n) b + 1/2^n$.

G-170 Lemma [fidelity_1-poly] ('main.tex:654'-'main.tex:664'), Case II of §8.2's ground-vs-guiding analysis: > In the context of the Local Hamiltonian problem, let there be (i) two > 'n'-qubit states ' $|\psi\rangle$ ' and ' $|\psi_0\rangle$ ', and (ii) the Hamiltonian (which forms > a POVM ' $\{H, I-H\}$ '). Assume the following three conditions hold > (simultaneously): ' $\langle\psi_0|H|\psi_0\rangle \leq a$ ' (YES) or ' $\langle\psi_0|H|\psi_0\rangle \geq b$ ' (NO);

main.tex:661

Lemma [fidelity_1-poly]: under $\langle\psi_0|H|\psi_0\rangle \leq a$ (YES) or $\geq b$ (NO), $b - a \geq 1/p_3(n)$ and $|\langle\psi|\psi_0\rangle| \geq 1 - 1/p_2(n)$, "one can directly infer the membership of the input H by estimating $\langle\psi|H|\psi\rangle$ in place of $\langle\psi_0|H|\psi_0\rangle$ if $1/p_2(n) < 1/p_3(n)$."

LEAN'S VERDICT (cont.)

FALSE lean / L2

lean/Qgadget/Findings/Section_8_GroundVsGuidingState.lean#energy_half_identity

```
theorem energy_half_identity {d : Type*} [Fintype d] [DecidableEq d]
  (ψ : EuclideanSpace ℂ d) (hψ : ||ψ|| = 1) :
  Qgadget.Borrowed.energy ((1 / 2 : ℂ) • (1 : Matrix d d ℂ)) ψ = 1 / 2 := by
```

'H = (1/2)I' gives energy '1/2' to every unit state, on any finite register. REAL: it could have failed for any Hamiltonian that is not a scalar multiple of the identity — the whole content is that ' $(c \bullet 1)$ ' acts as ' c ' on every vector, not merely on an eigenvector.

lean/Qgadget/Findings/Section_8_GroundVsGuidingState.lean#no_bound_fails

```
theorem no_bound_fails (n : ℕ) (ψ : EuclideanSpace ℂ (Qgadget.Borrowed.Qubits n))
  (hψ : ||ψ|| = 1) :
  Qgadget.Borrowed.energy ((1 / 2 : ℂ) • (1 : Matrix (Qgadget.Borrowed.Qubits n)
    (Qgadget.Borrowed.Qubits n) ℂ)) ψ < (1 - (1 / 2 : ℝ) ^ n) * (1 / 2) + (1 / 2 : ℝ) ^ n :=
  by
```

FINDING (G-167): the NO bound of 'main.tex:648' is a strict upper bound on every state's energy under 'H = (1/2)I', not a lower one — for EVERY qubit count 'n' and every unit state (the exact NO ground state included), the actual energy '1/2' sits strictly below the claimed bound ' $(1-1/2^n)(1/2) + 1/2^n$ '. REAL: ' $(1/2)^n > 0$ ' for every 'n' [...]

FALSE lean / L2

lean/Qgadget/Verbatim/Lemma_FidelityOnePoly.lean#energy_projKet1_psiG170YES

```
theorem energy_projKet1_psiG170YES :
  Qgadget.Borrowed.energy projKet1 psiG170YES = 5 / 9 := by
```

The YES instance's energy is '5/9'. 'H = projKet1', ground state 'ket0' (energy '0 = a'), witness 'psiG170YES' (overlap '2/3' with the ground state).

lean/Qgadget/Verbatim/Lemma_FidelityOnePoly.lean#overlap_ket0_psiG170YES

```
theorem overlap_ket0_psiG170YES : (inner (k := C) ket0 psiG170YES : ℂ) = 2 / 3 := by
```

The overlap with the ground state is exactly '2/3 = 1 - 1/p₂' at 'p₂ = 3'.

lean/Qgadget/Verbatim/Lemma_FidelityOnePoly.lean#fidelityOnePolyClaim_fails

```
theorem fidelityOnePolyClaim_fails : ¬ FidelityOnePolyClaim 0 (1 / 2) 3 2 := by
```

FINDING (G-170): at 'p₃ = 2', 'p₂ = 3' — meeting '1/p₂ < 1/p₃' — a YES instance's energy '5/9' exceeds a NO instance's energy '1/2', both instances meeting every hypothesis of 'main.tex:654'-'main.tex:664'. The YES instance: 'H = projKet1', ground state 'ket0' at energy '0 = a', gap 'b-a = 1/2 = 1/p₃', witness 'psiG170YES' at overlap [...]

THE DRAFT SAYS (cont.)	LEAN'S VERDICT (cont.)
G-181 §8.4 "Satisfying quantum ‘assignments’ and low energy subspaces" — the worked CSP example (‘main.tex:712’-‘main.tex:722’) and the general definition of the "satisfying assignment" set that follows it (‘main.tex:724’-‘main.tex:728’). main.tex:722 "For $x\rangle$, where x in $\{0,1\}^4$, the function $f(x) := \langle x H x\rangle$ counts the number of unsatisfied clauses for x with respect to $C(x_1, x_2, x_3, x_4)$."	TRUE lean / L2 lean/Qgadget/Findings/Section_8_SatisfyingAssignments.lean#H_diag_eq_numViolated theorem H_diag_eq_numViolated (x : Assign4) : H1eig x + H2eig x + H3eig x = (numViolated x : ℝ) := by FINDING (G-181): the sum of the three projector eigenvalues equals the number of violated clauses, at every one of the 16 assignments. REAL, a genuine 16-case computation: each ‘H_ieig’ is built from the ‘$(1\pm Z)/2$’ identity on TWO of the four variables (not from "violates ‘C_i’" directly), and the claim is that this projector construction [...]
G-182 §8.4 "Satisfying quantum ‘assignments’ and low energy subspaces" — the worked CSP example (‘main.tex:712’-‘main.tex:722’) and the general definition of the "satisfying assignment" set that follows it (‘main.tex:724’-‘main.tex:728’). main.tex:724 "For a general local Hamiltonian instance H in $L(C^{\sim\{2^n\}})$, the notion of satisfying assignment is a vector subspace (footnote: We include zero vector to the set.) of vector space $VS(C^{\sim\{2^n\}})$ given as $S_s := \{ \psi\rangle : \langle\psi H \psi\rangle \leq a\}$ for some specified parameter a in $\mathbb{R}$."	FALSE lean / L2 lean/Qgadget/Findings/Section_8_SatisfyingAssignments.lean#ket0G182_mem_satisfyingSet theorem ket0G182_mem_satisfyingSet : ket0G182 ∈ satisfyingSet (1 : Matrix (Fin 2) (Fin 2) ℂ) 1 := by ‘$0\rangle \in S_s$’ at ‘$H = I$’, ‘$a = 1$’: energy ‘$1 \leq 1$’. lean/Qgadget/Findings/Section_8_SatisfyingAssignments.lean#two_smul_ket0G182_not_mem_satisfyingSet theorem two_smul_ket0G182_not_mem_satisfyingSet : (2 : ℂ) • ket0G182 ∉ satisfyingSet (1 : Matrix (Fin 2) (Fin 2) ℂ) 1 := by ‘$2 0\rangle \notin S_s$’ at the same ‘H, a’: energy ‘$4 > 1$’. lean/Qgadget/Findings/Section_8_SatisfyingAssignments.lean#satisfyingSet_not_closed_under_smul theorem satisfyingSet_not_closed_under_smul : ¬ ∀ (c : ℂ) (ψ : EuclideanSpace ℂ (Fin 2)), ψ ∈ satisfyingSet (1 : Matrix (Fin 2) (Fin 2) ℂ) 1 → c • ψ ∈ satisfyingSet (1 : Matrix (Fin 2) (Fin 2) ℂ) 1 := by FINDING (G-182): ‘S_s’ is not closed under scalar multiplication, so ‘main.tex:724’'s claim that it is a vector subspace is FALSE — ‘$0\rangle \in S_s$’ and ‘$2 0\rangle \notin S_s$’ for the same ‘$H = I$’, ‘$a = 1$’. A set not closed under scalar multiplication is not a ‘Submodule’, and no ‘Submodule’ of ‘EuclideanSpace ℂ (Fin 2)’ can coincide with ‘S_s’ here.

THE DRAFT SAYS (cont.)

G-194 Section 9, the paragraph distinguishing MAX-2SAT’s YES/NO thresholds from the Local Hamiltonian problem’s, which states the numeric thresholds of the 3SAT-to-MAX2SAT reduction.

main.tex:851

"the MAX-2SAT emerging from the reduction of 3SAT has the YES-threshold of 7m, where m is the number of clauses in the input 3SAT. Similarly, for NO instance, any assignment should satisfy at most 6m clause, also called NO-threshold."

LEAN’S VERDICT (cont.)

FALSE lean / L2

lean/Qgadget/Findings/Section_9_MAX2SATThreshold.lean#card_violated_eq_one

theorem card_violated_eq_one (x : Fin 3 → Bool) :

(Finset.univ.filter (fun c : Fin 3 → Bool => ViolatesClause c x)).card = 1 := by

Every assignment violates EXACTLY ONE of the eight clauses. The eight clauses, one per sign pattern, partition the eight assignments: the set of clauses ‘x’ violates is the singleton ‘{x}’.

lean/Qgadget/Findings/Section_9_MAX2SATThreshold.lean#card_satisfied_eq_seven

theorem card_satisfied_eq_seven (x : Fin 3 → Bool) :

(Finset.univ.filter (fun c : Fin 3 → Bool => ¬ ViolatesClause c x)).card = 7 := by

The other seven clauses are satisfied. Immediate from ‘card_violated_eq_one’ and the total clause count ‘8 = 2³’.

lean/Qgadget/Findings/Section_9_MAX2SATThreshold.lean#witness_total_satisfied_two_clauses

theorem witness_total_satisfied_two_clauses (x : Fin 3 → Bool) (satVal violVal : ℕ)

(hsat : satVal = 7) (hviol : violVal = 6) :

satVal * (Finset.univ.filter (fun c : Fin 3 → Bool => ¬ ViolatesClause c x)).card
+ violVal * (Finset.univ.filter (fun c : Fin 3 → Bool => ViolatesClause c x)).card
= 55 := by

FINDING (G-194), first half: the best auxiliary choice satisfies ‘55’ of the ‘80’ 2SAT clauses, not ‘48’. Taking ‘main.tex:1681’'s own per-clause numbers — ‘7’ for a satisfied clause at its best value, ‘6’ for the one violated clause at its best — as given, and the exact counts ‘7’ and ‘1’ from above, the total is ‘55’.

lean/Qgadget/Findings/Section_9_MAX2SATThreshold.lean#witness_total_sum_le_55

theorem witness_total_sum_le_55 (x : Fin 3 → Bool) (v : (Fin 3 → Bool) → ℕ)

(hsat : ∀ c, ¬ ViolatesClause c x → v c ≤ 7)

(hviol : ∀ c, ViolatesClause c x → v c ≤ 6) :

∑ c, v c ≤ 55 := by

FINDING (G-194), second half: ‘55’ is not merely achieved but is an upper bound, for EVERY choice of auxiliary values — so ‘56’ is never reached, closing the ‘open_step’'s "no assignment satisfies 56". ‘v c’ is the number of the ten 2-clauses of clause ‘c’'s own gadget satisfied by whatever auxiliary value clause ‘c’ is given [...]

THE DRAFT SAYS (cont.)

G-202 Section 9, "Simulating ZX interaction with ZZ interaction under subspace restriction" — the sentence claiming ZX and ZZ interactions simulate each other on the computational basis.

main.tex:873

"one can easily see both ZX and ZZ simulate each other if we restrict to a part of the Hilbert space $|\psi\rangle = \{|i\rangle : i \text{ in } \{0,1\}^n\}$."

LEAN'S VERDICT (cont.)

FALSE lean / L2

lean/Qgadget/Findings/Section_9_ZXvsZZSimulation.lean#energy_x1TwoQ_basis

theorem energy_x1TwoQ_basis (x : Fin 4) :

Qgadget.Borrowed.energy x1TwoQ (EuclideanSpace.single x (1 : ℂ)) = 0 := by

Every off-diagonal two-qubit Pauli word above has ZERO diagonal, so its expectation vanishes on every computational basis state.

lean/Qgadget/Findings/Section_9_ZXvsZZSimulation.lean#energy_z1x2TwoQ_basis

theorem energy_z1x2TwoQ_basis (x : Fin 4) :

Qgadget.Borrowed.energy z1x2TwoQ (EuclideanSpace.single x (1 : ℂ)) = 0 := by

lean/Qgadget/Findings/Section_9_ZXvsZZSimulation.lean#energy_x1z2TwoQ_basis

theorem energy_x1z2TwoQ_basis (x : Fin 4) :

Qgadget.Borrowed.energy x1z2TwoQ (EuclideanSpace.single x (1 : ℂ)) = 0 := by

lean/Qgadget/Findings/Section_9_ZXvsZZSimulation.lean#no_affine_reproduces_product

theorem no_affine_reproduces_product (c p q : ℝ)

(h11 : c + p * 1 + q * 1 = (1 : ℝ) * 1)

(h1n : c + p * 1 + q * (-1) = (1 : ℝ) * (-1))

(hn1 : c + p * (-1) + q * 1 = (-1 : ℝ) * 1)

(hnn : c + p * (-1) + q * (-1) = (-1 : ℝ) * (-1)) :

False := by

FINDING (G-202): no constants 'c, p, q' make ' $c + p \cdot s_1 + q \cdot s_2 = s_1 s_2$ ' at all four sign pairs ' $s_1, s_2 \in \{1, -1\}$ '. So ' Z_{1Z_2} 's basis-expectation function — the product ' $s_1 s_2$ ' — is not the basis-expectation function of ANY 'ZX'-family Hamiltonian, whose basis expectation is always the affine function ' $c + p \cdot s_1 + q \cdot s_2$ ' (by [...])

lean/Qgadget/Findings/Section_9_ZXvsZZSimulation.lean#zx_does_not_simulate_zz_on_basis

theorem zx_does_not_simulate_zz_on_basis :

¬ ∃ c p q : ℝ,

(c + p * 1 + q * 1 = (1 : ℝ) * 1) ∧

(c + p * 1 + q * (-1) = (1 : ℝ) * (-1)) ∧

(c + p * (-1) + q * 1 = (-1 : ℝ) * 1) ∧

(c + p * (-1) + q * (-1) = (-1 : ℝ) * (-1)) := by

The same finding, packaged as a statement about the two Hamiltonian families: ' Z_{1Z_2} 's diagonal values at the four basis states are exactly the four sign products ' $s_1 s_2$ ', and by 'no_affine_reproduces_product' no affine ' $c + p \cdot s_1 + q \cdot s_2$ ' — the shape every 'ZX'-family basis expectation takes — matches all four. So 'ZX' does not simulate [...]

THE DRAFT SAYS (cont.)

G-1 Problem 5, the Guided Local Hamiltonian problem $\text{GLH}(k, \gamma, \delta)$
[cade2022complexity]

main.tex:881

The guiding state of the GLH problem is $|u\rangle = \sum_{i \in S} |i\rangle$ with $|S| \leq \text{poly}(n)$.

G-2 Problem 5, the Guided Local Hamiltonian problem $\text{GLH}(k, \gamma, \delta)$
[cade2022complexity]

main.tex:885

GLH promises fidelity with the ground space $\|\Pi_{\text{gs}} |u\rangle\|^2 \geq \gamma$, and that either $\lambda_0 \leq a$ or $\lambda_0 \geq b$.

LEAN'S VERDICT (cont.)

ILL_POSED lean / L2

lean/Qgadget/Borrowed/Verbatim/Problem_5_GLH.lean#norm_guidedStateAsWritten

theorem norm_guidedStateAsWritten (S : Finset (Fin N)) :

$\|\text{guidedStateAsWritten } S\| = \text{Real.sqrt } (S.\text{card} : \mathbb{R}) := \text{by}$

FINDING 1: the guided state of :881 is not a state, except when $|S| = 1$. Its norm is $\sqrt{|S|}$.

ILL_POSED lean / L2-WEAK

lean/Qgadget/Borrowed/Verbatim/Problem_5_GLH.lean#groundProjNormSq_guidedStateAsWritten

theorem groundProjNormSq_guidedStateAsWritten [NeZero N]

{H : Matrix (Fin N) (Fin N) $\mathbb{C}$ } (hH : H.IsHermitian)

(S : Finset (Fin N)) (hS : S.Nonempty) :

groundProjNormSq hH (guidedStateAsWritten S)

$= (S.\text{card} : \mathbb{R}) * \text{groundProjNormSq } hH (\text{subsetState } S \text{ hS}) := \text{by}$

FINDING 2: the fidelity promise of :885 is $|S|$ times the fidelity of Paper 01 :1261. With the unnormalised $|u\rangle$ of :881 the quantity the paper calls a fidelity is $|S| \cdot \|\Pi_{\text{gs}} |u\rangle\|^2$, so it ranges over $[0, |S|]$ and not over $[0, 1]$; the promise $\geq \gamma$ is correspondingly $|S|$ times weaker than the identical-looking promise on [...]

lean/Qgadget/Borrowed/Verbatim/Problem_12_GLH_SecondDefinition.lean#problem_12_promise_admits_low_squared_fidelity

theorem problem_12_promise_admits_low_squared_fidelity [NeZero N]

{H : Matrix (Fin N) (Fin N) $\mathbb{C}$ } (hH : H.IsHermitian)

(u : EuclideanSpace $\mathbb{C}$ (Fin N)) (hu : $0 < \text{groundProjNorm } hH \text{ u}$) (a b : $\mathbb{R}$) :

$\exists t : \text{GLH2Instance } N,$

$t.H = H \wedge \text{GLH2Fidelity } (1 / 2) \text{ t} \wedge \neg (1 / 2 \leq \text{groundProjNormSq } hH \text{ t.u}) := \text{by}$

FINDING 2: the appendix's 'GLH' admits guided states that §3's 'GLH' rejects at the same γ . From any u whose ground-space component is non-zero, the rescaling $u := (1/(2\|\Pi_{\text{gs}} u\|)) \cdot u$ has $\|\Pi_{\text{gs}} u\| = 1/2$.

THE DRAFT SAYS (cont.)	LEAN'S VERDICT (cont.)
G-13 main.tex:898 If a subset state $u\rangle$ over S with $S \leq \text{poly}(n)$ has $\langle u \psi_{\text{gs}} \rangle ^2 \geq 1/\text{poly}(n)$, then some i in S has $\langle i \psi_{\text{gs}} \rangle ^2 \geq 1/\text{poly}(n)$. Proof by contradiction: suppose no such i; then $\langle u \psi_{\text{gs}} \rangle ^2 < 1/\text{poly}(n)$.	TRUE_PROOF_INVALID lean / L2-PARTIAL lean/Qgadget/Borrowed/Analysis/Overlap.lean#exists_basis_overlap <pre>theorem exists_basis_overlap (S : Finset (Fin N)) (hS : S.Nonempty) (ψ : EuclideanSpace ℂ (Fin N)) (δ : ℝ) (h : δ ≤ ⟨subsetState S hS, ψ⟩_C ^ 2) : ∃ i ∈ S, δ / S.card ≤ ⟨EuclideanSpace.single i (1 : ℂ), ψ⟩_C ^ 2 := by</pre> P-23's corrected bound, quantitative (ledger row 'P-23', quoted by 'P-25'): a subset state's overlap with any 'ψ' distributes over its support with loss exactly 'S', never the paper's vacuous "$1/\text{poly}(n)$ implies $1/\text{poly}(n)$". If '$\delta \leq \langle \text{subsetState } S \text{ hS}, \psi \rangle ^2$', some '$i \in S$' has '$\langle i, \psi \rangle ^2 \geq \delta / S$'.
G-18 Problem 6, the Basis-GLH problem main.tex:919 Problem [Basis-GLH]: "The problem is the same as the GLH problem except that the subset state is replaced by a basis state $i\rangle$ with the promise that $\langle i \psi_{\text{gs}} \rangle ^2 \geq 1/\text{poly}(n)$."	ILL_POSED lean / L2-PARTIAL lean/Qgadget/Borrowed/Verbatim/Problem_6_BasisGLH.lean#basisGLH_is_singleton_GLH <pre>theorem basisGLH_is_singleton_GLH (i : Fin N) : guidedStateAsWritten ({i} : Finset (Fin N)) = EuclideanSpace.single i (1 : ℂ) ^ subsetState ({i} : Finset (Fin N)) ⟨i, Finset.mem_singleton_self i⟩ = EuclideanSpace.single i (1 : ℂ) := by</pre> The replacement of :920 is a restriction, not a new object, and at '$S = 1$' the paper's two incompatible normalisations agree. The GLH guided state of Paper 01 :1656 supported on '$\{i\}$' and Definition 3.2's subset state on '$\{i\}$' are both exactly the basis state '$i\rangle$'.
G-14 Problem 6, the Basis-GLH problem main.tex:926 For any constant $k \geq 2$ there are a, b in $[-1, 1]$ with $b - a$ in $\Omega(1/\text{poly}(n))$ such that Basis-GLH(k, δ) is BQP-complete. Containment half.	TRUE lean / – lean/Qgadget/Borrowed/Verbatim/Problem_6_BasisGLH.lean#basisGLH_fidelity_is_basis_overlap <pre>theorem basisGLH_fidelity_is_basis_overlap [NeZero N] (γ : ℝ) (t : BasisGLHInstance N) : GLHFidelity γ t.toGLH ↔ γ ≤ groundProjNormSq t.hH (EuclideanSpace.single t.i (1 : ℂ)) := by</pre> The GLH fidelity promise, read on the embedded instance, is the squared overlap of the basis state with the ground space. So Basis-GLH's promise at Paper 01 :1697 and GLH's at :885 are the same condition on the same vector, and P-24's "no new object" is discharged: membership in BQP for Basis-GLH follows from GLH's by the embedding, with [...]

THE DRAFT SAYS (cont.)	LEAN'S VERDICT (cont.)
G-15 main.tex:926 Basis-GLH(k, delta) is BQP-complete. Hardness half, offered at main.tex:917 as a consequence of the overlap lemma.	GAP lean / - lean/Qgadget/Borrowed/Analysis/ExistentialVsSelector.lean#existential_does_not_give_computable_selector <pre>theorem existential_does_not_give_computable_selector (n : ℕ) : (∀ c : Nat.Partrec.Code, ∃ i, HaltRel n c i) ∧ ¬ ∃ f : Nat.Partrec.Code → Bool, Computable f ∧ ∀ c, HaltRel n c (f c) :=</pre> The two together: a total relation whose existential holds and whose computable selector does not exist. The gap P-25 names is not vacuous.

G-3 Problem 7, the Guidable k-Local Hamiltonian problem GaLH [weggemans2023guidable]

main.tex:937

GaLH's extra promise: there exists a subset state $|u\rangle$ with $|\langle u | \psi_{gs} \rangle| \geq \gamma$.

ILL_POSED lean / L2-PARTIAL

lean/Qgadget/Borrowed/Verbatim/Problem_7_GaLH.lean#galh_extra_promise_weaker

```
theorem galh_extra_promise_weaker (S : Finset (Fin N)) (hS : S.Nonempty)
  (ψgs : EuclideanSpace C (Fin N)) (hψ : ‖ψgs‖ = 1) (γ : ℝ)
  (h : γ ≤ ‖⟨subsetState S hS, ψgs⟩_C‖ ^ 2) :
  γ ≤ ‖⟨subsetState S hS, ψgs⟩_C‖ := by
```

FINDING 1, first half: :937's promise is implied by Definition 3.2's and not conversely. Under the paper's own side conditions — ' $\gamma \leq 1$ ' from Paper 01 :1710, and ' $|\psi_{gs}\rangle$ ' a unit vector as the name "ground state" requires — the squared condition of Paper 01 :1261 and :885 implies the unsquared condition of :937.

lean/Qgadget/Findings/Section_9_SubsetVsCircuitGuidability.lean#norm_sq_subsetState_inner_univ

```
theorem norm_sq_subsetState_inner_univ (S : Finset (Fin N)) (hS : S.Nonempty)
  (hU : (Finset.univ : Finset (Fin N)).Nonempty) :
  ‖⟨subsetState S hS, subsetState (Finset.univ : Finset (Fin N)) hU⟩_C‖ ^ 2
  = (S.card : ℝ) / (N : ℝ) := by
```

FINDING (G-3), fidelity half: $|\langle ss(S) | ss(univ) \rangle|^2 = |S|/N$. At ' $N = 2^n$ ' this is exactly the row's ' $|\langle ss(S) | +^n \rangle|^2 = |S|/2^n$ '.

lean/Qgadget/Findings/Section_9_SubsetVsCircuitGuidability.lean#groundEnergy_sumXHam_eq_zero

```
theorem groundEnergy_sumXHam_eq_zero (n : N)
  (hU : (Finset.univ : Finset (Qgadget.Borrowed.Qubits n)).Nonempty) :
  Qgadget.Borrowed.groundEnergy (isHermitian_sumXHam n) = 0 := by
```

HALF A, closed: ' $\text{groundEnergy}(\text{sumXHam } n) = 0$ ', achieved at ' $|+\rangle^{\otimes n}$ '. ' $\text{energy_sumXHam_nonneg}$ ' gives every eigenvalue ' ≥ 0 ' (' $\text{energy_eigenvectorBasis}$ ' reads off each ' $hA.\text{eigenvalues } i$ ' as an energy), hence ' $\text{groundEnergy} \geq 0$ ' (' Finset.le_inf '); ' $\text{energy_sumXHam_subsetStateOnQubits_univ}$ ' plus ' $\text{norm_subsetStateOnQubits}$ ' give ' $\text{groundEnergy} \dots$ '

lean/Qgadget/Findings/Section_9_SubsetVsCircuitGuidability.lean#sumXHam_energy_zero_const

```
theorem sumXHam_energy_zero_const (n : N) (ψ : EuclideanSpace C (Qgadget.Borrowed.Qubits n))
  (hψ : ‖ψ‖ = 1) (hE : Qgadget.Borrowed.energy (sumXHam n) ψ = 0) :
  ∃ c : C, WithLp.ofLp ψ = fun _ => c := by
```

HALF B, closed: any unit ' ψ ' with ' $\text{energy}(\text{sumXHam } n) \psi = 0$ ' has CONSTANT amplitude — the row's uniqueness claim. Constant amplitude is exactly what ' $|+\rangle^{\otimes n}$ ' (the uniform subset state) IS; no other state, up to that constant's own value, achieves energy '0'.

THE DRAFT SAYS (cont.)	LEAN'S VERDICT (cont.)
G-19 Problem 8, the Basis-GaLH problem main.tex:948 Problem [Basis-GaLH]: "The problem is the same as the GaLH problem (\ref{prob: GaLHproblem}) except that the subset state is replaced by a basis state $i\rangle$ with the promise that $\langle i \psi_{\text{gs}}\rangle ^2 \geq 1/\text{poly}(n)$."	ILL_POSED lean / L2-PARTIAL lean/Qgadget/Borrowed/Verbatim/Problem_8_BasisGaLH.lean#basisGaLH_promise_squared_vs_galh_unsquared <pre>theorem basisGaLH_promise_squared_vs_galh_unsquared : ∃ (ψgs : EuclideanSpace ℂ (Fin 4)) (c : ℝ), ψgs = 1 ∧ GaLHExtraPromise c ψgs ∧ ¬ BasisGaLHPromise_readingB c ψgs := by</pre> FINDING 1: the replacement changes the promise's shape, not only its state. At one and the same threshold '$c = 1/2$', on four qubits' worth of register with '$\psi_{\text{gs}}\rangle$' the uniform subset state, Problem 7's unsquared promise at :937 holds and this problem's squared promise at Paper 01 :1726 fails — for every basis state at once, so the [...]
G-16 main.tex:955 For any constant $k \geq 2$ there are a, b in $[-1,1]$ with $b - a$ in $\Omega(1/\text{poly}(n))$ such that Basis-GaLH(k, δ) is QCMA-complete. Containment half.	TRUE lean / L2 lean/Qgadget/Borrowed/Analysis/GLH_OneSided.lean#basis_witness_never_accepts <pre>theorem basis_witness_never_accepts {n : ℕ} {A : Matrix (Fin (2 ^ n)) (Fin (2 ^ n)) ℂ} (hA : A.IsHermitian) {a b : ℝ} (hab : a < b) (hb : b ≤ Qgadget.Borrowed.groundEnergy hA) (i : Fin (2 ^ n)) (est : ℝ) (hest : est - Qgadget.Borrowed.energy A (EuclideanSpace.single i (1 : ℂ)) ≤ (b - a) / 4) : (a + b) / 2 < est := by</pre> Soundness for a BASIS witness, whatever the witness is ('main.tex:955', P-26 step 2). P-26 records Basis-GaLH $\in$ QCMA. lean/Qgadget/Findings/Section_9_BasisGaLHContainment.lean#basisGaLHPromise_readingB_implies_galhExtraPromise <pre>theorem basisGaLHPromise_readingB_implies_galhExtraPromise (c : ℝ) (ψgs : EuclideanSpace ℂ (Fin N)) (h : BasisGaLHPromise_readingB c ψgs) : GaLHExtraPromise (Real.sqrt c) ψgs := by</pre> FINDING (G-16): a basis witness for Basis-GaLH's promise is a subset-state witness for GaLH's. 'main.tex:1726's promise is squared; 'main.tex:937's is not; the two are reconciled by taking a square root.

THE DRAFT SAYS (cont.)	LEAN'S VERDICT (cont.)
G-17 Theorem 3.14 — Basis-GaLH(k, δ) is QCMA-complete main.tex:955 Basis-GaLH(k, δ) is QCMA-complete. Hardness half, proposed at main.tex:946 as being similar to Basis-GLH.	GAP lean / – lean/Qgadget/Borrowed/Verbatim/Theorem_3_14_BasisGaLH_QCMAComplete.lean#overlap_ratio_inverse_poly theorem overlap_ratio_inverse_poly (p q : ℝ) (hp : 0 < p) (hq : 0 < q) (s γ : ℝ) (hs : 0 < s) (hsp : s ≤ p) (hγ : 1 / q ≤ γ) : 1 / (p * q) ≤ γ / s := by Under a polynomial support bound, ‘γ/ S’ stays inverse-polynomial. REAL: the conclusion is an inequality between two rationals in ‘p’, ‘q’ and ‘S’, and it is exactly what reading A’s reduction needs.
G-206 Section 10’s Claim: a quantum polynomial-time reduction ‘$\mathcal{A}$’ from ‘GLH’ to the ‘{ZZ,Z}’-Hamiltonian problem, with the two-bullet success guarantee. main.tex:966 Claim: "There is a quantum polynomial-time reduction A that takes an instance of the guided local Hamiltonian and produces a ZZ Hamiltonian" with $P[A(x) \text{ in ZZ-Ham_yes}] \geq 2/3$ when x is in GLH_yes and $P[A(x) \text{ in ZZ-Ham_no}] \geq 2/3$ when x is in GLH_no.	TRUE lean / L2-PARTIAL lean/Qgadget/Findings/Section_10_ReductionClaim.lean#groundEnergy_zero_eq_zero theorem groundEnergy_zero_eq_zero {m : Type*} [Fintype m] [DecidableEq m] [Nonempty m] : Qgadget.Borrowed.groundEnergy (Matrix.isHermitian_zero (α := ℂ) (n := m)) = 0 := by The ground energy of the zero Hamiltonian is ‘0’. REAL: it could have failed had the zero matrix’s spectrum not been constant — ‘energy’ vanishing on the WHOLE space forces every eigenvalue, via ‘energy_eigenvectorBasis’, to be ‘0’, and only then does the ‘inf’ collapse to ‘0’ rather than to some other member of a nonconstant spectrum. lean/Qgadget/Findings/Section_10_ReductionClaim.lean#galhBQPZeroYesInstance_isYes theorem galhBQPZeroYesInstance_isYes : LHYes galhBQPZeroYesInstance := by ‘galhBQPZeroYesInstance’ is a YES instance: ‘$\lambda_0(0) = 0 \leq a = 0$’. lean/Qgadget/Findings/Section_10_ReductionClaim.lean#galhBQPZeroNoInstance_isNo theorem galhBQPZeroNoInstance_isNo : LHNo galhBQPZeroNoInstance := by ‘galhBQPZeroNoInstance’ is a NO instance: ‘$b = 0 \leq \lambda_0(0) = 0$’.
G-213 Section 10, "Step III" of the design steps for the LP — computing ‘$\langle u Z_{iX_j} u \rangle = \alpha_{ij}$’ for the guiding state of ‘GLH’. main.tex:1001 "Step III: Compute the expression $\langle u Z_{iX_j} u \rangle = \alpha_{ij}$ for all (i,j) in E in the given interaction graph. Note, the computation of α_{ij} requires BQP machine."	FALSE lean / L2 lean/Qgadget/Findings/Section_10_AlphaClassical.lean#energy_zOpQ_mul_xOpQ_guidedState theorem energy_zOpQ_mul_xOpQ_guidedState (i j : Fin n) (S : Finset (Qgadget.Borrowed.Qubits n)) : Qgadget.Borrowed.energy (zOpQ i * xOpQ j) (guidedStateOnQubits S) = $\sum y \in S.$filter (fun y => flipBit j y ∈ S), zSignQ i y := by FINDING (G-213): the classical formula (★). ‘$\alpha_{ij} = \langle u Z_{iX_j} u \rangle$’, for the raw guiding state of a finite set ‘S’, is the signed COUNT of elements ‘$y \in S$’ whose flip-‘j’ also lies in ‘S’ — a quantity a classical machine reads off in ‘$O(S ^2 n)$’ time by checking membership, against the draft’s claim that computing it "requires BQP [...]

THE DRAFT SAYS (cont.)

G-215 Section 10, "Step IV" of the design steps for the LP that replaces ‘ Z_{iX_j} ’ edges by ‘ $p_{ij} Z_{iZ_j} + q_{ij} Z_{iI_j} + r_{ij} I_{iZ_j} + sI$ ’, and the draft’s own blue diagnosis of it.

main.tex:1007

"Step IV: Compute the expression $\langle u | (p_{ij} Z_{iZ_j} + q_{ij} Z_{iI_j} + r_{ij} I_{iZ_j} + sI) | u \rangle = \alpha_{ij}$... (This is incorrect as the system of equations only possesses trivial solutions, which don’t induce the desired quantum reduction.)"

G-45

main.tex:1072

Eleven live sections are titled "Appendix IV", "Appendix V", "Appendix VIII", "Appendix IX", "Appendix XI", "Appendix XII", "Appendix XV", "Appendix XVI", "Appendix XVII", "Appendix XIX" and "Appendix XX", at main.tex:1072, 1152, 1430, 1459, 1581, 1880, 1916, 2222, 2259, 2580 and 2643.

LEAN’S VERDICT (cont.)

FALSE lean / L2

lean/Qgadget/Findings/Section_10_StepIV_LinearProgram.lean#stepIV_solvable

```
theorem stepIV_solvable :
  (0 : ℝ) * Qgadget.Borrowed.energy z1z2StepIV uStepIV +
  1 * Qgadget.Borrowed.energy z1iStepIV uStepIV +
  0 * Qgadget.Borrowed.energy iz2StepIV uStepIV +
  0 * Qgadget.Borrowed.energy (1 : Matrix (Fin 4) (Fin 4) ℂ) uStepIV =
  Qgadget.Borrowed.energy z1x2StepIV uStepIV := by
```

FINDING (G-215), first half: the displayed equation at ‘ $(i,j)=(1,2)$ ’ IS solvable. ‘ $q=1, p=r=0, s=0$ ’ solves ‘ $p \cdot 0 + q \cdot 2 + r \cdot 0 + s \cdot 2 = 2$ ’.

lean/Qgadget/Findings/Section_10_StepIV_LinearProgram.lean#stepIV_zero_not_solution

```
theorem stepIV_zero_not_solution :
  (0 : ℝ) * Qgadget.Borrowed.energy z1z2StepIV uStepIV +
  0 * Qgadget.Borrowed.energy z1iStepIV uStepIV +
  0 * Qgadget.Borrowed.energy iz2StepIV uStepIV +
  0 * Qgadget.Borrowed.energy (1 : Matrix (Fin 4) (Fin 4) ℂ) uStepIV ≠
  Qgadget.Borrowed.energy z1x2StepIV uStepIV := by
```

The zero vector is not a solution of Step IV’s own system: at ‘ $p=q=r=s=0$ ’ the left side is ‘0’, against ‘ $\alpha_{12} = 2$ ’. This is the sharp form of "the system only possesses trivial solutions" being false: even the trivial assignment fails it.

lean/Qgadget/Findings/Section_10_StepIV_LinearProgram.lean#stepIV_basis_system_trivial

```
theorem stepIV_basis_system_trivial (p q r s : ℝ)
  (h : ∀ x : Fin 4,
    Qgadget.Borrowed.energy z1x2StepIV (EuclideanSpace.single x (1 : ℂ)) =
    p * Qgadget.Borrowed.energy z1z2StepIV (EuclideanSpace.single x (1 : ℂ)) +
    q * Qgadget.Borrowed.energy z1iStepIV (EuclideanSpace.single x (1 : ℂ)) +
    r * Qgadget.Borrowed.energy iz2StepIV (EuclideanSpace.single x (1 : ℂ)) +
    s * Qgadget.Borrowed.energy (1 : Matrix (Fin 4) (Fin 4) ℂ)
      (EuclideanSpace.single x (1 : ℂ))) :
  p = 0 ∧ q = 0 ∧ r = 0 ∧ s = 0 := by
```

FINDING (G-215), second half: the all-basis-states system has only the zero solution. Requiring ‘ $\langle x | Z_{1X_2} | x \rangle = p \langle x | Z_{1Z_2} | x \rangle + q \langle x | Z_{1I} | x \rangle + r \langle x | Z_{2I} | x \rangle + s \langle x | I | x \rangle$ ’ at all four basis states gives the four sign equations ‘ $\pm p \pm q \pm r + s = 0$ ’ at the sign patterns ‘ $(+, +, +, +)$, ‘ $(-, +, -, +)$, ‘ $(-, +, +, +)$, ‘ $(+, -, -, +)$ ’ (‘diagZ1Z2, diagZ1I, diagIZ2’ at the four [...]

FALSE measured / –

method: tools/check_paper_integrity.py evidence: tools/paper.toml

THE DRAFT SAYS (cont.)	LEAN'S VERDICT (cont.)
G-8 Problem 11, the k-Local Hamiltonian problem k-LH(δ) main.tex:1075 Problem [k-Local Hamiltonian: k-LH(delta)]: tuple <H, a, b>; input a k-local H on n qubits with $r = \text{poly}(n)$ terms, each $\ H_j\ \leq 1$ with entries on $\text{poly}(n)$ bits, "and two constants a and b with $b-a \geq \delta > 0$"; promise $\lambda_0 \leq a$ on YES and $\lambda_0 \geq b$ on NO.	ILL_POSED lean / L2 lean/Qgadget/Borrowed/Verbatim/Problem_11_kLocalHamiltonian.lean#problem_11_cases_collide theorem problem_11_cases_collide [NeZero N] {H : Matrix (Fin N) (Fin N) ℂ} (hH : H.IsHermitian) (δ : ℝ) (hδ : 0 < δ) : $\exists t : \text{LHInstance } N, \text{LHGap } \delta \ t \wedge \text{LHPromise } t \wedge \text{LHYes } t \wedge \text{LHNo } t := \text{by}$ FINDING 1: 'k'-LH as stated at Paper 01 :2475 is not a promise problem, so its Output rule at :1084 — "Decide which is the case" — has no answer on some legal instances. For every Hermitian 'H' on a non-empty register and every '$\delta > 0$', the instance '$a := \lambda_0(H)$', '$b := \lambda_0(H) - \delta$' satisfies the Input condition '$b-a \geq \delta > 0$' of :1077 and [...]
G-7 Problem 12, the Guided Local Hamiltonian problem GLH(k, γ, δ) — the paper's SECOND definition of it [gharibian2022dequantizing] main.tex:1103 "Subset State (grilo2015qma): For any subset S of $[2^n]_1$, we define the subset state $S\rangle$ in $\mathbb{R}^{2^n}$, as the uniform superposition over the elements of S." The guided state is likewise placed in $\mathbb{R}^{2^n}$ at main.tex:1107.	FALSE lean / L2 lean/Qgadget/Borrowed/Verbatim/Problem_12_GLH_SecondDefinition.lean#qftEntry_not_real theorem qftEntry_not_real : (qftEntry 4 1 1).im $\neq 0 := \text{by}$ The draft's own Fourier transform has non-real entries, so a state space declared over '$\mathbb{R}$' is not closed under it. Witness '$N = 4$', '$x = y = 1$', where the entry is 'i'.
G-9 Problem 12, the Guided Local Hamiltonian problem GLH(k, γ, δ) — the paper's SECOND definition of it [gharibian2022dequantizing] main.tex:1111 Problem [gharibian2022dequantizing: GLH(k, gamma, delta)]: tuple <H; u>; (a,b)>; input includes "an efficient description of a semi-classical state u in $\mathbb{C}^{2^n}$"; promise "Fidelity with the ground space $\ \Pi_{\text{gs}} u\rangle\ \geq \gamma$" and either $\lambda_0 \leq a$ or $\lambda_0 \geq b$.	ILL_POSED lean / L2-WEAK lean/Qgadget/Borrowed/Verbatim/Problem_12_GLH_SecondDefinition.lean#problem_12_fidelity_scales theorem problem_12_fidelity_scales [NeZero N] {H : Matrix (Fin N) (Fin N) ℂ} (hH : H.IsHermitian) (v : EuclideanSpace ℂ (Fin N)) (c : ℝ) (hc : 0 ≤ c) : $\text{groundProjNorm } hH ((c : \mathbb{C}) \bullet v) = c * \text{groundProjNorm } hH v := \text{by}$ FINDING 1: :1116's fidelity is homogeneous of degree one in 'u', and :1113 does not normalise 'u'. Scaling the input state by '$c \geq 0$' scales the promised quantity by 'c'. lean/Qgadget/Borrowed/Verbatim/Problem_12_GLH_SecondDefinition.lean#problem_12_promise_admits_low_squared_fidelity theorem problem_12_promise_admits_low_squared_fidelity [NeZero N] {H : Matrix (Fin N) (Fin N) ℂ} (hH : H.IsHermitian) (u : EuclideanSpace ℂ (Fin N)) (hu : 0 < groundProjNorm hH u) (a b : ℝ) : $\exists t : \text{GLH2Instance } N,$ $t.H = H \wedge \text{GLH2Fidelity } (1 / 2) \ t \wedge \neg (1 / 2 \leq \text{groundProjNormSq } hH \ t.u) := \text{by}$ FINDING 2: the appendix's 'GLH' admits guided states that §3's 'GLH' rejects at the same 'γ'. From any 'u' whose ground-space component is non-zero, the rescaling '$u := (1/(2\ \Pi_{\text{gs}} u\)) \cdot u$' has '$\ \Pi_{\text{gs}} u\ = 1/2$'.

THE DRAFT SAYS (cont.)

G-11 Problem 13, the Guidable k-Local Hamiltonian problem $\text{GaLH}(k, \gamma, \delta)$ — the paper’s SECOND definition of it [weggemans2023guidable]

main.tex:1133

Problem [Guidable k-Local Hamiltonian (GaLH)]: tuple <H; (a,b)>; "Extra promise: Denote Π_{gs} for the projection on the subspace spanned by the ground state of H. Then, there exists a unitary V implemented by a quantum circuit composed of at most $T = O(\text{poly}(n))$ gates from a set of fixed gates G that produces the state $|\phi\rangle = V|0\rangle$, which has $\|\Pi_{\text{gs}} u\| \geq \gamma$."

G-221 Theorem (‘main.tex:1185’-‘main.tex:1186’): " $\{X, Y, Z\}$ -Hamiltonians is in P", for the problem of ‘main.tex:1175’-‘main.tex:1183’ ($H = \sum_k \alpha_k X_k + \beta_k Y_k + \gamma_k Z_k$, promise gap $1/\text{poly}(n)$). Two proofs are given: > First, construct the ground state, which is guaranteed to be a product > state over n qubits. [...] each one-particle state $|\psi_i\rangle$ can be computed

main.tex:1186

Theorem: " $\{X, Y, Z\}$ -Hamiltonians is in P." First proof: the ground state is a product state, and each one-qubit factor is found by minimising that qubit’s terms.

LEAN’S VERDICT (cont.)

FALSE lean / L2-PARTIAL

lean/Qgadget/Borrowed/Verbatim/Problem_13_GaLH_SecondDefinition.lean#problem_13_extra_promise_admits_non_subset_states

theorem problem_13_extra_promise_admits_non_subset_states :
 $\exists \varphi : \text{EuclideanSpace } \mathbb{C} \ (\text{Fin } 2), \|\varphi\| = 1 \wedge$
 $\forall (S : \text{Finset } (\text{Fin } 2)) \ (hS : S.\text{Nonempty}), \varphi \neq \text{subsetState } S := \text{by}$

FINDING 2: :137’s witness class is not :937’s. There is a unit vector that no subset state equals, so the state a circuit produces need not be the "subset state $|u\rangle$ " that Problem 7’s extra promise at :937 asks for.

TRUE lean / L2

lean/Qgadget/Verbatim/Theorem_XYZHam_inP.lean#groundEnergy_pauliCombo

theorem groundEnergy_pauliCombo (a b g : $\mathbb{R}$) :
 Qgadget.Borrowed.groundEnergy (pauliCombo_isHermitian a b g)
 = -Real.sqrt (a² + b² + g²) :=

FINDING (G-221): ‘main.tex:1186’’s Theorem holds, and the exact ground energy is $-\sqrt{\alpha^2 + \beta^2 + \gamma^2}$ — the closed form ‘main.tex:1188’-‘main.tex:1192’’s "First proof" only argues for in words ("each one-particle state can be computed efficiently").

lean/Qgadget/Support/TensorGroundEnergy.lean#groundEnergy_kron_add

theorem groundEnergy_kron_add [Nonempty n] [Nonempty m] (hA : A.IsHermitian)
 (hB : B.IsHermitian)
 (hK : (A $\otimes_k$ (1 : Matrix m m $\mathbb{C}$) + (1 : Matrix n n $\mathbb{C}$) $\otimes_k$ B).IsHermitian) :
 Qgadget.Borrowed.groundEnergy hK
 = Qgadget.Borrowed.groundEnergy hA + Qgadget.Borrowed.groundEnergy hB :=

THE ADDITIVITY THEOREM. ‘groundEnergy (A $\otimes$ I + I $\otimes$ B) = groundEnergy A + groundEnergy B’, for Hermitian ‘A’, ‘B’ on nonempty finite registers.

THE DRAFT SAYS (cont.)

G-222 Theorem (‘main.tex:1185’-‘main.tex:1186’): “{X, Y, Z}-Hamiltonians is in P”, for the problem of ‘main.tex:1175’-‘main.tex:1183’ ($H = \sum_k \alpha_k X_k + \beta_k Y_k + \gamma_k Z_k$, promise gap $1/\text{poly}(n)$). Two proofs are given: > First, construct the ground state, which is guaranteed to be a product > state over n qubits. [...] each one-particle state $|\psi_i\rangle$ can be computed

main.tex:1193

"An alternative proof for this is provided in [cubitt2016complexity]. For 1-local Hermitian matrices, there always exists a 1-qubit unitary matrix $U_{\{2 \times 2\}}$ such that $U^{\wedge \{n\}}$ can simultaneously diagonalize the input H , an instance of 1-LH."

G-228 Problem (‘main.tex:1200’-‘main.tex:1208’), the ‘{ZZ, Z}’-Hamiltonian problem: $H = \sum_{\{i,j\} \in E} \alpha_{\{ij\}} Z_i \otimes Z_j + \sum_{k \in V} \beta_k Z_k$ on an interaction graph $G=(V,E)$, diagonal in the computational basis. Theorem (‘main.tex:1215’-‘1216’): "The problem [...] is in P if edge parameter $m \leq O(\log(n))$, where $m := |E|$ ". Proof (‘main.tex:1218’): brute-force search

main.tex:1216

Theorem: "The problem \ref{prob:zz-ham} is in P if edge parameter $m \leq O(\log(n))$, where $m := |E|$ is the number of edges in the underlying interaction graph."

LEAN’S VERDICT (cont.)

FALSE lean / L2

lean/Qgadget/Verbatim/Theorem_XYZHam_inP.lean#no_unitary_diagonalizes

```
theorem no_unitary_diagonalizes (U : Matrix (Fin 2) (Fin 2) C)
  (hU1 : UH * U = 1) (hU2 : U * UH = 1) :
  ¬ ((UH ⊗k UH) * (pauliX2 ⊗k (1 : Matrix (Fin 2) (Fin 2) C)
    + (1 : Matrix (Fin 2) (Fin 2) C) ⊗k pauliZ2) * (U ⊗k U)).IsDiag := by
```

FINDING (G-222): ‘main.tex:1193’'s alternative proof is false as written — for EVERY 2×2 unitary ‘ U ’ (both defining equations carried as hypotheses), $(U \otimes U)^H (X \otimes I + I \otimes Z) (U \otimes U)$ is not diagonal. The paper’s own Hadamard-and-identity repair (‘main.tex:1193’'s corrected reading) is not a single ‘ $U^{\wedge \{2\}}$ ’; it is two DIFFERENT one-qubit unitaries [...]

lean/Qgadget/Verbatim/Theorem_XYZHam_inP.lean#pauliX2_mul_pauliZ2_ne

```
theorem pauliX2_mul_pauliZ2_ne : pauliX2 * pauliZ2 ≠ pauliZ2 * pauliX2 := by
```

‘ $XZ \neq ZX$ ’, checked at the ‘(0,1)’ entry: ‘ XZ ’ there is ‘-1’, ‘ ZX ’ there is ‘1’. REAL: it is the one fact that makes G-222’s counterexample work — drop it and every step of ‘no_unitary_diagonalizes’ still goes through vacuously.

TRUE lean / L2

lean/Qgadget/Verbatim/Theorem_ZZHam_LogEdges_inP.lean#ising_min_split

```
theorem ising_min_split {V : Type*} [Fintype V] [DecidableEq V] (W : Finset V)
  (f : (↑W → Bool) → ℝ) (β : V → ℝ) :
  Finset.univ.inf' (Finset.univ_nonempty (α := V → Bool))
```

FINDING (G-228): the exact separable-minimisation identity the paper’s proof of ‘main.tex:1216’ uses without stating. ‘ V ’ is the qubit set, ‘ W ’ the (at most ‘ $2m = O(\log n)$ ’) vertices touching some edge, ‘ f ’ the ‘ W ’-part of the energy (edge terms plus ‘ W ’'s own singleton terms — an arbitrary real function of the ‘ W ’-assignment, since this [...])

G-239

main.tex:1254

Theorem: "The problem \ref{prob:xyz-ham} with the edge parameter $m \leq O(\log(n))$ is in BQP."

Proof (main.tex:1256-1272): a reduction to GLH(γ, δ) whose guiding state is $|E\rangle_{\text{edge}} \otimes |S\rangle_{\text{singletons}}$.

TRUE_PROOF_INVALID lean / L2

lean/Qgadget/Support/TensorGroundEnergy.lean#groundEnergy_kron_add

```
theorem groundEnergy_kron_add [Nonempty n] [Nonempty m] (hA : A.IsHermitian)
  (hB : B.IsHermitian)
  (hK : (A  $\otimes_k$  (1 : Matrix m m  $\mathbb{C}$ ) + (1 : Matrix n n  $\mathbb{C}$ )  $\otimes_k$  B).IsHermitian) :
  Qgadget.Borrowed.groundEnergy hK
    = Qgadget.Borrowed.groundEnergy hA + Qgadget.Borrowed.groundEnergy hB :=
```

THE ADDITIVITY THEOREM. ‘groundEnergy $(A \otimes I + I \otimes B) = \text{groundEnergy } A + \text{groundEnergy } B$ ’, for Hermitian ‘A’, ‘B’ on nonempty finite registers.

lean/Qgadget/Verbatim/Theorem_XYZHam_LogEdges_inBQP.lean#groundEnergy_edgeSplit

```
theorem groundEnergy_edgeSplit {n : ℕ} (W : Finset (Fin n))
  {A : Matrix (LocalReg W) (LocalReg W)  $\mathbb{C}$ } {B : Matrix (CoReg W) (CoReg W)  $\mathbb{C}$ }
  (hA : A.IsHermitian) (hB : B.IsHermitian)
  (hK : (A  $\otimes_k$  (1 : Matrix (CoReg W) (CoReg W)  $\mathbb{C}$ )
    + (1 : Matrix (LocalReg W) (LocalReg W)  $\mathbb{C}$ )  $\otimes_k$  B).IsHermitian)
  (m c : ℕ) (hn : n  $\neq$  0) (hm : m  $\leq$  c * Nat.log 2 n) (hWcard : W.card  $\leq$  2 * m) :
  Qgadget.Borrowed.groundEnergy hK = Qgadget.Borrowed.groundEnergy hA +
  Qgadget.Borrowed.groundEnergy hB
     $\wedge$  Fintype.card (LocalReg W)  $\leq$  n  $^$  (2 * c) :=
```

FINDING (G-239): the problem’s TRUE content, and MORE than the paper states. ‘ H_W ’ acts on the (at most ‘ $2m$ ’, hence — by ‘ $m \leq O(\log n)$ ’ — polynomial-dimensional) register of the edges’ vertices; ‘ H_R ’ acts on the rest.

lean/Qgadget/Verbatim/Theorem_XYZHam_LogEdges_inBQP.lean#card_le_poly_of_edge_bound

```
theorem card_le_poly_of_edge_bound (n m c W : ℕ) (hn : n  $\neq$  0) (hm : m  $\leq$  c * Nat.log 2 n)
  (hW : W  $\leq$  2 * m) : 2  $^$  W  $\leq$  n  $^$  (2 * c) := by
```

‘ $m \leq O(\log n)$ ’ (with ‘ $|W| \leq 2m$ ’) forces ‘ $2^{W} \leq \text{poly}(n)$ ’. ‘ c ’ is the implicit constant of the paper’s ‘ $O(\log n)$ ’.

THE DRAFT SAYS (cont.)

G-240 The proof of Theorem ‘main.tex:1254’ (Appendix V, G-239’s theorem), claiming a guiding state for ‘log{XX+YY,X,Y,Z}Hamiltonian’: > We claim the following guided state for ‘log{XX+YY, X, Y, Z}’ Hamiltonian > will always be ‘1/poly(n)’ close to the ground state. > ‘|v> = |E>_edge ⊗ |S>_singletons’ (‘main.tex:1258’-‘1260’).

main.tex:1258

"We claim the following guided state for log{XX+YY, X, Y, Z} Hamiltonian will always be 1/poly(n) close to the ground state. |v> = |E>_edge ⊗ |S>_singletons", with |E>_edge taken as the uniform superposition |ss>_m (main.tex:1266).

G-241 The proof of Theorem ‘main.tex:1254’ (Appendix V, G-239’s theorem), claiming a guiding state for ‘log{XX+YY,X,Y,Z}Hamiltonian’: > We claim the following guided state for ‘log{XX+YY, X, Y, Z}’ Hamiltonian > will always be ‘1/poly(n)’ close to the ground state. > ‘|v> = |E>_edge ⊗ |S>_singletons’ (‘main.tex:1258’-‘1260’).

main.tex:1262

"|S>_singletons, the state of the 1-Local terms will be a product state whose exact form can be determined by inspecting the 1-Local Hamiltonian terms", used as a factor of the guiding state of a GLH(gamma, delta) instance (main.tex:1257-1260).

LEAN’S VERDICT (cont.)

FALSE lean / L2

lean/Qgadget/Findings/Section_13_XYZGuidingStateCounterexamples.lean#groundEnergy_hXXYY
theorem groundEnergy_hXXYY :

Qgadget.Borrowed.groundEnergy (m := Fin 2 × Fin 2) (A := hXXYY) hXXYY_isHermitian = -2 := by
‘groundEnergy hXXYY = -2’, attained at ‘vSinglet’. REAL: it combines the upper bound (‘vSinglet’, a specific unit vector, via the variational principle) with the lower bound (‘energy_hXXYY_ge’, over EVERY unit vector) — either alone would leave a gap.

lean/Qgadget/Findings/Section_13_XYZGuidingStateCounterexamples.lean#inner_uniform2_vSinglet
theorem inner_uniform2_vSinglet : ⟨⟨uniform2, vSinglet⟩⟩_C = 0 := by

FINDING (G-240): ‘main.tex:1258’'s claimed guide has ZERO overlap with ‘vSinglet’, which is at the TRUE ground energy ‘-2’. So the guide is not ‘1/poly(n)’-close to the ground state at this instance — it is orthogonal to it.

lean/Qgadget/Findings/Section_13_XYZGuidingStateCounterexamples.lean#hXXYY_ground_unique

theorem hXXYY_ground_unique (φ : EuclideanSpace ℂ (Fin 2 × Fin 2)) (hφ : ||φ|| = 1)
(henergy : Qgadget.Borrowed.energy hXXYY φ = -2) :

∃ c : ℂ, ||c|| = 1 ∧ φ = c • vSinglet := by

HALF B (G-240): ‘vSinglet’ is the UNIQUE ground state of ‘hXXYY’, up to phase. REAL: tracks equality through ‘energy_hXXYY_ge’'s own inequality chain — energy exactly ‘-2’ (not merely ‘≥ -2’) forces the Cauchy–Schwarz step, the AM/GM step, and the norm-constraint step all to be equalities simultaneously, which pins ‘φ_{00} = φ_{11} = 0’ [...]

FALSE lean / L2

lean/Qgadget/Findings/Section_13_XYZGuidingStateCounterexamples.lean#norm_inner_qSubsetState_signState_le

theorem norm_inner_qSubsetState_signState_le (S : Finset (Qubits n)) (hS : S.Nonempty) :
||⟨⟨qSubsetState S hS, signState n⟩⟩_C||
≤ Real.sqrt S.card * (Real.sqrt (Fintype.card (Qubits n)))⁻¹ := by

FINDING (G-241): ‘|⟨S|signState n⟩| ≤ √|S|·2^{-n/2}’ for every non-empty ‘S’. REAL: it uses only that every entry has the SAME modulus — dropping that (using, say, arbitrary signs of arbitrary MAGNITUDE) would let the sum’s modulus exceed this bound.

THE DRAFT SAYS (cont.)	LEAN'S VERDICT (cont.)
G-245 The proof of Theorem ‘main.tex:1254’ (Appendix V, G-239’s theorem), claiming a guiding state for ‘log{XX+YY,X,Y,Z}Hamiltonian’: > We claim the following guided state for ‘log{XX+YY, X, Y, Z}’ Hamiltonian > will always be ‘1/poly(n)’ close to the ground state. > ‘$v\rangle = E\rangle_{\text{edge}} \otimes S\rangle_{\text{singletons}}$’ (‘main.tex:1258’-‘1260’). main.tex:1266 "If we take a uniform superposition state $ss\rangle_m$ (on m-qubits) in place of $\psi\rangle$, then this state will be at least $1/\text{poly}(n)$ close to any vector in the subspace", with $\langle ss \psi\rangle \geq 1/\text{poly}(n)$ displayed at main.tex:1269.	FALSE lean / L2 lean/Qgadget/Findings/Section_13_XYZGuidingStateCounterexamples.lean#exists_unit_orthogonal_uniform <pre>theorem exists_unit_orthogonal_uniform (m : ℕ) (hm : 1 ≤ m) : ∃ u : EuclideanSpace ℂ (Fin (2 ^ m)), u = 1 ∧ ⟨⟨Qgadget.Borrowed.Verbatim.subsetState (Finset.univ : Finset (Fin (2 ^ m))) Finset.univ_nonempty, u⟩_C = 0 := by</pre> FINDING (G-245): for ‘$m \geq 1$’, some unit vector in ‘$\mathbb{C}^{2^m}$’ is orthogonal to ‘main.tex:1252’'s uniform subset state. REUSES ‘Qgadget.Borrowed.exists_unit_mem_inner_eq_zero’ (built for the paper’s ‘Π_{gs}’ vs.
G-246 Theorem (‘main.tex:1253’-‘1254’): "The problem $[\{XX+YY,X,Y,Z\}$-Hamiltonians, ‘main.tex:1240’-‘1247’] with the edge parameter $m \leq O(\log(n))$ is in BQP, where $m:= E$ is the number of edges in the underlying interaction graph." Proof (‘main.tex:1256’-‘1272’): a reduction to $\text{GLH}(\gamma,\delta)$ via a guiding state ‘$E\rangle_{\text{edge}} \otimes S\rangle_{\text{singletons}}$’. main.tex:1277 Heading (main.tex:1276): "Why problem \ref{thm:xyz-ham} is hard for P, BPP and NP machine". Body: "Yet there are no known BPP or NP algorithms for the problem. The key hurdle is the inverse-polynomial precision required to solve the problem."	FALSE lean / L2 lean/Qgadget/Verbatim/Theorem_XYZHam_LogEdges_inBQP.lean#groundEnergy_edgeSplit <pre>theorem groundEnergy_edgeSplit {n : ℕ} (W : Finset (Fin n)) {A : Matrix (LocalReg W) (LocalReg W) ℂ} {B : Matrix (CoReg W) (CoReg W) ℂ} (hA : A.IsHermitian) (hB : B.IsHermitian) (hK : (A ⊗_k (1 : Matrix (CoReg W) (CoReg W) ℂ) + (1 : Matrix (LocalReg W) (LocalReg W) ℂ) ⊗_k B).IsHermitian) (m c : ℕ) (hn : n ≠ 0) (hm : m ≤ c * Nat.log 2 n) (hWcard : W.card ≤ 2 * m) : Qgadget.Borrowed.groundEnergy hK = Qgadget.Borrowed.groundEnergy hA + Qgadget.Borrowed.groundEnergy hB ∧ Fintype.card (LocalReg W) ≤ n ^ (2 * c) :=</pre> FINDING (G-239): the problem’s TRUE content, and MORE than the paper states. ‘H_W’ acts on the (at most ‘$2m$’, hence — by ‘$m \leq O(\log n)$’ — polynomial-dimensional) register of the edges’ vertices; ‘H_R’ acts on the rest.

THE DRAFT SAYS (cont.)

G-256 Definition (‘main.tex:1312’-‘1318’), ‘l2’-norm sampling: $>$ We have an ‘l2’-norm sampling access to a quantum state ‘u’ if the $>$ probability to get the ‘i’-th entry of ‘u’ scales as $>$ ‘Pr[u_i] = |u_i| / ||u||_2’.

main.tex:1315

Definition l2-norm sampling: "the probability to get the i-th entry of u scales as Pr[u_i] = |u_i| / ||u||_2."

G-34 Definition 9.1 — parallel unitary embedding and the class QNC
[cite{moore2001parallel}]

main.tex:1433

Definition [moore2001parallel, QNC]: "QNC_k is the class of operators parallelizable to $O(\log^k n)$ depth with a polynomial number of ancilla. That is, F is in QNC^k if F(n) can be embedded in a circuit of depth at most $O(\log^k n)$, with at most $\text{poly}(n)$ ancilla. Then, QNC = $\bigcup_{k=0}^{\infty} \text{QNC}_k$."

LEAN’S VERDICT (cont.)

FALSE lean / L2

lean/Qgadget/Verbatim/Definition_L2NormSampling.lean#l1_ge_l2

theorem l1_ge_l2 {N : Type*} [Fintype N] (u : N → ℂ) :

Real.sqrt (∑ i, ||u i|| ^ 2) ≤ ∑ i, ||u i|| := by

‘||u||_1 ≥ ||u||_2’ for every ‘u : N → ℂ’. REAL: it could have failed had the cross terms ‘||u_i|| ||u_j||’ (‘i≠j’) in ‘(∑ ||u_i||)^2’ been negative rather than nonnegative — dropping to signed reals (or a non-absolute-value norm) would break the direction of the bound.

lean/Qgadget/Verbatim/Definition_L2NormSampling.lean#l1_gt_l2_of_two_ne_zero

theorem l1_gt_l2_of_two_ne_zero {N : Type*} [Fintype N] [DecidableEq N] (u : N → ℂ)
(i j : N) (hij : i ≠ j) (hi : u i ≠ 0) (hj : u j ≠ 0) :

Real.sqrt (∑ k, ||u k|| ^ 2) < ∑ k, ||u k|| := by

‘||u||_1 > ||u||_2’ strictly, whenever two DISTINCT entries of ‘u’ are both nonzero. Together with ‘l1_ge_l2’ this gives the row’s "equality exactly when at most one non-zero entry": equality forces every pair of entries to fail this hypothesis, i.e.

lean/Qgadget/Verbatim/Definition_L2NormSampling.lean#sum_div_norm_two_eq_sqrt_two

theorem sum_div_norm_two_eq_sqrt_two :

(∑ i, ||(![1,1] : Fin 2 → ℂ) i||) / Real.sqrt (∑ i, ||(![1,1] : Fin 2 → ℂ) i|| ^ 2)
= Real.sqrt 2 := by

FINDING (G-256): at ‘u=(1,1)’, ‘main.tex:1315’'s "probabilities" ‘|u_i|/||u||_2’ sum to ‘√2’, not ‘1’. The exact value the row’s ‘statement’ names.

ILL_POSED lean / L2-PARTIAL

lean/Qgadget/Borrowed/Verbatim/Definition_9_1_QNC.lean#definition_9_1_embedding_admits_nonunitary

theorem definition_9_1_embedding_admits_nonunitary

(n m : ℕ) (hm : 0 < m) (F : Matrix (Fin n → Bool) (Fin n → Bool) ℂ) :
∃ M : Matrix ((Fin n → Bool) × (Fin m → Bool)) ((Fin n → Bool) × (Fin m → Bool)) ℂ,
EmbeddedWithAncilla F M ∧ M^H * M ≠ 1 := by

The embedding relation of ‘main.tex:1434’ is satisfied by matrices that are not unitary. For every ‘n’, every ‘m ≥ 1’ and every ‘F’ whatever — in particular for every unitary ‘F(n)’, which is what :1434 declares ‘F(n)’ to be — there is an ‘M’ that is "embedded" in the paper’s sense and for which ‘M^H M ≠ 1’.

THE DRAFT SAYS (cont.)

G-360 the displayed definition of the QFT as a map on amplitude vectors. An unnumbered prose display, not a ‘theorem’-family environment: it advances no counter, so it has no number.

main.tex:1438

"QFT is a unitary operation that essentially performs the discrete Fourier transform on the amplitudes vector of a quantum state - the QFT maps the quantum state $\sum_{x=0}^{m-1} \alpha_x |x\rangle$ to the quantum state $\sum_{x=0}^{m-1} \beta_x |x\rangle$ where $\beta_x = (1/\sqrt{m}) \sum_{y=0}^{m-1} e^{i(2\pi/m)xy} \alpha_y$."

G-35 Theorem 9.2 — a log-depth circuit approximating the QFT [cite{cleve2000fast}]

main.tex:1451

Theorem [cleve2000fast]: "For any n and ϵ , there is a quantum circuit approximating the QFT modulo 2^n with precision ϵ that has size $O(n \log(n/\epsilon))$ and depth $O(\log n + \log \log(n/\epsilon))$."

G-37 the general definition of the QFT over a finite Abelian group, ‘\begin{equation}\label{eqn:qft}’ — a numbered ‘equation’, not a ‘theorem’-family environment, so it takes the equation counter and no shared-counter number.

main.tex:1465

Equation (eqn:qft), the general QFT over a finite Abelian group: $F_G = (1/\sqrt{|G|}) \sum_{x \in G} \sum_{y \in \hat{G}} \chi_y(x) |y\rangle\langle x|$.

LEAN’S VERDICT (cont.)

TRUE lean / L2

lean/Qgadget/Borrowed/Verbatim/Section_9_QFT_AmplitudeMap.lean#qftAmplitudeMatrix_unitary

theorem qftAmplitudeMatrix_unitary (m : ℕ) [NeZero m] :

(qftAmplitudeMatrix m)^H * qftAmplitudeMatrix m = 1 := by

The map of ‘main.tex:1448’ is unitary, for every ‘ $m \geq 1$ ’. This is the verdict on the only assertion the display makes: :1438 and Paper 01 :2905 call both ‘ $\Sigma \alpha_x |x\rangle$ ’ and ‘ $\Sigma \beta_x |x\rangle$ ’ "the quantum state", and that is true precisely because ‘ $F^\dagger F = 1$ ’, which is what is proved here.

ILL_POSED lean / L2

lean/Qgadget/Borrowed/Verbatim/Theorem_9_2_LogDepthQFT.lean#theorem_9_2_readingB_forces_empty_circuit_at_eps_eq_n

theorem theorem_9_2_readingB_forces_empty_circuit_at_eps_eq_n
(Circuit : Type) (size depth : Circuit → ℕ)
(approximatesQFT : ℕ → ℝ → Circuit → Prop)
(h : theorem_9_2_readingB Circuit size depth approximatesQFT)
(n : ℕ) (hn : n ≠ 0) :

∃ c : Circuit, approximatesQFT n (n : ℝ) c ∧ size c = 0 := by

Theorem 9.2, at ‘ $\epsilon = n$ ’, asserts a circuit with no gates that approximates the QFT modulo ‘ 2^n ’. "For any ‘ n ’ and ‘ ϵ ’" (‘main.tex:1452’) places no constraint on ‘ ϵ ’; at ‘ $\epsilon = n$ ’ the argument of the size bound’s logarithm is ‘1’, the logarithm is ‘0’, and the bound reads ‘size ≤ 0 ’.

ILL_POSED lean / L2-WEAK

lean/Qgadget/Borrowed/Verbatim/Section_10_QFT_FiniteAbelianGroup.lean#qftFiniteAbelianAsWritten_not_unitary_when_indices_differ

theorem qftFiniteAbelianAsWritten_not_unitary_when_indices_differ
(chi : Fin 1 → ZMod 2 → ℂ) :

(qftFiniteAbelianAsWritten chi)^H * qftFiniteAbelianAsWritten chi ≠ 1 := by

The identification ‘ $\hat{G} := G$ ’ is not free. With ‘ $\hat{G}$ ’ a one-element set and ‘ $G = \mathbb{Z}_2$ ’ — the Boolean group of :1470 at ‘ $n = 1$ ’, one of the two cases :1468 names — the display of :1465 fails ‘ $F^\dagger F = 1$ ’ for EVERY family of characters whatever, complete or not.

THE DRAFT SAYS (cont.)	LEAN'S VERDICT (cont.)
G-38 the general definition of the QFT over a finite Abelian group, ‘$\begin{equation}\label{eqn:qft}\end{equation}$’ — a numbered ‘equation’, not a ‘theorem’-family environment, so it takes the equation counter and no shared-counter number. main.tex:1468 "where Ghat : G -> C is a complete set of characters of G, and chi_y(x) denotes the yth character of G evaluated at x."	GAP lean / L2 lean/Qgadget/Borrowed/Verbatim/Section_10_QFT_FiniteAbelianGroup.lean#qftFiniteAbelian_not_unitary_when_characters_coincide theorem qftFiniteAbelian_not_unitary_when_characters_coincide : $(\text{qftFiniteAbelian } (G := \mathbb{Z}\text{Mod } 2) (\text{fun } _ _ => 1))^\text{H} *$ Without an orthogonality condition on ‘χ’, the display of ‘main.tex:1465’ does not define a unitary operator, so it does not define a Fourier transform. Witness: ‘$G = \mathbb{Z}_2$’ — the Boolean group of :1470 at ‘$n = 1$’, which :1468 names as one of its two crucial cases — and ‘$\chi_y \equiv 1$’ for every ‘y’, a family in which every member is a genuine [...]
G-364 "The characters of this group take the following form. ‘$\chi_y(x) = (-1)^{x \cdot y}$’ where ‘$x \cdot y = x_{1y_1} \oplus \dots \oplus x_{ny_n}$’." main.tex:1473 "The characters of this group take the following form. $\chi_y(x) = (-1)^{x \cdot y}$ where $x \cdot y = x_{1y_1} \text{ xor } \dots \text{ xor } x_{ny_n}$."	TRUE lean / L2 lean/Qgadget/Findings/Section_10_QFT_BooleanCharacterClassification.lean#booleanCharacter_existsUnique theorem booleanCharacter_existsUnique (n : ℕ) (chi : AddChar (Fin n → ℤMod 2) ℂ) : $\exists! y : \text{Fin } n \rightarrow \mathbb{Z}\text{Mod } 2, \forall x, \text{chi } x = \text{booleanCharacter } n \ y \ x := \text{by}$ Every additive character of ‘$\text{Fin } n \rightarrow \mathbb{Z}\text{Mod } 2$’ valued in ‘$\mathbb{C}$’ is ‘booleanCharacter n y’ for exactly one ‘y’ (‘main.tex:1473’, G-364).
G-31 §10.1.1 "Boolean Group" — the character formula, the claimed identity ‘$F_{\mathbb{Z}_2^n} = H^{\otimes n}$’, and the matrix given for ‘H’. Unnumbered prose displays; they take the ‘equation’ counter and no shared-counter number. main.tex:1481 "H = [[1, 1], [1, -1]]" given as the matrix representation of the Hadamard transform in the computational basis, immediately after the identity $F_{\mathbb{Z}_2^n} = (1/\sqrt{2^n}) \sum (-1)^{x \cdot y} y\rangle\langle x = H^{\otimes n}$ (main.tex:1477).	FALSE lean / L2 lean/Qgadget/Borrowed/Verbatim/Section_10_QFT_BooleanGroup.lean#hadamardAsWritten_not_unitary theorem hadamardAsWritten_not_unitary : hadamardAsWritten^H * hadamardAsWritten ≠ 1 := by The Hadamard matrix as printed at ‘main.tex:1481-1483’ is not unitary. Its columns have norm ‘$\sqrt{2}$’, so ‘$H^H H$’ has ‘2’ on the diagonal where the identity has ‘1’. lean/Qgadget/Borrowed/Verbatim/Section_10_QFT_BooleanGroup.lean#qftBoolean_ne_hadamardTensorPow theorem qftBoolean_ne_hadamardTensorPow (n : ℕ) (hn : 0 < n) : $\text{qftBoolean } n \neq \text{hadamardTensorPow } n := \text{by}$ ‘main.tex:1477’ is false for every ‘$n \geq 1$’. The witness is the ‘(0,0)’ entry: the left-hand side has ‘$2^{-n/2}$’ there and the right-hand side has ‘1’, because ‘$H^{\otimes n}$’ for the ‘H’ printed at :1481 has every entry ‘± 1’.

THE DRAFT SAYS (cont.)	LEAN'S VERDICT (cont.)
G-366 "$\omega_m := \exp(2\pi i/m)$ is a primitive m'th root of unity." main.tex:1491 "omega_m := exp(2 pi i/m) is a primitive mth root of unity."	TRUE lean / L2 lean/Qgadget/Findings/Section_10_QFT_OmegaPrimitiveRoot.lean#omega_isPrimitiveRoot theorem omega_isPrimitiveRoot (m : ℕ) (hm : m ≠ 0) : IsPrimitiveRoot (omega m) m := by $\omega_m := \exp(2\pi i/m)$ is a primitive m'th root of unity, for every $m \neq 0$ ('main.tex:1491', G-366).
G-367 The successive rewrites of $F_{2^n} x\rangle$, from the amplitude formula through the binary-digit substitution, the phase factorisation, and the bit-level simplification. main.tex:1493 "It implies $F_{2^n} x\rangle = (1/\sqrt{2^n}) \sum_{y \in Z_{2^n}} \omega_{2^n}^{xy} y\rangle$."	TRUE lean / L2 lean/Qgadget/Findings/Section_10_QFT_CyclicAmplitude.lean#qftCyclic_mulVec_single theorem qftCyclic_mulVec_single (n : ℕ) (x y : ZMod (2 ^ n)) : (qftCyclic n).mulVec (Pi.single x (1 : ℂ)) y = qftCyclicAmplitude n x.val y.val := by 'qftCyclic n' applied to the basis vector at 'x' has amplitude 'qftCyclicAmplitude n x y' at every 'y' ('main.tex:1493', G-367).
G-368 The successive rewrites of $F_{2^n} x\rangle$, from the amplitude formula through the binary-digit substitution, the phase factorisation, and the bit-level simplification. main.tex:1497 $= (1/\sqrt{2^n}) \sum_{y \in Z_{2^n}} \omega_{2^n}^{x(\sum_{k=0}^{n-1} y_k 2^k)} y_{n-1} \dots y_0\rangle.$	TRUE lean / L2 lean/Qgadget/Findings/Section_10_QFT_CyclicAmplitude.lean#digit_sum_eq theorem digit_sum_eq (n : ℕ) : ∀ y : ℕ, y < 2 ^ n → $y = \sum_{k \in \text{Finset.range } n, ((y / 2 ^ k) \% 2) * 2 ^ k} := \text{by}$ Every '$y < 2^n$' equals the sum of its binary digits, '$y = \sum_{k < n} ((y/2^k) \bmod 2) \cdot 2^k$' ('main.tex:1497', G-368, the paper's own substitution of 'y' by its bit representation '$y_{n-1} \dots y_0$'). lean/Qgadget/Findings/Section_10_QFT_CyclicAmplitude.lean#qftCyclicAmplitude_binary_digits theorem qftCyclicAmplitude_binary_digits (n x y : ℕ) (hy : y < 2 ^ n) : qftCyclicAmplitude n x y = qftCyclicAmplitude n x (∑ k ∈ Finset.range n, ((y / 2 ^ k) \% 2) * 2 ^ k) := by G-368's own claim: the amplitude of 'main.tex:1493' is unchanged when 'y' is replaced by the sum of its binary digits, at every 'x' and every '$y < 2^n$' — immediate from 'digit_sum_eq', since the two sides name the same natural number.

THE DRAFT SAYS (cont.)	LEAN'S VERDICT (cont.)
G-369 The successive rewrites of $'F_{\{2^n\} x}'$, from the amplitude formula through the binary-digit substitution, the phase factorisation, and the bit-level simplification. main.tex:1500 $= (1/\sqrt{2^n}) \sum_{y \text{ in } Z_{\{2^n\}}} \prod_{k=0}^{n-1} \omega_{\{2^n\}^{\{x y_k 2^k\}} y_{\{n-1\}} y \dots y_0}.$	TRUE lean / L2 lean/Qgadget/Findings/Section_10_QFT_CyclicAmplitude.lean#omega_pow_mul_sum_eq_prod <pre>theorem omega_pow_mul_sum_eq_prod (n x : ℕ) (d : ℕ → ℕ) (m : ℕ) : (omega (2 ^ n)) ^ (x * ∑ k ∈ Finset.range m, d k * 2 ^ k) = ∏ k ∈ Finset.range m, (omega (2 ^ n)) ^ (x * d k * 2 ^ k) := by</pre> The phase factors over the binary digits, $\omega^{\{x \cdot \sum_k d_k 2^k\}} = \prod_k \omega^{\{x \cdot d_k \cdot 2^k\}}$ ($'main.tex:1500'$, G-369), for an arbitrary digit sequence $'d'$.
G-41 main.tex:1497/:1500 write the QFT's output basis vector as $' y_{\{n-1\}} \dots y_0>'$, most significant bit first; main.tex:1504 writes the SAME vector as $'\otimes_{k=0}^{n-1} y_k\rangle = y_0\rangle \otimes y_1\rangle \otimes \dots \otimes y_{\{n-1\}}\rangle'$, least significant bit first — joined to the earlier displays by $'='$. main.tex:1504 $= (1/\sqrt{2^n}) \text{tensor}_{\{k=0\}^{n-1}} \sum_{y_k \text{ in } Z_2} \omega_{\{2^n\}^{\{x y_k 2^k\}} y_k}, \text{ joined by "="}$ to the display at main.tex:1500 whose ket is written $y_{\{n-1\}} y \dots y_0\rangle$.	ILL_POSED lean / L2 lean/Qgadget/Findings/Section_10_QFT_BitOrderReversal.lean#stateACoeff_ne_stateBCoeff <pre>theorem stateACoeff_ne_stateBCoeff : stateACoeff 2 1 (![true, false]) ≠ stateBCoeff 2 1 (![true, false]) := by</pre> $'main.tex:1500'$ and $'main.tex:1504'$ denote DIFFERENT vectors at $'n = 2'$, $'x = 1'$ (G-41): the register state $'(true, false)'$ gets a different coefficient under the two conventions the two displays use for the same bound variable $'y'$.
G-39 §10.1.2 "Cyclic Group" — the QFT over $'\mathbb{Z}_{\{2^n\}}'$, the primitive root $'\omega_m'$, the bitwise factorisation, and the claim about which bits each output qubit depends on. Un-numbered prose displays; they take the $'equation'$ counter and no shared-counter number. main.tex:1508 $F x\rangle = (1/\sqrt{2^n}) \text{tensor}_{\{k=0\}^{n-1}} (0\rangle + \omega_{\{2^n\}^{\{x 2^k\}} 1\rangle) = (1/\sqrt{2^n}) \text{tensor}_{\{k=0\}^{n-1}} (0\rangle + \omega_{\{2^n\}^{\{x 2^k\}} x\rangle 1\rangle).$	TRUE lean / L2-PARTIAL lean/Qgadget/Borrowed/Verbatim/Section_10_QFT_CyclicGroup.lean#cyclic_omega_exponent_collapse <pre>theorem cyclic_omega_exponent_collapse (n k x : ℕ) (hk : k ≤ n) : (omega (2 ^ n)) ^ (x * 2 ^ k) = (omega (2 ^ (n - k))) ^ x := by</pre> The second equality of $'main.tex:1508'$ holds: the $'k'$-th single-qubit factor's phase, written $'\omega_{\{2^n\}^{\{x 2^k\}}'$ on the left and $'\omega_{\{2^n\}^{\{x 2^{n-k}\}}^{\{x\}}'$ on the right, is the same number for every $'k \leq n'$ and every $'x'$. REAL, and it could have failed: the two sides are exponentials of $'2\pi i \cdot x \cdot 2^k / 2^n'$ and $'2\pi i \cdot x / 2^{n-k}'$, and they agree because $'2^n \dots'$
G-370 The successive rewrites of $'F_{\{2^n\} x}'$, from the amplitude formula through the binary-digit substitution, the phase factorisation, and the bit-level simplification. main.tex:1508 "Thus, QFT can be simplified to bit level as $F x\rangle = (1/\sqrt{2^n}) \text{tensor}_{\{k=0\}^{n-1}} (0\rangle + \omega_{\{2^n\}^{\{x 2^k\}} 1\rangle)$", the first equality, following main.tex:1504's $\text{tensor}_{\{k\}} \sum_{y_k \text{ in } Z_2} \omega_{\{2^n\}^{\{x y_k 2^k\}} y_k$.	TRUE lean / L2 lean/Qgadget/Findings/Section_10_QFT_CyclicAmplitude.lean#qft_bit_sum_eq_pair <pre>theorem qft_bit_sum_eq_pair (n k x : ℕ) : (fun j : ℤMod 2 => ∑ b : ℤMod 2, (omega (2 ^ n)) ^ (x * b.val * 2 ^ k) * (Pi.single b (1 : ℂ) : ℤMod 2 → ℂ) j) = (fun j : ℤMod 2 => if j.val = 0 then (1 : ℂ) else (omega (2 ^ n)) ^ (x * 2 ^ k)) := by</pre> The $'k'$-th tensor factor's two-term sum collapses to the paper's pair $' 0\rangle + \omega^{\{x 2^k\}} 1\rangle'$ ($'main.tex:1508'$, G-370, first equality only — its own scope, since the SECOND equality of the same display is G-39/G-509). The amplitude vector is a function $'\mathbb{ZMod } 2 \rightarrow \mathbb{C}';$ no tensor-product register is modelled (G-41's note).

THE DRAFT SAYS (cont.)	LEAN'S VERDICT (cont.)
G-509 main.tex:1508 "omega_m := exp(2 pi i/m) is a primitive mth root of unity" (main.tex:1491), under which the second equality of main.tex:1508 reads $\omega_{2^n}^{x \cdot 2^k} = \omega_{2^{n-k}}^x$.	TRUE lean / L2 lean/Qgadget/Probes/CyclicCollapseWithoutRangeHypothesis.lean#omega_collapse_every_k theorem omega_collapse_every_k (n k x : ℕ) : $(\omega_{2^n})^x = (\omega_{2^{n-k}})^{x \cdot 2^k} := \text{by}$ The transcribed collapse of ‘main.tex:1508’ needs no range hypothesis.
G-40 §10.1.2 "Cyclic Group" — the QFT over $\mathbb{Z}_{2^n}$, the primitive root ω_m, the bitwise factorisation, and the claim about which bits each output qubit depends on. Un-numbered prose displays; they take the ‘equation’ counter and no shared-counter number. main.tex:1510 "Alternatively, we can interpret $F x\rangle$ as a tensor product of single-qubit states, where the k-th qubit only depends on the n-k least significant bits of x."	TRUE lean / L2-PARTIAL lean/Qgadget/Borrowed/Verbatim/Section_10_QFT_CyclicGroup.lean#cyclic_qubit_depends_on_low_bits theorem cyclic_qubit_depends_on_low_bits (n k x q : ℕ) (hk : k ≤ n) : $(\omega_{2^n})^x = (\omega_{2^{n-k}})^{x \cdot 2^k} := \text{by}$ The claim of ‘main.tex:1510’ holds: "the ‘k’-th qubit only depends on the ‘n-k’ least significant bits of ‘x’". Two inputs that agree modulo 2^{n-k} — that is, whose ‘n-k’ least significant bits coincide — give the same ‘k’-th factor $0\rangle + \omega_{2^n}^{x \cdot 2^k} 1\rangle$, because the phase is unchanged when ‘x’ is shifted by any multiple ‘q · [...]’
G-32 §10.1.3 "General Finite Abelian Group" — the QFT over $\mathbb{Z}_N$, the claim that it "strictly requires $N = 2^n$", the two-step generalisation plan, the transformation $x, 0\rangle \mapsto x, x\rangle \mapsto 0, x\rangle$, the shift operator, and the diagonalisation identity. Unnumbered prose displays; they take the ‘equation’ counter and no shared-counter number. main.tex:1517 Of the QFT over $\mathbb{Z}_N$ displayed at main.tex:1515: "This definition strictly requires $N = 2^n$ for some n in $\mathbb{Z}$."	FALSE lean / L2 lean/Qgadget/Borrowed/Verbatim/Section_10_QFT_GeneralFiniteAbelianGroup.lean#qft_ZN_unitary_for_every_N theorem qft_ZN_unitary_for_every_N (N : ℕ) [NeZero N] : $(\text{qftZN } N)^H * \text{qftZN } N = 1 := \text{by}$ ‘main.tex:1517’ is false. The display of :1515 is defined, and unitary, for every ‘$N \geq 1$’; no power-of-two hypothesis appears in this statement or in its proof.
G-378 "The key observation is that $x\rangle$, the Fourier basis, is an eigenstate of the unitary shift operator $U := \sum_{x \in \mathbb{Z}_N} x+1\rangle\langle x$, which adds ‘1’ modulo ‘N’." main.tex:1526 "The key observation is that $x\rangle$, the Fourier basis, is an eigenstate of the unitary shift operator $U := \sum_{x \in \mathbb{Z}_N} x+1\rangle\langle x$, which adds 1 modulo N."	TRUE lean / L2 lean/Qgadget/Findings/Section_10_QFT_ShiftEigenstate.lean#shiftOperator_mulVec_qftZN_col theorem shiftOperator_mulVec_qftZN_col (N : ℕ) [NeZero N] (x : ℤMod N) : $(\text{shiftOperator } N).mulVec (\text{fun } i \Rightarrow \text{qftZN } N \ i \ x) = (\mathbb{ZMod}.stdAddChar \ (-x) : \mathbb{C}) \bullet (\text{fun } i \Rightarrow \text{qftZN } N \ i \ x) := \text{by}$ The Fourier-basis column ‘$i \mapsto F_{\mathbb{Z}_N}[x](i)$’ is an eigenvector of the shift operator, with eigenvalue ω_N^{-x} (‘main.tex:1526’, G-378).

THE DRAFT SAYS (cont.)

G-377 §10.1.3 "General Finite Abelian Group" — the QFT over $\mathbb{Z}_N$, the claim that it "strictly requires $N = 2^n$ ", the two-step generalisation plan, the transformation $|x, 0\rangle \mapsto |x, x\rangle \mapsto |0, x\rangle$, the shift operator, and the diagonalisation identity. Unnumbered prose displays; they take the ‘equation’ counter and no shared-counter number.

main.tex:1532

"the QFT operator diagonalizes the above shift operator, that is, $F_{\mathbb{Z}_N}^\dagger U F_{\mathbb{Z}_N} = \sum_{x \in \mathbb{Z}_N} \omega_N^{-x} |x\rangle\langle x|$."

G-42 "Context for QPE" — the input promise and the task of quantum phase estimation, as restated in Appendix IX. Unnumbered prose; it advances no counter.

main.tex:1539

"Context for QPE: We have a sparse unitary operator on which a controlled- U^j operation can be efficiently performed for j in $\mathbb{Z}$. Furthermore, we are given an eigenstate $|\eta\rangle$ of U with eigenvalue $e^{2\pi i \phi_\eta}$. The task is to find $\tilde{\phi}_\eta$ such that $||\tilde{\phi}_\eta - \phi_\eta|| \leq \epsilon$."

G-33 The caption of the Quantum Phase/Eigenvalue Estimation figure, and in particular its closing Remark

main.tex:1554

Figure caption: "The unitary $U = e^{-iH}$ is implemented using the technique of Hamiltonian simulation ... Remark: $H^{\otimes t}$ is the Hadamard transform, which is a Fourier transform over the Boolean group $(\mathbb{Z}/2\mathbb{Z})^t$."

LEAN'S VERDICT (cont.)

TRUE lean / L2

lean/Qgadget/Borrowed/Verbatim/Section_10_QFT_GeneralFiniteAbelianGroup.lean#qft_diagonalises_shift

```
theorem qft_diagonalises_shift (N : ℕ) [NeZero N] :
  (qftZN N)H * shiftOperator N * qftZN N
    = Matrix.diagonal (fun x : ZMod N => (ZMod.stdAddChar (-x) : ℂ)) := by
```

The diagonalisation asserted at ‘main.tex:1532’ holds, for every $N \geq 1$: ‘ $F_{\mathbb{Z}_N}^\dagger U F_{\mathbb{Z}_N}$ ’ is the diagonal matrix with entries ω_N^{-x} . :1526’s claim that the Fourier basis vector ‘ $F_{\mathbb{Z}_N}|x\rangle$ ’ is an eigenstate of the shift operator is the same statement read column by column, with the eigenvalue ω_N^{-x} that :1532 records.

ILL_POSED lean / L2-WEAK

lean/Qgadget/Borrowed/Verbatim/Section_10_QPE_Context.lean#qpe_context_task_is_ill_posed

```
theorem qpe_context_task_is_ill_posed (U : E →I[C] E) (algorithm : E → ℝ) (ε : ℝ)
  (hε : ε < 1 / 2) (η : E) (hη : U η = η) : ¬ qpe_context_task U algorithm ε := by
```

The task stated at ‘main.tex:1539’ cannot be performed, by any algorithm whatever, whenever ‘ $\epsilon < 1/2$ ’. Not because the algorithm is hard to build, but because the specification does not determine an answer: the phase ‘ φ_η ’ is recoverable from the eigenvalue ‘ $e^{2\pi i \varphi_\eta}$ ’ only modulo ‘1’, so a vector fixed by ‘ U ’ is an eigenstate "with [...]

ILL_POSED lean / L2-PARTIAL

lean/Qgadget/Borrowed/Verbatim/Caption_3017_QPE_HadamardRemark.lean#p71_readings_differ

```
theorem p71_readings_differ (t : ℕ) (ht : 0 < t) :
  matTensorPow pauliZ t ≠ hadamardTensorPow t := by
```

The caption’s two readings of ‘ $H^{\otimes t}$ ’ give different matrices, for every ‘ $t \geq 1$ ’. Under the binding the caption states twice — ‘ H ’ the input Hamiltonian — the symbol denotes ‘ $\text{matTensorPow } H \ t$ ’; under the Remark it denotes the Hadamard layer.

THE DRAFT SAYS (cont.)	LEAN'S VERDICT (cont.)
G-380 QPE step 3, "Apply controlled-U^j: $2^{-\{m/2\}} \sum_j j\rangle \eta\rangle x\rangle \rightarrow 2^{-\{m/2\}} \sum_j j\rangle U^j \eta\rangle x\rangle = 2^{-\{m/2\}} \sum_{j=0}^{2^m-1} e^{\{2\pi i j \varphi_{\eta}\}} j\rangle \eta\rangle x\rangle$." main.tex:1562 "3. Apply controlled-U^j: $2^{-\{m/2\}} \sum_j j\rangle \eta\rangle x\rangle \rightarrow 2^{-\{m/2\}} \sum_j j\rangle U^j \eta\rangle x\rangle = 2^{-\{m/2\}} \sum_{j=0}^{2^m-1} e^{\{2\pi i j \varphi_{\eta}\}} j\rangle \eta\rangle x\rangle$."	TRUE lean / L2 lean/Qgadget/Findings/Section_10_QPEPhaseKickback.lean#phaseKickback_pow <pre>theorem phaseKickback_pow {E : Type*} [AddCommGroup E] [Module ℂ E] (U : Module.End ℂ E) (eta : E) (phi : ℝ) (hU : U eta = Complex.exp (2 * (Real.pi : ℂ) * Complex.I * phi) • eta) : ∀ j : ℕ, (U ^ j) eta = (Complex.exp (2 * (Real.pi : ℂ) * Complex.I * phi)) ^ j • eta := by Phase kickback iterates: if '$U \eta = e^{\{2\pi i \varphi\}} \eta$', then '$U^j \eta = e^{\{2\pi i j \varphi\}} \eta$' for every '$j : \mathbb{N}$'** ('main.tex:1562', G-380).</pre>
G-381 QPE step 4, "Apply the inverse Fourier transform: $2^{-\{m/2\}} \sum_j e^{\{2\pi i j \varphi_{\eta}\}} j\rangle \eta\rangle x\rangle \rightarrow \varphi_{\eta}\rangle \eta\rangle x\rangle$." main.tex:1563 "4. Apply the inverse Fourier transform: $2^{-\{m/2\}} \sum_{j=0}^{2^m-1} e^{\{2\pi i j \varphi_{\eta}\}} j\rangle \eta\rangle x\rangle \rightarrow \varphi_{\eta}\rangle \eta\rangle x\rangle$."	FALSE lean / L2 lean/Qgadget/Findings/Section_10_QPEInverseFourierStep.lean#hadamard_mulVec_stepThree <pre>theorem hadamard_mulVec_stepThree : hadamard2.mulVec stepThreeState = ![(1 + Complex.I) / 2, (1 - Complex.I) / 2] := by Applying the inverse Fourier transform to step 3's register gives '$((1+i) 0\rangle + (1-i) 1\rangle)/2$' — not a basis state.</pre> lean/Qgadget/Findings/Section_10_QPEInverseFourierStep.lean#hadamard_off_grid_not_basis_state <pre>theorem hadamard_off_grid_not_basis_state : ∀ j : Fin 2, (hadamard2.mulVec stepThreeState) j = (Real.sqrt 2)⁻¹ := by 'main.tex:1563's arrow is FALSE at 'm = 1', '$\varphi = 1/4$' (G-381): the output of applying the inverse Fourier transform has BOTH entries of modulus '$1/\sqrt{2}$', not '1' at some single index and '0' elsewhere — so it is not the basis state '$\varphi_{\eta}\rangle$' the display prints, and measuring returns either outcome with probability '$1/2$'.</pre>
G-36 Lemma 10.1 — the quantum eigenvalue estimation lemma [Nielsen & Chuang, ch. 5 §2, cited in prose without a key] main.tex:1567 Lemma [Quantum eigenvalue estimation]: "Let $H = \sum_{i=1}^m H_i$ be a $O(\log n)$-local Hamiltonian acting on n qubits, with eigenvectors $\psi_j\rangle$ and the corresponding eigenvalues λ_j in $[0,1]$. Then there is a quantum algorithm that, given an input eigenvector $\psi_j\rangle$, will output with probability at least p and epsilon-approximation of λ_j ... in time $\text{poly}(m, 1/\epsilon)$."	GAP lean / L2-PARTIAL lean/Qgadget/Borrowed/Verbatim/Lemma_10_1_QuantumEigenvalueEstimation.lean#lemma_10_1_accuracy_clause_is_vacuous_for_large_eps <pre>theorem lemma_10_1_accuracy_clause_is_vacuous_for_large_eps (ε lam : ℝ) (hε : 1 / 2 ≤ ε) (h0 : 0 ≤ lam) (h1 : lam ≤ 1) : (1 / 2 : ℝ) - lam ≤ ε := by The accuracy clause of 'main.tex:1568' is satisfied by a constant whenever '$\epsilon \geq 1/2$'. The eigenvalues are promised to lie in '[0,1]'; the midpoint '$1/2$' is within '$1/2$' of every point of that interval; and 'ϵ' is never required to be smaller.</pre>

THE DRAFT SAYS (cont.)

G-279 Example [MAX-3-LIN] (‘main.tex:1645’-‘1651’): the input is a system of equations ‘ $x_i \oplus x_j \oplus x_k = b_l$ ’ over ‘GF(2)’, and the goal is to maximise ‘ $S(x) = \sum_{l \in [m]} (x_l^1 \oplus x_l^2 \oplus x_l^3)$ ’.

main.tex:1648

MAX-3-LIN: the input is a system of equations $x_i \text{ xor } x_j \text{ xor } x_k = b_l$ over GF(2), and the task is to maximise $S(x) = \sum_{l \in [m]} (x_l^1 \text{ xor } x_l^2 \text{ xor } x_l^3)$.

G-282 Example [7-Gadget for 3SAT to MAX-2SAT reduction] (‘main.tex:1676’-‘1682’): for ‘ $C_k = (x_1 \vee x_2 \vee x_3)$ ’, the ten 2SAT clauses of ‘main.tex:1678’ form a gadget with the property that a satisfying assignment of ‘ C_k ’ lets some ‘ y^k ’ satisfy seven of the ten, and an unsatisfying one lets at most six.

main.tex:1677

Example [7-gadget for 3SAT to MAX-2SAT]: for $C_k = (x_1 \text{ or } x_2 \text{ or } x_3)$ the gadget is the ten 2SAT clauses $x_1, x_2, x_3, (\text{not } x_1 \text{ or not } x_2), (\text{not } x_2 \text{ or not } x_3), (\text{not } x_3 \text{ or not } x_1), y^k, (x_1 \text{ or not } y^k), (x_2 \text{ or not } y^k), (x_3 \text{ or not } y^k)$. If C_k is satisfied some y^k satisfies at least seven; if not, every y^k satisfies at most six.

G-286 Proposition, item 2 (‘main.tex:1723’-‘1728’): with ‘ $S_2 = \{0,1\}^k$ ’, “an S_2 -partial α -gadget reducing f to F is a strict α -gadget reducing f to F .”

main.tex:1726

Proposition, item 2: with $S_2 = \{0,1\}^k$, “an S_2 -partial α -gadget reducing f to F is a strict α -gadget reducing f to F ”.

LEAN’S VERDICT (cont.)

FALSE lean / L2

lean/Qgadget/Verbatim/Example_Max3Lin.lean#maxThreeLin_objective_ignores_rhs

```
theorem maxThreeLin_objective_ignores_rhs :
  maxThreeLinSatisfies false false false false = true ∧
  maxThreeLinObjective false false false = 0 ∧
  maxThreeLinSatisfies true false false false = false ∧
  maxThreeLinObjective true false false = 1 ∧

  maxThreeLinObjective false false false < maxThreeLinObjective true false false := by
```

FINDING (G-279): the objective ‘ S ’ never reads ‘ b_l ’, so maximising it need not satisfy the system. For ‘ $b_l = \text{false}$ ’: ‘000’ satisfies the equation and scores ‘ $S = 0$ ’; ‘100’ violates it and scores ‘ $S = 1$ ’, the unique maximum of a one-equation objective.

TRUE lean / L2

lean/Qgadget/Verbatim/Example_7Gadget.lean#sevenGadget_isGadget

```
theorem sevenGadget_isGadget : sevenGadget.IsGadget 7 := by
```

FINDING (G-282): the ten clauses of ‘main.tex:1678’, unit-weighted, are a 7-gadget reducing ‘ $C_k = x_1 \vee x_2 \vee x_3$ ’ to ‘{literals, 2-clauses}’, in the formal sense of ‘main.tex:1655’-‘1667’. REAL: each of the three conjuncts is an obligation the ten clauses (and the number 7) could have failed — a single mistranscribed clause, or $\alpha = 6$ [...]

FALSE lean / L2

lean/Qgadget/Verbatim/Proposition_S2PartialStrict.lean#sevenGadget_isSPartialGadget_univ

```
theorem sevenGadget_isSPartialGadget_univ :
  sevenGadget.IsSPartialGadget (Set.univ : Set (Fin 3 → Bool)) 7 := by
```

‘sevenGadget’ is an ‘ S ’-partial 7-gadget at ‘ $S = \text{Set.univ}$ ’ — the draft’s ‘ S_2 ’ (‘main.tex:1723’). The first and third conditions (‘CondUpperOnSat’/‘CondUpperOnUnsat’) are the unrestricted ones already proved (‘sevenGadget_isGadget’, G-282); the two “attains” conditions restricted to ‘Set.univ’ are no weaker than their unrestricted [...]

lean/Qgadget/Verbatim/Proposition_S2PartialStrict.lean#sevenGadget_not_strict

```
theorem sevenGadget_not_strict : ¬ sevenGadget.IsStrictGadget 7 := by
```

FINDING (G-286): ‘sevenGadget’ is NOT a strict 7-gadget — refuting the Proposition’s item 2 at the object of item 1’s own ‘ S_2 ’-partial reading. ‘main.tex:1668’-‘1671’ demands ‘ $\forall b$ ’, and at ‘ $a = (0,0,0)$ ’, ‘ $b = \text{true}$ ’ (‘ $y = \text{true}$ ’), the count is 4, not ‘ $\alpha - 1 = 6$ ’.

THE DRAFT SAYS (cont.)

G-288 Example [Witness matrix for 3SAT to MAX-2SAT] (‘main.tex:1742’-‘1780’): the 8×4 witness matrix ‘W_b’ built from the 7-gadget (G-282), and the truth table (‘main.tex:1764’-‘1780’) it is "constructed from."

main.tex:1746

Example: the witness matrix W_b for the 7-gadget, with S = {0,1}^3, has y = 0 on 000, 001, 010 and 100, y = 0/1 on 110, 101 and 011, and y = 1 on 111.

G-289 Example [Witness matrix for 3SAT to MAX-2SAT] (‘main.tex:1742’-‘1780’): the 8×4 witness matrix ‘W_b’ built from the 7-gadget (G-282), and the truth table (‘main.tex:1764’-‘1780’) it is "constructed from."

main.tex:1760

"This witness matrix was constructed from the following truth table. The choice of the auxiliary variable y is made such that the criteria of the witness matrix are fulfilled."

LEAN’S VERDICT (cont.)

TRUE lean / L2

lean/Qgadget/Verbatim/Example_WitnessMatrix.lean#witnessMatrix_all_rows_correct

```
theorem witnessMatrix_all_rows_correct :
  sevenCount false false false false = 6 ∧
  sevenCount false false true false = 7 ∧
  sevenCount false true false false = 7 ∧
  sevenCount true false false false = 7 ∧
  (sevenCount true true false false = 7 ∧ sevenCount true true false true = 7) ∧
  (sevenCount true false true false = 7 ∧ sevenCount true false true true = 7) ∧
  (sevenCount false true true false = 7 ∧ sevenCount false true true true = 7) ∧
  sevenCount true true true true = 7 := by decide
```

FINDING (G-288): every row of ‘W_b’, under every choice its entry allows, attains ‘ $\alpha = 7$ ’ on a satisfying ‘x’ and ‘ $\alpha - 1 = 6$ ’ at ‘x = 000’. Packaged as one statement over all eight rows (with the three "0/1" rows each carrying both choices).

FALSE lean / L2

lean/Qgadget/Verbatim/Example_WitnessMatrix.lean#tableChoice_y1_fails_at_001

```
theorem tableChoice_y1_fails_at_001 : sevenCount false false true true = 6 := by decide
```

FINDING (G-289): at ‘x = 001, 010, 100’, the truth table’s own choice ‘y = 1’ (‘main.tex:1770’-‘1772’) reaches only 6 of the ten clauses, not ‘ $\alpha = 7$ ’ — so it does not fulfil ‘eqn: witness1’ (‘main.tex:1733’), the criterion the table is said to satisfy, at any of the three rows.

lean/Qgadget/Verbatim/Example_WitnessMatrix.lean#tableChoice_y1_fails_at_010

```
theorem tableChoice_y1_fails_at_010 : sevenCount false true false true = 6 := by decide
```

lean/Qgadget/Verbatim/Example_WitnessMatrix.lean#tableChoice_y1_fails_at_100

```
theorem tableChoice_y1_fails_at_100 : sevenCount true false false true = 6 := by decide
```

THE DRAFT SAYS (cont.)

G-290 Example [Witness matrix for 3SAT to MAX-2SAT] ('main.tex:1742'-'1780'): the 8×4 witness matrix 'W_b' built from the 7-gadget (G-282), and the truth table ('main.tex:1764'-'1780') it is "constructed from."

main.tex:1764

The displayed truth table: columns x_1 , x_2 , x_3 , y , (not x_1 or not x_2), (not x_2 or not x_3), (not x_1 or not x_3), (x_1 or not y), (x_2 or not y), (x_3 or not y), with eight rows of values.

LEAN'S VERDICT (cont.)

FALSE lean / L2

lean/Qgadget/Verbatim/Example_WitnessMatrix.lean#twelve_entries_wrong_thirteen_conjuncts

```
theorem twelve_entries_wrong_thirteen_conjuncts :
- row 001 (main.tex:1770), y=1: printed (x1 $\vee$  $\neg$ y) = 1; actual 0.
  clauseOrNotY1 false true  $\neq$  true  $\wedge$ 
- row 010 (main.tex:1771), y=1: printed (x1 $\vee$  $\neg$ y, x2 $\vee$  $\neg$ y, x3 $\vee$  $\neg$ y) = (1,0,1);
  - actual (0,1,0).
  clauseOrNotY1 false true  $\neq$  true  $\wedge$ 
  clauseOrNotY2 true true  $\neq$  false  $\wedge$ 
  clauseOrNotY3 false true  $\neq$  true  $\wedge$ 
- row 100 (main.tex:1772), y=1: printed (x3 $\vee$  $\neg$ y) = 1; actual 0.
  clauseOrNotY3 false true  $\neq$  true  $\wedge$ 
- row 011 (main.tex:1773): printed negated pairs (0,1,0); actual (1,0,1).
  clauseNegOr12 false true  $\neq$  false  $\wedge$ 
  clauseNegOr23 true true  $\neq$  true  $\wedge$ 
  clauseNegOr13 false true  $\neq$  false  $\wedge$ 
- row 011: printed (x2 $\vee$  $\neg$ y) = 0; actual 1 (x2 = true dominates).
  clauseOrNotY2 true false  $\neq$  false  $\wedge$  clauseOrNotY2 true true  $\neq$  false  $\wedge$ 
- row 101 (main.tex:1774): printed negated pairs (0,0,1); actual (1,1,0).
  clauseNegOr12 true false  $\neq$  false  $\wedge$ 
  clauseNegOr23 false true  $\neq$  false  $\wedge$ 
  clauseNegOr13 true true  $\neq$  true := by decide
```

FINDING (G-290): twelve of the forty-eight printed clause entries ('main.tex:1765'-'1779') disagree with the clause evaluated at the row's own printed '(x1,x2,x3,y)'. THIRTEEN CONJUNCTS witness the twelve wrong cells: eleven cells each get one conjunct naming the printed cell (the row's assignment, from 'main.tex', and the printed value) [...]

lean/Qgadget/Verbatim/Example_WitnessMatrix.lean#clause_defs_match_gadget

```
theorem clause_defs_match_gadget :  $\forall$  x1 x2 x3 y : Bool,
  (clauseNegOr12 x1 x2, clauseNegOr23 x2 x3, clauseNegOr13 x1 x3,
   clauseOrNotY1 x1 y, clauseOrNotY2 x2 y, clauseOrNotY3 x3 y) =
  (sevenGadgetC 3 ![x1, x2, x3] ![y], sevenGadgetC 4 ![x1, x2, x3] ![y],
   sevenGadgetC 5 ![x1, x2, x3] ![y], sevenGadgetC 7 ![x1, x2, x3] ![y],
   sevenGadgetC 8 ![x1, x2, x3] ![y], sevenGadgetC 9 ![x1, x2, x3] ![y]) := by
```

The six hand-transcribed clause functions above agree with 'Qgadget.Support.sevenGadgetC's clauses 4-9 at every input — they are an independent restatement of the SAME six functions, not a fresh definition the row's finding could be an artifact of.

THE DRAFT SAYS (cont.)

G-292 Definition [canonical witness matrix] (‘main.tex:1786’-‘1788’): if ‘F’ is complementation closed, "a canonical witness matrix is the $(S, (2^{|S|} + k - k')_S)$ matrix W whose ... remaining columns are all possible column vectors that are distinct up to complementation from one another and the columns corresponding to the primary variables," ‘k’_S’

main.tex:1787

Definition [canonical witness matrix]: if F is complementation closed, "a canonical witness matrix is the $(S, (2^{|S|} + k - k')_S)$ matrix W whose first k columns correspond to the primary variables and whose remaining columns are all possible column vectors that are distinct up to complementation from one another and the columns corresponding to the primary variables", k’_S being the number of primary columns distinct up to complementation.

LEAN’S VERDICT (cont.)

FALSE lean / L2

lean/Qgadget/Verbatim/Definition_CanonicalWitnessMatrix.lean#card_complClasses_eq_eight

theorem card_complClasses_eq_eight :

(Finset.image complClassRep (Finset.univ : Finset (Fin 4 → Bool))).card = 8 := by decide

The 16 length-4 Boolean vectors fall into exactly 8 complementation classes — the image of ‘complClassRep’ over all 16 vectors.

lean/Qgadget/Verbatim/Definition_CanonicalWitnessMatrix.lean#card_primary_classes_eq_three

theorem card_primary_classes_eq_three :

(Finset.image complClassRep

({primaryCol1, primaryCol2, primaryCol3} : Finset (Fin 4 → Bool))).card = 3 := by decide

The three primary columns occupy exactly THREE distinct complementation classes (‘k’_S = 3’) — none is its own complement, and no two of the three are complementary to each other.

lean/Qgadget/Verbatim/Definition_CanonicalWitnessMatrix.lean#five_classes_remain_after_removing_primary

theorem five_classes_remain_after_removing_primary :

((Finset.image complClassRep (Finset.univ : Finset (Fin 4 → Bool))) \ (Finset.image complClassRep

({primaryCol1, primaryCol2, primaryCol3} : Finset (Fin 4 → Bool))).card = 5 := by

decide

FINDING (G-292): removing the three primary columns’ complementation classes from the eight total classes leaves FIVE, not the thirteen the printed formula $2^{|S|} + k - k'_S = 16 + 3 - 3 = 16$ columns would need (‘16 - 3’ further columns). The corrected count $2^{|S|-1} + k - k'_S = 8 + 3 - 3 = 8$ total columns (‘5’ further, plus the [...]

THE DRAFT SAYS (cont.)

G-293 Fact [Problem with Non-hereditary F] (‘main.tex:1791’-‘1793’): for ‘ $f(a) = a$ ’ and ‘ $F = PC_1$ ’, the equal-weight constraints ‘ $X \oplus Y_i \oplus Y_j$ ’ (‘ $1 \leq i < j \leq k$ ’) "yields a ‘ $(2 - 2/k)$ ’-gadget for even ‘ k ’; thus, the performance tends to 2 in the limit," and separately, "any gadget with ‘ k ’ auxiliary variables has performance at most ‘ $2 - 2^{\{1-k\}}$ ’."

main.tex:1792

Fact [Problem with non-hereditary F]: for $f(a) = a$ and $F = PC_1$, the constraints $X \text{ xor } Y_i \text{ xor } Y_j$, $1 \leq i < j \leq k$, with equal weights, "for an appropriate choice of this weight ... yields a $(2 - 2/k)$ -gadget for even k ; thus, the performance tends to 2 in the limit."

G-294 Fact [Problem with Non-hereditary F] (‘main.tex:1791’-‘1793’): for ‘ $f(a) = a$ ’ and ‘ $F = PC_1$ ’, the equal-weight constraints ‘ $X \oplus Y_i \oplus Y_j$ ’ (‘ $1 \leq i < j \leq k$ ’) "yields a ‘ $(2 - 2/k)$ ’-gadget for even ‘ k ’; thus, the performance tends to 2 in the limit," and separately, "any gadget with ‘ k ’ auxiliary variables has performance at most ‘ $2 - 2^{\{1-k\}}$ ’."

main.tex:1792

"On the other hand, it can be shown that any gadget with k auxiliary variables has performance at most $2 - 2^{\{1-k\}}$ ".

LEAN’S VERDICT (cont.)

FALSE lean / L2

lean/Qgadget/Verbatim/Fact_NonHereditaryPC1.lean#no_weight_gives_three_halves

theorem no_weight_gives_three_halves :

$\neg \exists (w : \mathbb{R}) (hw : 0 \leq w), (pc1Gadget\ w\ hw).IsGadget\ (3 / 2) :=$ by

FINDING (G-293): no non-negative weight makes the ‘ $k=4$ ’ equal-weight gadget a ‘ $3/2$ ’-gadget — refuting ‘main.tex:1792’s “‘ $(2-2/k)$ ’-gadget for even ‘ k ’” at ‘ $k = 4$ ’ (‘ $2 - 2/4 = 3/2$ ’). The argument: ‘CondUpperOnUnsat’ at ‘ $Y = (0,0,1,1)$ ’ forces ‘ $w \cdot 4 \leq 3/2 - 1 = 1/2$ ’, i.e.

lean/Qgadget/Verbatim/Fact_NonHereditaryPC1.lean#pc1Gadget_isGadget_half

theorem pc1Gadget_isGadget_half : (pc1Gadget (1/2) (by norm_num)).IsGadget 3 := by

FINDING (G-294, and G-293’s positive half): the ‘ $k=4$ ’ equal-weight gadget at ‘ $w = 1/2$ ’ is a genuine 3-gadget — ‘main.tex:1792’s claimed universal bound ‘ $2 - 2^{\{1-4\}} = 15/8$ ’ is exceeded by an ACTUAL gadget with ‘ $k = 4$ ’ auxiliary variables (‘ $3 > 15/8$ ’), refuting "any gadget with ‘ k ’ auxiliary variables has performance at most ‘ $2 - [\dots]$ ’"

FALSE lean / L2

lean/Qgadget/Verbatim/Fact_NonHereditaryPC1.lean#pc1Gadget_isGadget_half

theorem pc1Gadget_isGadget_half : (pc1Gadget (1/2) (by norm_num)).IsGadget 3 := by

FINDING (G-294, and G-293’s positive half): the ‘ $k=4$ ’ equal-weight gadget at ‘ $w = 1/2$ ’ is a genuine 3-gadget — ‘main.tex:1792’s claimed universal bound ‘ $2 - 2^{\{1-4\}} = 15/8$ ’ is exceeded by an ACTUAL gadget with ‘ $k = 4$ ’ auxiliary variables (‘ $3 > 15/8$ ’), refuting "any gadget with ‘ k ’ auxiliary variables has performance at most ‘ $2 - [\dots]$ ’"

THE DRAFT SAYS (cont.)

G-297 Fact [Importance of Hereditary F] (‘main.tex:1795’-‘1797’): if two auxiliary columns ‘Y_1, Y_2’ of a witness matrix ‘W’ are identical, replace ‘Y_2’ by ‘Y_1’ in every constraint; "By the hereditary property of F, all the resulting constraints are from F. And, the resulting instance satisfies all the properties of α -gadgets."

main.tex:1796

Fact [Importance of Hereditary F]: if two auxiliary columns Y_1, Y_2 of a witness matrix W are identical, replace Y_2 by Y_1 in every constraint; "By the hereditary property of F, all the resulting constraints are from F. And, the resulting instance satisfies all the properties of alpha-gadgets."

LEAN’S VERDICT (cont.)

FALSE lean / L2

lean/Qgadget/Verbatim/Fact_HereditaryImportance.lean#twoSat_hereditary

```
theorem twoSat_hereditary : ∀ j : Fin 8,
  (∃ i : Fin 2, restrictDiag (twoSatArity2 j) = twoSatArity1 i) ∨
  restrictDiag (twoSatArity2 j) = (fun _ => true) ∨
  restrictDiag (twoSatArity2 j) = (fun _ => false) := by decide
```

‘2SAT’ is hereditary (‘main.tex:1690’) at arity 2→1: every one of the eight members, restricted at its two arguments identified, is EITHER a member of ‘2SAT’'s arity-1 family OR a constant function.

lean/Qgadget/Verbatim/Fact_HereditaryImportance.lean#restrictDiag_negOr_eq_const_true

```
theorem restrictDiag_negOr_eq_const_true :
  restrictDiag (twoSatArity2 5) = (fun _ => true) := by decide
```

FINDING (G-297): (‘Y_1 ∨ ¬Y_2’) (‘twoSatArity2 5’, ‘x ∨ ¬y’) restricted at ‘Y_1 = Y_2’ is the CONSTANT function ‘true’.

lean/Qgadget/Verbatim/Fact_HereditaryImportance.lean#restrictDiag_negOr_not_in_twoSatArity1

```
theorem restrictDiag_negOr_not_in_twoSatArity1 :
  ∀ i : Fin 2, restrictDiag (twoSatArity2 5) ≠ twoSatArity1 i := by decide
```

The constant ‘true’ is NOT a member of ‘2SAT’'s arity-1 family — so the restricted constraint is not "from ‘F’" as ‘main.tex:1797’ claims, though it IS in ‘ $F \cup \{0,1\}$ ’ as the Hereditary Property (‘main.tex:1690’) actually allows. REAL: ‘twoSat_hereditary’ (all eight land in ‘ $F \cup \{0,1\}$ ’) and this theorem (one specific member of the eight [...])

THE DRAFT SAYS (cont.)

G-300 "As a fact from Boolean analysis, there are two nonconstant functions on a single variable, and twelve nonconstant functions on pairs of variables (out of $2^{\{2^2\}}$ functions), so that there are $2 \cdot 7 + 12 \cdot C(7,2) = 266$ functions to consider." ('main.tex:1843'-'1845')

main.tex:1844

"As a fact from Boolean analysis, there are two nonconstant functions on a single variable, and twelve nonconstant functions on pairs of variables (out of $2^{\{2^2\}}$ functions), so that there are $2 \times 7 + 12 \times C(7,2) = 266$ functions to consider."

G-303 'main.tex:1850'-'1860', the LP-based construction of a gadget reducing Parity Check ('PC', 'main.tex:1630') to '2SAT': "the solver outputs the following twelve choices of constraint functions" ('main.tex:1852'-'1858', twelve conjunctions ' $(X_i \wedge \neg Y_j)$ '/' $(X_i \wedge Y_j)$ '/etc.), and "The linear program yields ' $\alpha = 5$ ', so it is a '5'-gadget"

main.tex:1852

"When they solve the LP, the solver outputs the following twelve choices of constraint functions": $(X_1 \text{ and not } Y_1)$, $(X_1 \text{ and } Y_2)$, ..., $(\text{not } X_3 \text{ and not } Y_4)$, in a gadget "for reducing an element of PC to 2SAT family of constraints".

LEAN'S VERDICT (cont.)

FALSE lean / L2

lean/Qgadget/Verbatim/BooleanFunctionCounts.lean#card_nonconstant_fn2_eq_fourteen

theorem card_nonconstant_fn2_eq_fourteen :

(Finset.univ.filter (fun g : Bool → Bool → Bool => ¬ IsConstantFn2 g)).card = 14 := by decide

FINDING (G-300), first half: 14 of the 16 functions are nonconstant — matching the draft's OWN count of nonconstant pair-functions (12) plus the 2 single-variable nonconstant functions the draft counts separately, only if pair-functions and single-variable functions were disjoint categories; they are not, since a function of one variable [...]

lean/Qgadget/Verbatim/BooleanFunctionCounts.lean#card_dependsOnBoth_eq_ten

theorem card_dependsOnBoth_eq_ten :

(Finset.univ.filter (fun g : Bool → Bool → Bool => DependsOnBoth g)).card = 10 := by decide

FINDING (G-300), second half: exactly 10 of the 16 functions depend on BOTH arguments — not the 12 the draft states as "nonconstant functions on pairs of variables." The other 4 nonconstant functions ('14 - 10 = 4') depend on exactly one argument: the identity and negation in each coordinate.

FALSE lean / L2

lean/Qgadget/Verbatim/LPSolverOutput_PC.lean#pcConstraint1_not_twoSat

theorem pcConstraint1_not_twoSat : ∀ i : Fin 8, twoSatForms i ≠ pcConstraint1 := by decide

FINDING (G-303): ' $X_1 \wedge \neg Y_1$ ' equals none of the eight '2SAT' forms. The other eleven printed constraints are conjunctions of the same shape ' $(X_i \wedge Y_j)$ ' up to negating either literal) and fail identically, by the same case analysis; this one instance is the row's own witness.

lean/Qgadget/Verbatim/LPSolverOutput_PC.lean#all_twelve_constraints_are_conjunctions

theorem all_twelve_constraints_are_conjunctions :

(∀ i : Fin 8, twoSatForms i ≠ (fun x y : Bool => x && !y)) ∧ - X1 ∧ ¬Y1
(∀ i : Fin 8, twoSatForms i ≠ (fun x y : Bool => x && y)) ∧ - X1 ∧ Y2 (shape)
(∀ i : Fin 8, twoSatForms i ≠ (fun x y : Bool => !x && y)) ∧ - ¬X1 ∧ Y3
(∀ i : Fin 8, twoSatForms i ≠ (fun x y : Bool => !x && !y)) := by decide

All twelve printed constraints ('main.tex:1852,1855,1858'), as conjunctions of one primary literal and one auxiliary literal, matching none of the eight '2SAT' forms — the general shape 'pc-Constraint1' witnesses.

THE DRAFT SAYS (cont.)	LEAN'S VERDICT (cont.)
G-304 main.tex:1860 "The linear program yields $\alpha = 5$, so it is a 5-gadget." (for the twelve constraints of main.tex:1852-1858, reducing PC_0).	FALSE measured / – method: tools/probes/appendix_xi_pc_gadget_alpha_lp.py evidence: tools/probes/appendix_xi_pc_gadget_alpha_lp.py
G-308 the two matrix-analysis lemmas opening Appendix XV's spectral toolkit ('main.tex:1936'-'1949'): "For a Normal matrix N, the operator norm is the same as the spectral norm. They are related to the spectral radius (maximum magnitude eigenvalue) by $\ N\ _{\text{op}} = \ N\ _2 = \max_{\ x\ _2 \neq 0} \ Nx\ _2 / \ x\ _2 = (\text{maximum eigenvalue of } N^\dagger N)^{1/2}$." — and Weyl's perturbation inequality main.tex:1940 Lemma: "For a Normal matrix N, the operator norm is the same as the spectral norm": $\ N\ _{\text{op}} = \ N\ _2 = \max_{\ x\ _2 \neq 0} \ Nx\ _2 / \ x\ _2 = (\text{maximum eigenvalue of } N^\dagger N)^{1/2}$.	TRUE lean / L2 lean/Qgadget/Findings/Section_15_OperatorNormEigenvalue.lean#topRQ_isGreatest_eigenvalue theorem topRQ_isGreatest_eigenvalue (A : Matrix n n ℂ) : IsGreatest {μ : ℝ Module.End.HasEigenvalue (Matrix.toEuclideanLin (A^H * A)) (μ : ℂ)} (topRQ A) := by FINDING (G-308): 'topRQ A' is the largest eigenvalue of 'A^H * A' — it IS an eigenvalue of 'toEuclideanLin (A^H * A)', and it DOMINATES every eigenvalue — the j = 1 case of Courant-Fischer, for an arbitrary complex square matrix 'A' (no normality hypothesis). lean/Qgadget/Findings/Section_15_OperatorNormEigenvalue.lean#l2_opNorm_eq_sqrt_topRQ theorem l2_opNorm_eq_sqrt_topRQ (A : Matrix n n ℂ) : $\ A\ = \text{Real.sqrt (topRQ A)}$:= by FINDING (G-308): '$\ A\ = \text{sqrt(topRQ A)}$', the paper's displayed identity, with NO normality hypothesis on 'A'.
G-322 ‘\begin{lemma}’ ('main.tex:2058'-'2070', Appendix XV, "We will use the following result about the inverse of a block matrix"): $> [[A, B], [C, D]]^{-1} = [[(A - BD^{-1}C)^{-1}, -A B (D - C A^{-1}B)^{-1}], [-D^{-1}C (A - BD^{-1}C)^{-1}, (D - C A^{-1}B)^{-1}]]$’. main.tex:2066 Lemma (block inverse): $[[A, B], [C, D]]^{-1} = [[(A - B D^{-1} C)^{-1}, -A B (D - C A^{-1} B)^{-1}], [-D^{-1} C (A - B D^{-1} C)^{-1}, (D - C A^{-1} B)^{-1}]]$.	FALSE lean / L2 lean/Qgadget/Verbatim/Lemma_BlockMatrixInverse.lean#blockMatrix_inv theorem blockMatrix_inv : $\text{blockMatrix}^{-1} = !![1 / 2, -1 / 2; 0, 1]$:= by 'blockMatrix⁻¹ = $[[1/2, -1/2], [0, 1]]$', computed directly. REAL: the inverse of a specific invertible matrix is a computed fact (a singular matrix, or a different one, would give a different or no inverse). lean/Qgadget/Verbatim/Lemma_BlockMatrixInverse.lean#printed_top_right_ne_true theorem printed_top_right_ne_true : $(-2 : \mathbb{Q}) * 1 * (1 - 0 * (1 / 2 : \mathbb{Q}) * 1)^{-1} \neq \text{blockMatrix}^{-1} 0 1$:= by FINDING (G-322): the printed top-right block '-AB(D-CA⁻¹B)⁻¹' ('main.tex:2066') disagrees with the true inverse's top-right entry at 'A=2, B=1, C=0, D=1'. Printed: '$-2 \cdot 1 \cdot (1-0)^{-1} = -2$'.

THE DRAFT SAYS (cont.)

G-328 the self-energy substitution steps ('main.tex:2079'-'2093', Appendix XV, "Finally, we can represent ' $\Sigma_{-}(z)$ ' using the matrix series expansion ... ' $\Sigma_{-}(z) := zI_{--} - G_{-}^{-1}(z) = zI_{--} - (G_{-}^{-1} - V_{-} - V_{-} + (G_{-}^{-1} - V_{-} + V_{-})^{-1} V_{-})^{-1}$ ' ... ' $= H_{-} + V_{-} + V_{-} + (G_{-}^{-1} - V_{-} + V_{-})^{-1} V_{-} + \dots$ ' ... ' $\Sigma_{-}(z) = H_{-} + V_{-} + V_{-} + G_{-} + V_{-} + V_{-} + G_{-} + V_{-} + G_{-} + \dots$ '.")

main.tex:2093

"Evoking the matrix series expansion, we finally get $\Sigma_{-}(z) = H_{-} + V_{-} + V_{-} + G_{-} + V_{-} + V_{-} + G_{-} + V_{-} + G_{-} + \dots$ ".

G-338 the displayed projected block ('main.tex:2143'-'2145', Appendix XV, "The above series requires computing projections for V , which is as follows"): $\langle V_{1-} \rangle = [Y + AZ_{-i}I_{-j} + D(I_{-i}X_{-j} + 1)] \otimes |0\rangle \langle 0|_{-k}$.

main.tex:2144

$V_{1-} = [Y + A Z_{-i} I_{-j} + D (I_{-i} X_{-j} + 1)] (x) |0\rangle \langle 0|_{-k}$, for $V_{-1} = [Y + D(I_{-i} X_{-j} I_{-k} + I_{-i} I_{-j} I_{-k})] - (A/2)(Z_{-i} I_{-j} I_{-k} + Z_{-i} I_{-j} Z_{-k})$.

LEAN'S VERDICT (cont.)

FALSE lean / L2

lean/Qgadget/Findings/Section_15_SelfEnergySubstitution.lean#printed_fourth_term_entry

theorem printed_fourth_term_entry :

(Pm2 * Vwit * Pp2) * Gpp0 * (Pp2 * Vwit * Pp2) * Gpp0 = !![0, 1 / 25; 0, 0] := by

The printed fourth term, ' $V_{-}\{+\}G_{-}\{++\}(z)V_{-}\{++\}G_{-}\{++\}(z)$ ' ('main.tex:2093', missing its trailing ' $V_{-}\{+\}$ '), has a nonzero ' $(-,+)$ ' entry, ' $1/25$ '. REAL: this is a computed fact about the specific matrix product — the entry could have come out ' 0 ' (which is what ' $\Sigma_{-}(z)$ ', an operator on ' L_{-} ' alone, would require of every term in its own [...])

lean/Qgadget/Findings/Section_15_SelfEnergySubstitution.lean#printed_fourth_term_off_diagonal_nonzero

theorem printed_fourth_term_off_diagonal_nonzero :

((Pm2 * Vwit * Pp2) * Gpp0 * (Pp2 * Vwit * Pp2) * Gpp0) 0 1 $\neq$ 0 := by

FINDING (G-328): the printed fourth term's ' $(-,+)$ ' entry is nonzero, so it is NOT an operator on ' L_{-} ' alone — ' $\Sigma_{-}(z)$'s own expansion ('main.tex:2093's left side) must be.

FALSE lean / L2

lean/Qgadget/Findings/Section_15_ZXGadgetProjectedBlocks.lean#printed_V1mm_ne_true_at_A1

theorem printed_V1mm_ne_true_at_A1 :

(k3 Z2 I2 P0 : Matrix (Fin 2 $\times$ Fin 2 $\times$ Fin 2) (Fin 2 $\times$ Fin 2 $\times$ Fin 2) $\mathbb{Q}$)

$\neq$ Pm * V1 1 0 0 0 * Pm := by

FINDING (G-338): the draft's printed ' $V_{1-} = [Y + AZ_{-i}I_{-j} + D(I_{-i}X_{-j} + 1)] \otimes |0\rangle \langle 0|_{-k}$ ' ('main.tex:2144') disagrees with ' $\Pi_{-} V_{-1} \Pi_{-}$ ' at ' $A=1, Y=0, D=0$ '. The printed expression there is ' $Z_{-i} \otimes |0\rangle \langle 0|_{-k}$ '; the true value ('V1mm_true_at_A1') is its negation, and the two are genuinely different matrices (checked at index '((0,0,0),(0,0,0))', where [...])

THE DRAFT SAYS (cont.)

G-339 the "Generating ZX interaction terms from (ZZ, XX) interaction" gadget ('main.tex:2100'-'2189', Appendix XV). Three qubits '(i, j, k)', 'k' the ancilla; ' $H = \delta|1\rangle\langle 1|_k$ ' ('main.tex:2112'), and the perturbation ' $V = V_1 + V_2 + V_3$ ' ('main.tex:2119'-'2132'): ' $V_1 = [Y + D(X_j + 1)] - (A/2)(Z_i + Z_i Z_k)$ ', ' $V_2 = B(X_j X_k + X_k)$ '

main.tex:2148

$V_{2-+} = B(I_i X_j + 1) (x) |0\rangle\langle 1|_k$; $V_{2+-} = B(I_i X_j + 1) (x) |1\rangle\langle 0|_k$; $V_{3++} = V_3$; $V_{++} = [Y + C Z_i I_j + D(I_i X_j + 1)] (x) |1\rangle\langle 1|_k$.

G-340 the "Generating ZX interaction terms from (ZZ, XX) interaction" gadget ('main.tex:2100'-'2189', Appendix XV). Three qubits '(i, j, k)', 'k' the ancilla; ' $H = \delta|1\rangle\langle 1|_k$ ' ('main.tex:2112'), and the perturbation ' $V = V_1 + V_2 + V_3$ ' ('main.tex:2119'-'2132'): ' $V_1 = [Y + D(X_j + 1)] - (A/2)(Z_i + Z_i Z_k)$ ', ' $V_2 = B(X_j X_k + X_k)$ '

main.tex:2179

"The third term $V_{-+} G_{++}(z) V_{+-} = B^2/(1 - \delta) (X_j + 1)^2 (x) |0\rangle\langle 0|_k$."

LEAN'S VERDICT (cont.)

TRUE lean / L2

lean/Qgadget/Support/ZXGadgetOperators.lean#V2_mp_eq

theorem V2_mp_eq (B : $\mathbb{Q}$) : Pm * V2 B * Pp = B • k3 I2 (X2 + I2) K01 := by

' $\Pi_- V_2 \Pi_+ = B(X_j + 1) \otimes |0\rangle\langle 1|_k$ ' ('main.tex:2148').

lean/Qgadget/Support/ZXGadgetOperators.lean#V2_pm_eq

theorem V2_pm_eq (B : $\mathbb{Q}$) : Pp * V2 B * Pm = B • k3 I2 (X2 + I2) K10 := by

' $\Pi_+ V_2 \Pi_- = B(X_j + 1) \otimes |1\rangle\langle 0|_k$ ' ('main.tex:2152').

lean/Qgadget/Support/ZXGadgetOperators.lean#V3_pp_eq

theorem V3_pp_eq (C : $\mathbb{Q}$) : Pp * V3 C * Pp = V3 C := by

' $\Pi_+ V_3 \Pi_+ = V_3$ ' ('main.tex:2156'): ' V_3 ' already lives entirely in ' L_+ '. REAL: ' $\Pi_+ Z_k \Pi_+ = -\Pi_+$ ' (qubit 'k' in ' $|1\rangle$ ', ' Z_2 ' acts as '-1') turns ' $(C/2)(Z_i - Z_i Z_k)$ ' into ' $(C/2)(Z_i - (-Z_i)) = C Z_i$ ', entirely supported on ' P_1 '; getting that sign wrong would instead annihilate ' V_3 ' on ' L_+ '.

lean/Qgadget/Support/ZXGadgetOperators.lean#Vpp_eq

theorem Vpp_eq (A B C D y1 y2 : $\mathbb{Q}$) :
Pp * Vsum A B C D y1 y2 * Pp
= y1 • k3 Z2 Z2 P1 + y2 • k3 X2 X2 P1 + C • k3 Z2 I2 P1
+ D • (k3 I2 X2 P1 + k3 I2 I2 P1) := by

' $\Pi_+ V \Pi_+ = Y_{\{ij\}} + C Z_i + D(X_j + 1)$ ', projected onto ' L_+ ' ('main.tex:2160'): ' V_2 ' contributes nothing (' $\Pi_+ X_k \Pi_+ = 0$ ') and ' V_1 's ' A -term cancels (' $\Pi_+ Z_k \Pi_+ = -\Pi_+$ ' flips ' $-(A/2)(Z_i + Z_i Z_k)$ ' to ' $-(A/2)(Z_i - Z_i) = 0$ '), leaving exactly the printed ' C ', not ' A '.

FALSE lean / L2

lean/Qgadget/Support/ZXGadgetOperators.lean#term3_op_eq

theorem term3_op_eq (B : $\mathbb{Q}$) :

(Pm * V2 B * Pp) * Pp * (Pp * V2 B * Pm) = (B ^ 2) • k3 I2 ((X2 + I2) * (X2 + I2)) P0 := by

The third self-energy term's operator part, 'main.tex:2179': ' $V_{\{-+\}} \Pi_+ V_{\{+-\}} = B^2(X_j + 1)^2 \otimes |0\rangle\langle 0|_k$ ', general in 'B'. Only ' V_2 ' contributes to ' $V_{\{-+\}}$ '/' $V_{\{+-\}}$ ' ('V2_mp_eq'/'V2_pm_eq'), so this needs no hypothesis on 'A, C, D, Y'.

THE DRAFT SAYS (cont.)

G-341 the third and fourth terms of the self-energy expansion (‘main.tex:2177’-‘2185’, Appendix XV, "Now, we plug the values of above projected operator into eqn \ref{eqn: self-energy-expansion}"): > Third term: ‘ $V_{\{-++\}}(z)V_{\{+-\}} = B^2/(1-\delta)(X_{j+1})^2 \otimes |0\rangle\langle 0|_k$ ’. > Fourth term: ‘ $V_{\{-+\}}G_{\{++\}}(z)V_{\{++\}}G_{\{++\}}(z)V_{\{+-\}} > = B^2/(z-\delta)^2(X_{j+1})Z_i(X_{j+1}) \otimes |0\rangle\langle 0|_k$ ’

main.tex:2183

Fourth term, first summand: $V_{-+}G_{++}(z)V_{++}G_{++}(z)V_{+-}$ contains $B^2/(z-\delta)^2(X_j+1)Z_i(X_j+1)(x)|0\rangle\langle 0|_k$.

LEAN’S VERDICT (cont.)

FALSE lean / L2

lean/Qgadget/Findings/Section_15_SelfEnergyThirdFourthTerms.lean#term4_at_C3_eq

```
theorem term4_at_C3_eq :
  (Pm * Vsum 0 2 3 0 0 0 * Pp) * Pp * (Pp * Vsum 0 2 3 0 0 0 * Pp) * Pp *
  (Pp * Vsum 0 2 3 0 0 0 * Pm)
  = (24 : ℚ) • k3 Z2 (X2 + I2) P0 := by
```

The fourth term’s operator part at ‘A=0, B=2, C=3, D=0, y1=y2=0’ (‘C=3, B=2, D=0, Y=0’ in the row’s naming).

lean/Qgadget/Findings/Section_15_SelfEnergyThirdFourthTerms.lean#term4_is_3x_printed_at_C3

```
theorem term4_is_3x_printed_at_C3 :
  (Pm * Vsum 0 2 3 0 0 0 * Pp) * Pp * (Pp * Vsum 0 2 3 0 0 0 * Pp) * Pp *
  (Pp * Vsum 0 2 3 0 0 0 * Pm)
  = (3 : ℚ) • ((2 : ℚ) ^ 2 • k3 Z2 ((X2 + I2) * (X2 + I2)) P0) := by
```

FINDING (G-341): at ‘C=3, B=2, D=0, Y=0’ the fourth term’s operator part is 3 times the draft’s printed first summand (‘printed_first_summand_at_B2’, which is the fourth term’s own shape at the implicit ‘C=1’).

THE DRAFT SAYS (cont.)

G-342 the "Generating ZX interaction terms from (ZZ, XX) interaction" gadget ('main.tex:2100'-'2189', Appendix XV). Three qubits '(i, j, k)', 'k' the ancilla; ' $H = \delta|1\rangle\langle 1|_k$ ' ('main.tex:2112'), and the perturbation ' $V = V_1 + V_2 + V_3$ ' ('main.tex:2119'-'2132'): ' $V_1 = [Y + D(X_j + 1)] - (A/2)(Z_i + Z_i Z_k)$ ', ' $V_2 = B(X_j X_k + X_k)$ '

main.tex:2183

Fourth term, third summand: " $4DB^2/(z - \delta)^2 (X_j + 1)^3 (x) |0\rangle\langle 0|_k$ ".

G-356 '\begin{definition}' ('main.tex:2249'-'2252', Appendix XVI, "The reduction is based on a property of the Fourier transform for an Abelian group, which is summarised in the following theorem"): > Consider an Abelian group 'G' and a function 'f: $G \mapsto \mathbb{C}$ ' that is > constant on the cosets of the subgroup 'H' of 'G'. Then the Fourier > transform 'f^' is constant on the dual subgroup ' $H^\perp$ '.

main.tex:2251

"summarised in the following theorem" (a definition environment): for an Abelian group G and f: $G \rightarrow \mathbb{C}$ constant on the cosets of a subgroup H, "the Fourier transform f^ is constant on the dual subgroup $H^\perp$."

LEAN'S VERDICT (cont.)

FALSE lean / L2

lean/Qgadget/Support/ZXGadgetOperators.lean#cube_X2_add_I2

theorem cube_X2_add_I2 :

$(X2 + I2) * (X2 + I2) * (X2 + I2) = (4 : \mathbb{Q}) \bullet (X2 + I2) := \text{by}$

' $(X + 1)^3 = 4(X + 1)$ ' for the Pauli 'X', the fact 'main.tex:2183's third summand needs and, printed as ' $4D(X_j+1)^3$ ', gets wrong by a further factor of '4' (G-342: ' $4 \cdot 4 = 16$ ' against the true '4').

lean/Qgadget/Findings/Section_15_SelfEnergyThirdFourthTerms.lean#term4_at_D1_eq

theorem term4_at_D1_eq :

$(Pm * Vsum\ 0\ 2\ 0\ 1\ 0\ 0 * Pp) * Pp * (Pp * Vsum\ 0\ 2\ 0\ 1\ 0\ 0 * Pp) * Pp * (Pp * Vsum\ 0\ 2\ 0\ 1\ 0\ 0 * Pm)$
 $= (16 : \mathbb{Q}) \bullet k3\ I2\ (X2 + I2)\ P0 := \text{by}$

The fourth term's operator part at 'A=0, B=2, C=0, D=1, y1=y2=0' ('D=1, B=2, C=0, Y=0' in the row's naming).

lean/Qgadget/Findings/Section_15_SelfEnergyThirdFourthTerms.lean#printed_third_summand_at_B2D1

theorem printed_third_summand_at_B2D1 :

$((4 : \mathbb{Q}) * 1 * (2 : \mathbb{Q})^2) \bullet k3\ I2\ ((X2 + I2) * (X2 + I2) * (X2 + I2))\ P0$
 $= (64 : \mathbb{Q}) \bullet k3\ I2\ (X2 + I2)\ P0 := \text{by}$

The draft's printed third summand, ' $4DB^2(X_j+1)^3 \otimes |0\rangle\langle 0|_k$ ' ('main.tex:2183'), at 'D=1, B=2'. Uses 'cube_X2_add_I2': ' $(X_j+1)^3 = 4(X_j+1)$ ', so the printed ' $4 \cdot 1 \cdot 2^2 \cdot (X_j+1)^3 = 4 \cdot 4 \cdot 4 \bullet (X_j+1) = 64 \bullet (X_j+1)$ '.

FALSE lean / L2

lean/Qgadget/Verbatim/Definition_FourierCosetSupport.lean#fhat4_not_const_on_Hperp

theorem fhat4_not_const_on_Hperp : fhat4 0 $\neq$ fhat4 2 := by

FINDING (G-356), first half: 'f^' is NOT constant on ' $H^\perp = \{0, 2\}$ ' — it takes the two different values '2' and '-2' there, refuting the draft's printed conclusion at this instance.

lean/Qgadget/Verbatim/Definition_FourierCosetSupport.lean#fhat4_vanishes_off_Hperp

theorem fhat4_vanishes_off_Hperp : fhat4 1 = 0 $\wedge$ fhat4 3 = 0 :=

FINDING (G-356), second half: 'f^' vanishes OFF ' $H^\perp = \{1, 3\}$ ', matching the corrected statement ('f^' is supported on ' $H^\perp$ ').

THE DRAFT SAYS (cont.)	LEAN'S VERDICT (cont.)
G-22 Definition 12.1 (polynomial-time many-one reduction), and the sentence immediately after it asserting that the relation is reflexive and transitive main.tex:2263 Definition [poly-time many-one reductions, AB09]: "Let $A, B \subseteq \Sigma^*$ which encodes computational problems, we say that problem A poly-time many-one reduces to B, denoted $A \leq^p_m B$, if there exists a polynomial time computable function $f: \Sigma^* \rightarrow \Sigma^*$ such that $x \in A$ iff $f(x) \in B$." The next sentence, main.tex:2266: "This relation is reflexive and transitive."	TRUE lean / – lean/Qgadget/Borrowed/Verbatim/Definition_12_1_ManyOneTransitivity.lean#polyTimeReducible_trans_of_comp <pre>theorem polyTimeReducible_trans_of_comp (hcomp : ∀ f g : List Bool → List Bool, Complexity.PolyTimeComputable f → Complexity.PolyTimeComputable g → Complexity.PolyTimeComputable (fun x => g (f x))) {S T U : Complexity.Lang} (h1 : Complexity.PolyTimeReducible S T) (h2 : Complexity.PolyTimeReducible T U) : Complexity.PolyTimeReducible S U := by</pre> Transitivity of '$\leq_p$', from the composition of poly-time functions. The hypothesis is Mathlib's 'proof_wanted TM2ComputableInPolyTime.comp', stated here as a named assumption rather than an 'axiom' — CLAUDE.md section 5.
G-23 main.tex:2268 Definition [Poly-time Turing Reduction]: "we say that problem A poly-time Turing reduces to B, denoted $A \leq^p_T B$, if there exists a polynomial-time oracle Turing machine M that has oracle access to the set B and decides membership of strings in A."	TRUE measured / – method: tools/check_standard_definitions.py evidence: docs/standard-definitions.md
G-24 Theorem 12.3, whose conclusion is about a symbol its statement never introduces; and Definition 12.7, hardness and completeness main.tex:2272 Theorem: "For any language $A \subseteq \Sigma^*$, if $A \leq^p_T B$, then $A \in P^B$. In particular, if $B = \text{SAT}$, then $L' \in P^{\{NP\}}$."	FALSE lean / – lean/Qgadget/Borrowed/Verbatim/Theorem_12_3_and_Definition_12_7.lean#theorem_12_3_readings_differ <pre>theorem theorem_12_3_readings_differ (A : Lang) : ∃ C : Set Lang, theorem_12_3_intended C A ∧ ¬ theorem_12_3_literal C := by</pre> The two readings are not the same statement. REAL: it exhibits a class and a language for which the intended reading HOLDS and the literal one FAILS, so no amount of charity makes the printed sentence say what it means.
G-25 Proposition [the chain '$\leq^p_m \implies \leq^p_{tt} \implies \leq^p_T$'], together with the three reduction notions it relates main.tex:2277 Definition [Poly-time Truth Table Reduction]: "we say that problem A poly-time Truth Table reduces to B, denoted $A \leq^p_{tt} B$, if there exists a polynomial-time non-adaptive oracle Turing machine M which has oracle access to the set B and decides membership of strings in A."	TRUE lean / – lean/Qgadget/Borrowed/Verbatim/Proposition_12_5_ReductionChain.lean#adaptiveStep_not_truthTable <pre>theorem adaptiveStep_not_truthTable {A B : Lang Bool} (r : TruthTable A B) : ttStep r true ≠ adaptiveStep := by</pre> No truth-table reduction produces the adaptive step function. The converse direction of P-117's step 3, and the one that makes the word load-bearing rather than merely descriptive: 'adaptiveStep' asks a query that depends on the answer received, which 'ttStep_query_depends_only_on_length' forbids.

THE DRAFT SAYS (cont.)	LEAN'S VERDICT (cont.)
G-26 Proposition [the chain '$\leq^P_m \implies \leq^P_{tt} \implies \leq^P_T$'], together with the three reduction notions it relates main.tex:2281 Proposition: "Let A, B subset Sigma[*] encode computational problems. Then A $\leq^P_m$ B implies A $\leq^P_{tt}$ B implies A $\leq^P_T$ B."	TRUE lean / L2-PARTIAL lean/Qgadget/Borrowed/Verbatim/Proposition_12_5_ReductionChain.lean#ttStep_run <pre>theorem ttStep_run {A B : Lang X} (r : TruthTable A B) (x : X) : ∀ (k n : ℕ), n ≤ (r.queries x).length → (r.queries x).length - n < k → runOracle (ttStep r) B k x ((r.queries x).take n).map B) = r.combine x ((r.queries x).map B) := by</pre>
G-27 Definition [poly-time isomorphic sets] main.tex:2286 Definition [poly-time isomorphic sets]: "Two sets A and B are polynomial-time isomorphic if there exist functions f: A → B such that: x in A iff f(x) in B; f is a bijection from A to B, hence f⁻¹ exists; both f and f⁻¹ are polynomial-time computable."	FALSE lean / L2 lean/Qgadget/Borrowed/Verbatim/Definition_12_6_PolyTimeIsomorphic.lean#isoConditionAsTyped_always <pre>theorem isoConditionAsTyped_always (A B : Set X) : isoConditionAsTyped A B :=</pre>
G-28 main.tex:2297 Definition [Randomized reductions]: "Let A = {A_{yes}, A_{no}} and B = {B_{yes}, B_{no}} be promise problems. We say that A $\leq^P_r$ B if there exists a polynomial-time probabilistic Turing machine M such that: Completeness: if x in A_{yes} then Pr_z[M(x,z) in B_{yes}] ≥ 2/3. Soundness: if x in A_{no} then Pr_z[M(x,z) in B_{no}] ≥ 2/3."	TRUE measured / – method: tools/check_standard_definitions.py evidence: docs/standard-definitions.md
G-29 Definition [BP-Operator] and Definition [BQ-Operator] main.tex:2305 Definition [BP-Operator]: "Let C be a class of (promise) problems. Then BP.C consists of all (promise) problems A = (A_{yes}, A_{no}) that can be reduced to a complete (promise) problem B = (B_{yes}, B_{no}) in C by a polynomial-time randomised reduction with success probability ≥ 2/3, i.e., BP.C = {A : A $\geq^P_q$ B}."	FALSE lean / L2-PARTIAL lean/Qgadget/Borrowed/Verbatim/Definition_12_8_PromiseOperators.lean#asWritten_depends_on_free_B <pre>theorem asWritten_depends_on_free_B : ∃ (R : Lang Bool → Lang Bool → Prop) (B₁ B₂ : Lang Bool), operatorAsWritten R B₁ ≠ operatorAsWritten R B₂ := by</pre> The display is not a definition: its value depends on the free 'B'. lean/Qgadget/Borrowed/Verbatim/Definition_12_8_PromiseOperators.lean#display_differs_from_prose <pre>theorem display_differs_from_prose : ∃ (R : Lang Bool → Lang Bool → Prop) (Complete : Lang Bool → Prop) (B : Lang Bool), operatorIntended R Complete ≠ operatorAsWritten R B := by</pre> The display and the prose define different classes, even for a fixed 'B': the prose quantifies existentially over complete problems and runs the reduction from 'A', the display runs it from 'B'.

THE DRAFT SAYS (cont.)	LEAN'S VERDICT (cont.)
G-21 main.tex:2312 Definition [Quantum Reductions], whose two bullets read "Completeness: if x in A_{yes} implies $\Pr[z \text{ in } B_{\text{yes}}] \geq 2/3$" and "Completeness: if x in A_{no} implies $\Pr[z \text{ in } B_{\text{no}}] \geq 2/3$".	FALSE lean / L2-PARTIAL lean/Qgadget/Borrowed/Verbatim/ReadingsDiffer.lean#p53_readings_differ <pre>theorem p53_readings_differ : quantumRed_byLabel (fun b => b) (fun _ => true) ∧ ¬ quantumRed_byContent (fun b => b) (fun _ => true) := by</pre> The two readings are different relations. Dropping the soundness clause admits a reduction of the identity language to the constantly-true one, which the content reading refuses.
G-20 Definition [BP-Operator] and Definition [BQ-Operator] main.tex:2321 Definition [BQ-Operator]: "Let C be a class of (promise) problems. Then $BQ.C$ consists of all (promise) problems $A=(A_{\text{yes}}, A_{\text{no}})$ that can be reduced to a complete (promise) problem B in C by a polynomial-time randomised reduction with success probability $\geq 2/3$, i.e., $BP.C = \{A : A \geq_q B\}$."	FALSE lean / L2-PARTIAL lean/Qgadget/Borrowed/Verbatim/Definition_12_8_PromiseOperators.lean#randomized_operator_ne_quantum_operator <pre>theorem randomized_operator_ne_quantum_operator : ∃ (Rrand Rquant : Lang Bool → Lang Bool → Prop) (Complete : Lang Bool → Prop), operatorIntended Rrand Complete ≠ operatorIntended Rquant Complete := by</pre> Copying the 'BP' body cannot define 'BQ'. The 'BQ'-Operator at 'main.tex:2321' reproduces 'main.tex:2306' word for word, including "by a polynomial-time randomised reduction".
G-30 Theorem 12.3, whose conclusion is about a symbol its statement never introduces; and Definition 12.7, hardness and completeness main.tex:2328 Definition: "A language L subseq Σ^* is hard for a class C under poly time many-one reduction if and only if every L' in C, $L' \leq_m L$. If, in addition, L is in the complexity class C, then L is defined to be complete for the class C."	TRUE lean / L2-PARTIAL lean/Qgadget/Borrowed/Verbatim/Theorem_12_3_and_Definition_12_7.lean#complete_strictly_stronger <pre>theorem complete_strictly_stronger (L : Lang) : Hard (∅ : Set Lang) L ∧ ¬ Complete (∅ : Set Lang) L := by</pre> Completeness is strictly stronger than hardness. REAL, and it is the obligation that could have failed: had 'Hard' already forced membership, the draft's "if, in addition" would be redundant and the two notions would coincide.
G-389 main.tex:2585 Table "The 16 Boolean Functions over $\{P, Q\}$ in CNF" (main.tex:2583-2607): sixteen rows, each a named function and its truth "Vector" (Contradiction 0000, AND 1000, Inhibition (P and not Q) 0100, Identity (P) 1100, ..., Tautology 1111).	TRUE measured / – method: tools/probes/appendix_xix_boolean_tables.py evidence: tools/probes/appendix_xix_boolean_tables.py
G-390 main.tex:2585 The same table's "CNF Representation" column, for example AND = (not P or Q) and (P or not Q) and (P or Q), XOR = (not P or not Q) and (P or Q), Tautology = 1 (or P or not P).	TRUE measured / – method: tools/probes/appendix_xix_boolean_tables.py evidence: tools/probes/appendix_xix_boolean_tables.py

THE DRAFT SAYS (cont.)	LEAN’S VERDICT (cont.)
G-391 main.tex:2612 Table "Correspondence between Boolean Functions and Pauli Hamiltonians ($x \rightarrow (I-Z)/2$)" (main.tex:2609-2641): AND $\rightarrow (1/4)(I - Z_1 - Z_2 + Z_1 Z_2)$, XOR $\rightarrow (1/2)(I - Z_1 Z_2)$, OR $\rightarrow (1/4)(3I - Z_1 - Z_2 - Z_1 Z_2)$, and 13 more. Note: "$Z_1$ acts on variable P, and Z_2 acts on variable Q. Eigenvalue 1 corresponds to Boolean True."	TRUE measured / – method: tools/probes/appendix_xix_boolean_tables.py evidence: tools/probes/appendix_xix_boolean_tables.py
G-395 main.tex:2654 "\item \exists $y \in \{0,1\}^N$ such that $P(y) \geq a$, or" and "\item \forall $y \in \{0,1\}^N$ we have $P(y) < a$." (main.tex:2654-2655).	FALSE measured / – method: tools/check_paper_integrity.py evidence: tools/paper.toml

4 Commented-out claims

Rows citing a line inside a LaTeX comment block. This material is not currently in the paper.

THE DRAFT SAYS	LEAN'S VERDICT
G-400 The commented "Quantum Phase/Eigenvalue Estimation" circuit figure (<code>\label{fig:qpe}</code>), whose registers are labelled with unconditional outputs: ancilla measured to λ_0, witness register left in $\psi_0\rangle$. <code>main.tex:1017</code> Commented Qcircuit "Quantum Phase/Eigenvalue Estimation" (main.tex:1011-1025): ancilla $0^t\rangle \rightarrow H^{\{x\}t} \rightarrow \text{control} \rightarrow \text{QFT}^\dagger(Z/2^tZ) \rightarrow \text{measure}$, output labelled $\lambda_0\rangle$; system input $u\rangle$, output labelled $\psi_0\rangle$.	FALSE lean / L2 <code>lean/Qgadget/Findings/Section_10_QPEOutputLabels.lean#qpe_output_prob_ne_one</code> <code>theorem qpe_output_prob_ne_one :</code> $\ (\text{inner } (k := \mathbb{C}) \text{ ket0 ketPlus : } \mathbb{C}) \ ^ 2 \neq 1 := \text{by}$ FINDING (G-400), first half: the printed outcome $\langle 0, 0\rangle\rangle$ has probability $1/2$, not 1. Reuses G-164's <code>overlap_ket0_ketPlus_sq</code> (<code>'ketPlus'</code> IS this row's <code>' u'<</code>) rather than re-deriving it; the new content is the negative comparison against the figure's implicit certainty. <code>lean/Qgadget/Findings/Section_8_GroundVsGuidingState.lean#overlap_ket1_ketPlus_sq</code> <code>theorem overlap_ket1_ketPlus_sq : \ (inner (k := C) ket1 ketPlus : C) \ ^ 2 = 1 / 2 := by</code> $\alpha_2 ^2 = \langle \psi_0^\perp \psi \rangle ^2 = 1/2$. <code>lean/Qgadget/Findings/Section_10_QPEOutputLabels.lean#qpe_probs_sum_to_one</code> <code>theorem qpe_probs_sum_to_one :</code> $\ (\text{inner } (k := \mathbb{C}) \text{ ket0 ketPlus : } \mathbb{C}) \ ^ 2 + \ (\text{inner } (k := \mathbb{C}) \text{ ket1 ketPlus : } \mathbb{C}) \ ^ 2 = 1 :=$ <code>by</code> The two probabilities exhaust the outcome space, so nothing is lost by the two computations above: the figure's two labelled outcomes are the only ones and together account for probability 1, split evenly rather than concentrated on $\psi_0\rangle$ as printed.

THE DRAFT SAYS (cont.)	LEAN'S VERDICT (cont.)
G-424 the "Generating ZX interaction terms from (ZZ, XX) interaction" gadget ('main.tex:2100'-'2189', Appendix XV). Three qubits '(i, j, k)', 'k' the ancilla; '$H = \delta 1\rangle\langle 1 _k$' ('main.tex:2112'), and the perturbation '$V = V_{-1} + V_{-2} + V_{-3}$' ('main.tex:2119'-'2132'): '$V_{-1} = [Y + D(X_j + 1)] - (A/2)(Z_i + Z_i Z_k)$', '$V_{-2} = B(X_j X_k + X_k)$' main.tex:2164 Commented sentence: "If we examine the terms V_{-1}-, V_{-2}+, V_{-2}-, and V_{-3}++ above carefully, we can see that each of them is a local Hamiltonian that includes a ZX interaction."	FALSE lean / L2-PARTIAL lean/Qgadget/Support/ZXGadgetOperators.lean#V3_pp_eq theorem V3_pp_eq (C : $\mathbb{Q}$) : Pp * V3 C * Pp = V3 C := by '$\Pi_{-} + V_{-3} \Pi_{-} = V_{-3}$' ('main.tex:2156'): '$V_{-3}$' already lives entirely in '$L_{-}+$'. REAL: '$\Pi_{-} + Z_k \Pi_{-} = -\Pi_{-}+$' (qubit 'k' in '$1\rangle$', '$Z_{-2}$' acts as '-1') turns '$(C/2)(Z_i - Z_i Z_k)$' into '$(C/2)(Z_i - (-Z_i)) = C Z_i$', entirely supported on 'P_{-1}'; getting that sign wrong would instead annihilate 'V_{-3}' on '$L_{-}+$'. lean/Qgadget/Findings/Section_15_CommentBlockZXClaims.lean#V3_eq_Z2_I2_P1 theorem V3_eq_Z2_I2_P1 (C : $\mathbb{Q}$) : V3 C = C • k3 Z2 I2 P1 := by '$V_{-3} = C \cdot Z_i \otimes I_j \otimes 1\rangle\langle 1 _k$': since '$1\rangle\langle 1 = (1-Z)/2$', this closed form is built entirely from '{I, Z}' factors on every qubit, so 'V_{-3}' carries no 'X'- or 'Y'-Pauli component (a reading of this closed form, not a further Lean obligation — see the module header). REAL: the closed form itself is a computed fact ('$I_k - Z_k$' cancelling to [...])
G-425 the comment block 'main.tex:2163'-'2172' (Appendix XV, 'region: comment' — this text is inside '$\backslash\begin{comment}...\end{comment}$' and does not typeset): > If we examine the terms 'V_{-1}+', 'V_{-2}+', 'V_{-2}+', and 'V_{-3}++' above > carefully, we can see that each of them is a local Hamiltonian that > includes a 'ZX' interaction. Although none among 'V_{-1}', 'V_{-2}' and 'V_{-3}' main.tex:2164 Same sentence: "each of them [V_{-1}-, V_{-2}+, V_{-2}-, V_{-3}++] is a local Hamiltonian".	FALSE lean / L2 lean/Qgadget/Findings/Section_15_CommentBlockZXClaims.lean#V2mp_not_symm_at_B1 theorem V2mp_not_symm_at_B1 : $(Pm * V2 \ 1 * Pp) \neq (Pm * V2 \ 1 * Pp)^T :=$ by FINDING (G-425): at '$B=1$', 'V_{-2}+' differs from its transpose (every entry here is rational, so "conjugate transpose" is the plain transpose). 'V_{-2}+' = k3 I2 (X2+I2) K01' ('V2_mp_eq' at '$B=1$') and '$K01 \neq K01^T = K10$'.

THE DRAFT SAYS (cont.)

G-427 the displayed projected block (‘main.tex:2143’-‘2145’, Appendix XV, "The above series requires computing projections for V , which is as follows"): $> V_{\{1-\}} = [Y + AZ_{iI_j} + D(I_{iX_j+1})] \otimes |0\rangle \langle 0|_k$.

main.tex:2167

Commented display: $V_{1-} = [Y + A Z_{iI_j} + D(I_{iX_j+1})] (x) |0\rangle \langle 0|_k$.

G-428 the comment block ‘main.tex:2163’-‘2172’ (Appendix XV, ‘region: comment’ — this text is inside ‘\begin{comment}... \end{comment}’ and does not typeset): $>$ If we examine the terms ‘ $V_{\{1-\}}$ ’, ‘ $V_{\{2+\}}$ ’, ‘ $V_{\{2+-\}}$ ’, and ‘ $V_{\{3++\}}$ ’ above $>$ carefully, we can see that each of them is a local Hamiltonian that $>$ includes a ‘ZX’ interaction. Although none among ‘ V_{1-} ’, ‘ V_{2-} ’ and ‘ V_{3-} ’

main.tex:2170

Commented display: $V_{1-} = Y(x) (1 + Z_k)/2 + A Z_{iI_j}(x) (1 + Z_k)/2 + D I_{iX_j}(x) (1 + Z_k)/2 + 1(x) (1 + Z_k)/2$, expanding main.tex:2167.

LEAN’S VERDICT (cont.)

FALSE lean / L2

lean/Qgadget/Findings/Section_15_ZXGadgetProjectedBlocks.lean#V1mm_true_at_A1

theorem V1mm_true_at_A1 :

$Pm * V1 \ 1 \ 0 \ 0 \ 0 * Pm = (-1 : \mathbb{Q}) \bullet k3 \ Z2 \ I2 \ P0 := by$

The true projected block at ‘ $A=1, Y=0, D=0$ ’ is ‘ $-Z_{iI_j} \otimes |0\rangle \langle 0|_k$ ’. Specialises ‘ $V1_{mm_eq}$ ’ (a fact about ALL coefficients ‘ $A, D, y1, y2$ ’) to the row’s instance; REAL because ‘ $V1_{mm_eq}$ ’ itself is (its proof is a genuine computation of how the ancilla projector acts on ‘ Z_k ’, which could have come out with the opposite sign).

lean/Qgadget/Findings/Section_15_ZXGadgetProjectedBlocks.lean#printed_V1mm_ne_true_at_A1

theorem printed_V1mm_ne_true_at_A1 :

$(k3 \ Z2 \ I2 \ P0 : Matrix (Fin \ 2 \times Fin \ 2 \times Fin \ 2) (Fin \ 2 \times Fin \ 2 \times Fin \ 2) \mathbb{Q})$

$\neq Pm * V1 \ 1 \ 0 \ 0 \ 0 * Pm := by$

FINDING (G-338): the draft’s printed ‘ $V_{\{1-\}} = [Y + AZ_{iI_j} + D(I_{iX_j+1})] \otimes |0\rangle \langle 0|_k$ ’ (‘main.tex:2144’) disagrees with ‘ $\Pi_- V_{1-} \Pi_-$ ’ at ‘ $A=1, Y=0, D=0$ ’. The printed expression there is ‘ $Z_{iI_j} \otimes |0\rangle \langle 0|_k$ ’; the true value (‘ $V1mm_true_at_A1$ ’) is its negation, and the two are genuinely different matrices (checked at index ‘ $((0,0,0),(0,0,0))$ ’, where [...])

FALSE lean / L2

lean/Qgadget/Findings/Section_15_CommentBlockZXClaims.lean#expansion_true

theorem expansion_true (A D y1 y2 : $\mathbb{Q}$) :

$(Yop \ y1 \ y2 + A \bullet k3 \ Z2 \ I2 \ I2 + D \bullet (k3 \ I2 \ X2 \ I2 + k3 \ I2 \ I2 \ I2)) * Pm$
 $= (1 / 2 : \mathbb{Q}) \bullet (Yop \ y1 \ y2 * (k3 \ I2 \ I2 \ I2 + k3 \ I2 \ I2 \ Z2))$
 $+ (A / 2 : \mathbb{Q}) \bullet (k3 \ Z2 \ I2 \ I2 * (k3 \ I2 \ I2 \ I2 + k3 \ I2 \ I2 \ Z2))$
 $+ (D / 2 : \mathbb{Q}) \bullet (k3 \ I2 \ X2 \ I2 * (k3 \ I2 \ I2 \ I2 + k3 \ I2 \ I2 \ Z2))$
 $+ (D / 2 : \mathbb{Q}) \bullet (k3 \ I2 \ I2 \ I2 + k3 \ I2 \ I2 \ Z2) := by$

‘main.tex:2170’ correctly expanded: ‘ $[Y + AZ_{iI_j} + D(I_{iX_j+1})] \otimes |0\rangle \langle 0|_k = Y \otimes (1 + Z_k)/2 + AZ_{iI_j} \otimes (1 + Z_k)/2 + DI_{iX_j} \otimes (1 + Z_k)/2 + D \cdot 1 \otimes (1 + Z_k)/2$ ’ (the printed form’s last term lacks its ‘ D ’).

lean/Qgadget/Findings/Section_15_CommentBlockZXClaims.lean#printed_expansion_ne_true_at_D2

theorem printed_expansion_ne_true_at_D2 :

$(1 / 2 : \mathbb{Q}) \bullet (k3 \ I2 \ I2 \ I2 + k3 \ I2 \ I2 \ Z2)$

$\neq (2 / 2 : \mathbb{Q}) \bullet (k3 \ I2 \ I2 \ I2 + k3 \ I2 \ I2 \ Z2) := by$

FINDING (G-428): at ‘ $D=2, A=0, Y=0$ ’ the draft’s printed expansion (‘main.tex:2170’, without ‘ D ’ on the last term) disagrees with ‘main.tex:2167’s display it is expanding. The printed expansion’s last term is ‘ $1 \otimes (1 + Z_k)/2$ ’ (coefficient ‘ 1 ’, not ‘ D ’); at ‘ $D=2$ ’ the true last term is ‘ $D \otimes (1 + Z_k)/2 = 2 \otimes (1 + Z_k)/2$ ’, twice the printed one, so [...]

THE DRAFT SAYS (cont.)

G-444 the commented "Quantum Mechanics" primer's recipe for Schrödinger's equation ('main.tex:2361'-'2446', 'region: comment' — inside '\begin{comment}...\end{comment}', never typeset), step 5 ('main.tex:2380'): > Use Schrödinger's equation to compute the state by solving the associated > ODE with a boundary condition. > ' $|\psi(t)\rangle = \sum_j \alpha_j(0) \exp(-i E_j/\hbar) |E_j\rangle$ '

main.tex:2380

Recipe step 5: "Use Schrödinger's equation to compute the state by solving the associated ODE with a boundary condition." |psi(t)> = sum_j alpha_j(0) exp(-i E_j/hbar) |E_j>.

G-447 the commented "Quantum Mechanics" primer's §"Meaning of Uncertainty Principle" ('main.tex:2388'-'2445', inside the same '\begin{comment}...\end{comment}' block as G-444, 'main.tex:2361'-'2446').

main.tex:2416

Commented display $(\Delta A)^2 = \sum_a (a^2 + \langle A \rangle^2 - 2a\langle A \rangle)^2 \cdot P(a)$.

LEAN'S VERDICT (cont.)

FALSE lean / L2

lean/Qgadget/Findings/Section_QM_SchrodingerRecipe.lean#printed_ne_true_at_zero

theorem printed_ne_true_at_zero (t : ℝ) : printedPsi t ≠ truePsi 0 := by

FINDING (G-444): the printed vector, evaluated at the witness, differs at 't = 0' from ' $|\psi(0)\rangle$ ' itself. 'printedPsi's second component is ' $(1/\sqrt{2})\cdot\exp(-i)$ '; the true ' $|\psi(0)\rangle$'s second component is ' $(1/\sqrt{2})\cdot 1$ '.

lean/Qgadget/Findings/Section_QM_SchrodingerRecipe.lean#true_evolves

theorem true_evolves : truePsi 0 ≠ truePsi Real.pi := by

The TRUE solution genuinely evolves (the contrast the row's own note draws): at the witness, ' $|\psi(0)\rangle \neq |\psi(\pi)\rangle$ '.

FALSE lean / L2

lean/Qgadget/Findings/Section_QM_UncertaintyPrinciple.lean#printed_sum_eq_16

theorem printed_sum_eq_16 :
((2 : ℂ) ^ 2 + (braketDiag psiPlus0 psiPlus1 2 (-2)) ^ 2
- 2 * 2 * (braketDiag psiPlus0 psiPlus1 2 (-2))) ^ 2
* ((starRingEnd ℂ) psiPlus0 * psiPlus0)
+ ((-2 : ℂ) ^ 2 + (braketDiag psiPlus0 psiPlus1 2 (-2)) ^ 2
- 2 * (-2) * (braketDiag psiPlus0 psiPlus1 2 (-2))) ^ 2
* ((starRingEnd ℂ) psiPlus1 * psiPlus1) = 16 := by

FINDING (G-447): 'main.tex:2416's own display, at the witness, is '16', not the variance '4' — the printed formula is left symbolic in ' $\langle A \rangle$ ' and only THEN evaluated via 'mean_eq_zero', so this states the paper's own formula rather than a pre-simplified rewrite of it.

lean/Qgadget/Findings/Section_QM_UncertaintyPrinciple.lean#variance_eq_4

theorem variance_eq_4 :
((2 : ℂ) - braketDiag psiPlus0 psiPlus1 2 (-2)) ^ 2 * ((starRingEnd ℂ) psiPlus0 * psiPlus0)
+ ((-2 : ℂ) - braketDiag psiPlus0 psiPlus1 2 (-2)) ^ 2
* ((starRingEnd ℂ) psiPlus1 * psiPlus1) = 4 := by

The true variance, at the same witness, is '4', the value 'main.tex:2416' claims to compute and does not.

THE DRAFT SAYS (cont.)

G-450 the commented "Quantum Mechanics" primer's §"Meaning of Uncertainty Principle" ('main.tex:2388'-'2445', inside the same '\begin{comment}...\end{comment}' block as G-444, 'main.tex:2361'-'2446').

main.tex:2437

"or, $2 \sqrt{|\langle \psi | A^2 | \psi \rangle|} \cdot \sqrt{|\langle \psi | B^2 | \psi \rangle|} = 2 \sqrt{\langle A^2 \rangle \langle B^2 \rangle} = |\langle \psi | [A, B] | \psi \rangle|$ ", presented as a restatement of main.tex:2433.

G-463 the commented "Appendix D: Primer for Abstract Algebra" ('main.tex:2451'-'2570', inside '\begin{comment}...\end{comment}', never typeset).

main.tex:2504

Commented theorem: "Z₂ (+) Z₃ is isomorphic to Z₆."

G-470 the commented "Appendix D: Primer for Abstract Algebra" ('main.tex:2451'-'2570', inside '\begin{comment}...\end{comment}', never typeset).

main.tex:2545

"An integral domain has three essential properties- (i) commutativity, (ii) existence of unity and (iii) cancellation property."

LEAN'S VERDICT (cont.)

FALSE lean / L2

lean/Qgadget/Findings/Section_QM_UncertaintyPrinciple.lean#anticommutator_expect_zero

```
theorem anticommutator_expect_zero :
  bracketFull psiPlusI0 psiPlusI1
    ((PauliX * PauliZ + PauliZ * PauliX) 0 0) ((PauliX * PauliZ + PauliZ * PauliX) 0 1)
    ((PauliX * PauliZ + PauliZ * PauliX) 1 0) ((PauliX * PauliZ + PauliZ * PauliX) 1 1) = 0 :=
```

by

' $\langle \psi | \{X, Z\} | \psi \rangle = 0$ ' at ANY state, since ' $\{X, Z\}$ ' is the zero operator ('XZ_add_ZX_eq_zero') — this is 'main.tex:2433's right side.

lean/Qgadget/Findings/Section_QM_UncertaintyPrinciple.lean#commutator_expect_abs_eq_two

```
theorem commutator_expect_abs_eq_two :
  ||bracketFull psiPlusI0 psiPlusI1
    ((PauliX * PauliZ - PauliZ * PauliX) 0 0) ((PauliX * PauliZ - PauliZ * PauliX) 0 1)
    ((PauliX * PauliZ - PauliZ * PauliX) 1 0) ((PauliX * PauliZ - PauliZ * PauliX) 1 1)|| = 2 :=
```

by

The modulus of 'main.tex:2437's right side at the witness is '2'.

TRUE lean / L2

lean/Qgadget/Findings/Section_AlgebraPrimer_CommentBlock.lean#oneOne_hasOrder_six

```
theorem oneOne_hasOrder_six : addOrderOf ((1, 1) : ZMod 2 × ZMod 3) = 6 := by
```

The row's own witness: '(1,1)' has additive order '6' in 'ZMod 2 × ZMod 3'. Real content, by the group-theoretic route (not a finite-case 'decide'): the order of a pair is the 'lcm' of the two components' orders ('Prod.addOrderOf_mk'), and '1' has additive order 'n' in 'ZMod n' ('ZMod.addOrderOf_one'), so this reduces to 'lcm 2 3 = 6' — [...]

FALSE lean / L2

lean/Qgadget/Findings/Section_AlgebraPrimer_CommentBlock.lean#quaternions_noncommutative

```
theorem quaternions_noncommutative : qi * qj ≠ qj * qi := by
```

The real quaternions, a 'DivisionRing' (hence satisfying 'main.tex:2542's cancellation property), are not commutative. Real content: 'i*j = k' and 'j*i = -k' ('qi_mul_qj', 'qj_mul_qi'), and reading off the 'imK' component of 'k' versus '-k' gives '1 = -1' in ' $\mathbb{R}$ ' — false, since ' $\mathbb{R}$ ' does not have characteristic '2' — so 'i*j ≠ j*i'.

lean/Qgadget/Findings/Section_AlgebraPrimer_CommentBlock.lean#even_two_has_no_unity

```
theorem even_two_has_no_unity : ∀ e : ℤ, Even e → e * 2 ≠ 2 := by
```

No even integer is a multiplicative unity for '2'. Real content: 'e * 2 = 2' forces 'e = 1' (an integer-arithmetic fact that could have come out differently, e.g.

5 Figure claims

Rows citing one of the draft’s figures, ordered by image path.

THE DRAFT SAYS	LEAN’S VERDICT
G-487 HHL schematic figure: (a) phase estimation ‘$H^{\otimes n}$’, controlled ‘U’, ‘$FT^\dagger$’; (b) rotation ‘$R(\lambda^{-1})$’ on the ancilla; (c) "Uncompute": ‘FT’, controlled ‘U’, ‘$H^{\otimes n}$’; outputs ‘$1\rangle$’, clock ‘$0\rangle^{\otimes n}$’, input ‘$x\rangle$’. pics/HHL-Algorithm-Schematic-1.ppm HHL schematic: (a) phase estimation $H^{\otimes n}$, controlled U, $FT^\dagger$; (b) $R(\lambda^{-1})$ rotation on ancilla S; (c) "Uncompute": FT, controlled U, $H^{\otimes n}$; outputs $1\rangle$, clock $0\rangle^{\otimes n}$, input $x\rangle$.	FALSE lean / L2 lean/Qgadget/Findings/Section_Figures_HHLSchematicUncompute.lean#boxC_printed_eq_scaled_e1 <pre>theorem boxC_printed_eq_scaled_e1 : had4.mulVec (ctrlUPlus.mulVec (qft4.mulVec clockAfterBoxA)) = (4 : ℂ) • (![0, 1, 0, 0] : Fin 4 → ℂ) := by</pre> Box (c) AS PRINTED (‘FT’, controlled ‘U’, ‘$H^{\otimes 2}$’) leaves the clock at ‘$4 \bullet 01\rangle$’, not ‘$00\rangle$’. REAL: this is a genuine ‘4×4’ complex matrix computation over the actual gates — replacing ‘$ctrlUPlus$’ by any other diagonal changes the entries the sum below reduces to, and the vanishing of the ‘0’, ‘2’, ‘3’ components is not forced by the [...] lean/Qgadget/Findings/Section_Figures_HHLSchematicUncompute.lean#boxC_corrected_eq_scaled_e0 <pre>theorem boxC_corrected_eq_scaled_e0 : had4.mulVec (ctrlUMinus.mulVec (qft4.mulVec clockAfterBoxA)) = (4 : ℂ) • (![1, 0, 0, 0] : Fin 4 → ℂ) := by</pre> The same boxes with ‘$ctrlUMinus$’ (controlled ‘$U^\dagger$’) in place of ‘$ctrlUPlus$’ return the clock to ‘$4 \bullet 00\rangle$’. REAL, by the same obligation as ‘$boxC_printed_eq_scaled_e1$’, at the ONE entry that differs (‘$ctrlUMinus$’ vs. lean/Qgadget/Findings/Section_Figures_HHLSchematicUncompute.lean#boxC_printed_ne_boxC_corrected <pre>theorem boxC_printed_ne_boxC_corrected : had4.mulVec (ctrlUPlus.mulVec (qft4.mulVec clockAfterBoxA)) ≠ had4.mulVec (ctrlUMinus.mulVec (qft4.mulVec clockAfterBoxA)) := by</pre> FINDING (G-487): box (c) as printed and its corrected form (‘controlled $U^\dagger$’) do not agree at this instance — the printed uncompute leaves the clock at ‘$01\rangle$’, the corrected one at ‘$00\rangle$’, and ‘$4 \bullet [0,1,0,0] \neq 4 \bullet [1,0,0,0]$’ since their ‘$0$’-th entries ‘$0$’ and ‘$4$’ differ in ‘$\mathbb{C}$’.

THE DRAFT SAYS (cont.)

G-510 HHL circuit: phase estimation ($H^{\otimes m}$), one controlled- U , $QFT^\dagger$, a controlled rotation R on the top register (one control dot on the middle/clock register), the inverse block (QFT , controlled- $U^\dagger$, $H^{\otimes m}$). Printed outputs: top $|1\rangle$, middle $|0\rangle^{\otimes m}$, bottom $|x\rangle$ (the solution of $A|x\rangle = |b\rangle$). The image fixes neither m nor the spectrum of

pics/HHL-circuit.png

HHL circuit: the bottom register's output is labelled $|x\rangle$, the solution of the linear system, with no precision or resolution hypothesis on the image.

LEAN'S VERDICT (cont.)

FALSE lean / L2

lean/Qgadget/Findings/Section_Figures_HHLCircuitOffGrid.lean#runCircuit_off_drawn

theorem runCircuit_off_drawn :

runCircuit U_OFF 1 1 DRAWN = ![![(4 : $\mathbb{C}$), 2], ![0, -2 * Complex.I]] := by

The drawn circuit's OFF-grid output, at $|b\rangle = (1,1)$ unnormalised and the drawn $\beta = (0,1)$: row = clock, column = bottom. REAL: a genuine seven-gate composition over the actual instance — every entry below is the accumulated product of the drawn gates, and the vanishing $(1,0)$ entry and the pure-imaginary $(1,1)$ entry are not typed [...]

lean/Qgadget/Findings/Section_Figures_HHLCircuitOffGrid.lean#runCircuit_on_drawn

theorem runCircuit_on_drawn :

runCircuit U_ON 1 1 DRAWN = ![![(4 : $\mathbb{C}$), 4], ![0, 0]] := by

The drawn circuit's ON-grid output ($A = \pi \cdot I$, both eigenphases $1/2$), same $|b\rangle$ and same drawn β . REAL, by the same obligation: replacing U_OFF by U_ON is a genuine change to the second controlled-diagonal gate, and the resulting cancellation to a REAL vector (no i survives) is a fact of this instance, not of the shape of the [...]

lean/Qgadget/Findings/Section_Figures_HHLCircuitOffGrid.lean#bottom_ratio_off_ne_printed_ratio

theorem bottom_ratio_off_ne_printed_ratio :

(runCircuit U_OFF 1 1 DRAWN) 0 1 * X_PRINTED 0

$\neq$ (runCircuit U_OFF 1 1 DRAWN) 0 0 * X_PRINTED 1 := by

FINDING (G-510), off the grid: the bottom register conditioned on the middle register reading $|0\rangle$ is NOT proportional to the printed $|x\rangle$. Stated cross-multiplied to avoid dividing in $\mathbb{C}$: the bottom register is $(4, 2)$, of ratio $1/2$; $|x\rangle$ is printed $(1, 2)$, of ratio 2 — were the two proportional, $2 \cdot 1$ would equal $4 \cdot 2$ [...]

THE DRAFT SAYS (cont.)	LEAN'S VERDICT (cont.)
G-511 HHL circuit: phase estimation ($H^{\otimes m}$), one controlled-U, $QFT^\dagger$, a controlled rotation R on the top register (one control dot on the middle/clock register), the inverse block (QFT, controlled-$U^\dagger$, $H^{\otimes m}$). Printed outputs: top $1\rangle$, middle $0\rangle^{\otimes m}$, bottom $x\rangle$ (the solution of $A x\rangle = b\rangle$). The image fixes neither m nor the spectrum of pics/HHL-circuit.png HHL circuit: the middle register's output is labelled $0\rangle^{\otimes m}$, the clock register returned to its initial state by the inverse phase estimation, with no precision or resolution hypothesis on the image.	FALSE lean / L2 lean/Qgadget/Findings/Section_Figures_HHLCircuitOffGrid.lean#clockPurity_off_eq theorem clockPurity_off_eq : purity2 (clockDensity (runCircuit U_OFF 1 1 DRAWN)) = 7 / 9 := by FINDING (G-511), off the grid: the middle register's reduced state has purity $7/9$, so it is not $0\rangle^{\otimes m}$ — not even a pure state. REAL: the four entries of clockDensity are read off $\text{runCircuit_off_drawn}$'s actual output, and $7/9 \neq 1$ is the refutation; a pure $0\rangle\langle 0$ output would give purity 1 exactly, as clockPurity_on_eq [...] lean/Qgadget/Findings/Section_Figures_HHLCircuitOffGrid.lean#clockPurity_on_eq theorem clockPurity_on_eq : purity2 (clockDensity (runCircuit U_ON 1 1 DRAWN)) = 1 := by On the grid, the middle register's reduced state has purity 1 ($0\rangle\langle 0$).
G-483 pics/boolean-cubes.png Labelled Boolean cubes for $n = 1, 2, 3, 4$, each vertex a bit string and each edge joining two vertices.	FALSE measured / – method: tools/probes/figures_cube_sphere_hhl_witnesses.py evidence: tools/probes/figures_cube_sphere_hhl_witnesses.py
G-486 pics/bqp-npCartesianProduct.png A sphere $x^2 + y^2 + z^2 = 1$ drawn $=$ a segment $z^2 \leq 1$ x a circle $x^2 + y^2 = 1$.	FALSE measured / – method: tools/probes/figures_cube_sphere_hhl_witnesses.py evidence: tools/probes/figures_cube_sphere_hhl_witnesses.py
G-479 §8.2 "Difference between Ground State and Guiding State" — the Case I/II fidelity-versus-energy analysis. The proof of Lemma [exp-fidelity] (main.tex:612–main.tex:651) writes the decomposition $\psi\rangle = \alpha_0 \psi_0\rangle + \alpha_1 \psi_0^\perp\rangle$ (main.tex:628), the resulting expansion of $\langle\psi H \psi\rangle$ (main.tex:632), the amplitude bound on $\psi_0^\perp\rangle$, the fidelity pics/qpe-generic01.png The live figure included at main.tex:582 and main.tex:1552: ancilla $0\rangle \rightarrow H(x)t \rightarrow \text{control} \rightarrow QFT^\dagger(Z/2^tZ) \rightarrow \text{measure}$, output labelled $\lambda\rangle$; register "Witness: $u\rangle$", output labelled $\psi_0\rangle$; register "Instance: $x\rangle$".	FALSE lean / L2 lean/Qgadget/Findings/Section_8_GroundVsGuidingState.lean#overlap_ket0_ketPlus_sq theorem overlap_ket0_ketPlus_sq : $\ (inner (k := C) ket0 ketPlus : C)\ ^2 = 1 / 2 := by$ $\alpha_1 ^2 = \langle\psi_0 \psi\rangle ^2 = 1/2$. lean/Qgadget/Findings/Section_10_QPEOutputLabels.lean#qpe_output_prob_ne_one theorem qpe_output_prob_ne_one : $\ (inner (k := C) ket0 ketPlus : C)\ ^2 \neq 1 := by$ FINDING (G-400), first half: the printed outcome $(0, 0\rangle)$ has probability $1/2$, not 1. Reuses G-164's $\text{overlap_ket0_ketPlus_sq}$ (ketPlus IS this row's $u\rangle$) rather than re-deriving it; the new content is the negative comparison against the figure's implicit certainty.

THE DRAFT SAYS (cont.)

G-505 QPE step 4, "Apply the inverse Fourier transform: $e^{2\pi i j \varphi_{\eta}} |j\rangle |\eta\rangle |x\rangle \rightarrow |\varphi_{\eta}\rangle |\eta\rangle |x\rangle$."

[pics/qpe_tex_qz.png](#)

Standard QPE: t ancillas $|0\rangle^{\otimes t}$, H on each, controls $U^{2^{t-1}}$ down to U^{2^0} on $|\psi\rangle$, QFT[†], measurement; the ancilla output is labelled $|2^t \theta\rangle$.

LEAN'S VERDICT (cont.)

FALSE lean / L2

[lean/Qgadget/Findings/Section_10_QPEInverseFourierStep.lean#hadamard_mulVec_stepThree](#)

```
theorem hadamard_mulVec_stepThree :  
  hadamard2.mulVec stepThreeState  
    = ![(1 + Complex.I) / 2, (1 - Complex.I) / 2] := by
```

Applying the inverse Fourier transform to step 3's register gives $((1+i)|0\rangle + (1-i)|1\rangle)/2$ — not a basis state.

[lean/Qgadget/Findings/Section_10_QPEInverseFourierStep.lean#hadamard_off_grid_not_basis_state](#)

```
theorem hadamard_off_grid_not_basis_state :  
  ∀ j : Fin 2, ||(hadamard2.mulVec stepThreeState) j|| = (Real.sqrt 2)-1 := by
```

'main.tex:1563's arrow is FALSE at ' $m = 1$ ', ' $\varphi = 1/4$ ' (G-381): the output of applying the inverse Fourier transform has BOTH entries of modulus $1/\sqrt{2}$, not '1' at some single index and '0' elsewhere — so it is not the basis state $|\varphi_{\eta}\rangle$ the display prints, and measuring returns either outcome with probability $1/2$.

6 Support files of the frozen source

One row citing supporting material outside main.tex – a preamble or assistant-file line, not the draft’s own prose.

THE DRAFT SAYS	LEAN’S VERDICT
G-46 manish_assist/commands.tex:67 \newcommand{\log}{\mathsf{log}}, so that \log typesets as a sans-serif "log".	FALSE measured / – method: tools/check_paper_integrity.py evidence: tools/paper.toml

7 Census

Every row of the ledger appears here once, whether or not it carries a Lean witness. Grouped by verdict; ordered by id within each group.

TRUE

G-14 main.tex:926 lean holds, given the cited BQP-completeness
G-16 main.tex:955 lean holds; soundness needs no fidelity
G-22 main.tex:2263 lean the standard definition, correctly stated
G-23 main.tex:2268 measured correctly stated
G-25 main.tex:2277 lean correctly stated; the one word that matters is present
G-26 main.tex:2281 lean both implications hold
G-28 main.tex:2297 measured correctly stated, and correctly labelled
G-30 main.tex:2328 lean the standard definitions, both correct
G-39 main.tex:1508 lean the bitwise phase collapse is correct, and sharp in $k \leq n$
G-40 main.tex:1510 lean the k -th qubit really does depend only on the low $n-k$ bits
G-44 main.tex:1145 prose accurate, where the sentence it annotates overstates
G-55 main.tex:99 cited agrees with the draft's own gadget definition, which is finite
G-75 main.tex:143 cited TSSW's Definition 3.10 is exactly the LP the draft describes
G-80 main.tex:166 prose every gadget under the draft's definition is finite
G-84 main.tex:183 prose the standard definition, with exactly q bits read rather than at most q
G-87 main.tex:212 prose the worked example does reduce ZX terms to ZZ and XX terms
G-88 main.tex:216 prose agrees with the frozen KKR extract
G-92 main.tex:241 cited TSSW's LP always minimizes α , and Lemma 4.9's gadgets are proven optimal
G-96 main.tex:264 prose true for an orthogonal projector, which the word projector means
G-97 main.tex:268 prose true under either definition; the source states it as a lemma
G-98 main.tex:272 prose a tensor product of Hermitian unitaries is a Hermitian unitary
G-100 main.tex:284 prose the linear system problem, correctly stated
G-113 main.tex:338 lean holds, but from positivity as well as the norm bound
G-116 main.tex:375 prose all eight rows are correct
G-117 main.tex:384 prose the projector onto the one violating assignment, correctly factored
G-142 main.tex:506 prose true once W_j exists, that is for $0 \leq H_j \leq I$
G-147 main.tex:532 lean holds for a unit η once $r = m$ and $0 \leq H_j \leq I$
G-153 main.tex:586 prose true: guided ground-energy estimation is in BQP by phase estimation
G-157 main.tex:599 prose true: the product of the two outcomes has mean $\langle \psi | Z_1 X_2 | \psi \rangle$
G-177 main.tex:703 prose true: every case has inverse-polynomial fidelity, and Case III bypasses QPE
G-180 main.tex:720 prose true: the zero-energy states span the satisfying basis states
G-181 main.tex:722 lean true once the identity factors are placed on the right qubits
G-184 main.tex:806 prose true of the draft's gadgets; the generalised definition is unwritten (G-48)
G-206 main.tex:966 lean true but empty: decide GLH in BQP, then output a fixed ZZ instance
G-221 main.tex:1186 lean each qubit's term has least eigenvalue $-\sqrt{\alpha^2 + \beta^2 + \gamma^2}$

G-228 main.tex:1216 lean brute force over at most $2m$ edge vertices, closed form for the rest

G-250 main.tex:1283 prose for constant fidelity, Cade et al.'s hardness and containment cover it

G-251 main.tex:1283 prose hardness for overlaps below $1/\sqrt{2}$ is proved, containment noted, by GL22

G-258 main.tex:1322 prose with correct l_2 sampling, one sample and one row of H_u give λ_0

G-265 main.tex:1364 prose without the guide it is k -LH, QMA-complete for constant $k \geq 2$ by KKR

G-267 main.tex:1369 prose true for the worst case: Cade et al. prove hardness up to fidelity $1 - 1/\text{poly}$

G-270 main.tex:1588 prose a well-formed definition, used consistently in the appendix

G-272 main.tex:1592 prose the clause is a function from $\{0,1\}^3$ to $\{0,1\}$

G-273 main.tex:1611 prose the eight rows are the truth table of x_1 or x_2 or x_3

G-274 main.tex:1617 prose a well-formed definition for a finite family

G-275 main.tex:1621 prose two well-defined 3-ary constraint functions

G-277 main.tex:1632 prose a well-formed definition of a constraint, under the title of another term

G-278 main.tex:1636 prose a well-formed weighted optimisation problem

G-280 main.tex:1655 prose the three conditions define a gadget without contradiction

G-282 main.tex:1677 lean the best y satisfies 7 clauses when C_k holds and 6 when it fails

G-283 main.tex:1690 prose a well-formed closure condition, with constants allowed

G-284 main.tex:1694 prose four well-formed conditions; the equalities are demanded only on S

G-285 main.tex:1725 prose with S the satisfying set the four conditions are the gadget's three

G-287 main.tex:1731 prose a well-formed definition; its S is written as a strict subset

G-288 main.tex:1746 lean each listed y attains 7 on satisfying rows and 6 on 000

G-295 main.tex:1792 cited TSSW state this near-verbatim, crediting Karloff and Zwick, not themselves

G-296 main.tex:1792 prose identifying two arguments of PC_1 gives the projection onto the third

G-302 main.tex:1848 prose 2^7 bounds over all (a, b) plus one equality per satisfying assignment

G-308 main.tex:1940 lean the displayed chain holds for every square matrix; normality is unused

G-310 main.tex:1945 prose true as Weyl's inequality, in finite dimension with multiplicities counted

G-311 main.tex:1955 prose KKR's Lemma 1, transcribed faithfully; λ is left undefined

G-314 main.tex:1971 prose KKR's resolvent, correct for Hermitian H with an orthonormal eigenbasis

G-316 main.tex:1984 prose both spaces are spanned by eigenvectors of H , so both are invariant

G-317 main.tex:2027 prose KKR's definition of self-energy, transcribed exactly

G-320 main.tex:2047 prose the frozen source says so of its own application

G-321 main.tex:2049 prose the inverse of $G^{-1} - V$, written in the $(+, -)$ blocks

G-324 main.tex:2073 prose the bottom-right block, which the printed lemma has right

G-327 main.tex:2089 prose the factorisation $(I - V_{++} G_+) G_+^{-1}$, inverted

G-333 main.tex:2112 prose the projector onto $|1\rangle$ is $(1 - Z)/2$

G-334 main.tex:2119 prose the two eigenspaces of $\delta |1\rangle\langle 1|_k$, under a garbled set-builder

G-337 main.tex:2140 prose the self-energy series to third order, with the fourth-order remainder

G-339 main.tex:2148 lean the other four projections of V are correct

G-345 main.tex:2186 prose the three perturbation parts hold ZZ, XX and one-qubit terms, no Z-X pair

G-351 main.tex:2232 prose the code generated by G and its dual, standard definitions

G-353 main.tex:2239 prose a well-formed average-case problem: a nonzero low-weight kernel vector

G-358 main.tex:1432 prose the source proposes QNC as NC's quantum analogue, through embedding

G-360 main.tex:1438 lean the amplitude map is unitary for every $m \geq 1$

G-361 main.tex:1454 prose the cited paper by Cleve and Watrous proves the theorem
 G-363 main.tex:1463 prose the draft uses the QFT at :341 and :2249-2251; essential is a judgement
 G-364 main.tex:1473 lean the characters of Z_2^n are exactly the maps $x \rightarrow (-1)^{x \cdot y}$
 G-366 main.tex:1491 lean $\exp(2\pi i/m)$ has order exactly m for every $m \geq 1$
 G-367 main.tex:1493 lean the column of F_{2^n} at x , under $\omega = \omega_{2^n}$
 G-368 main.tex:1497 lean y is rewritten through its binary digits, most significant first
 G-369 main.tex:1500 lean a power of a sum of exponents is the product of the powers
 G-370 main.tex:1508 lean each factor's sum over y_k in $\{0, 1\}$ is $|0\rangle + \omega^{x \cdot 2^k} |1\rangle$
 G-372 main.tex:1520 prose true of every finite Abelian group, the subsection's subject
 G-375 main.tex:1526 prose creating the Fourier state beside x is easy for every modulus N
 G-377 main.tex:1532 lean the Fourier transform diagonalises the shift, with eigenvalues ω^{-x}
 G-378 main.tex:1526 lean each Fourier basis vector is an eigenvector of the unitary shift
 G-380 main.tex:1562 lean controlled- U^j on the eigenstate kicks back the phase $e^{2\pi i j \phi}$
 G-386 main.tex:2260 prose reductions and completeness recur throughout the live draft
 G-389 main.tex:2585 measured all 16 truth vectors match their names; the order is $(P,Q) = 11,10,01,00$
 G-390 main.tex:2585 measured each CNF is the conjunction of the maxterms of its false positions
 G-391 main.tex:2612 measured all 16 operators have eigenvalue $f(P,Q)$ on $|PQ\rangle$, as the note says
 G-403 main.tex:1022 prose the drawn inputs are the guided state and an instance register
 G-413 main.tex:1172 prose five tensor factors, two of them non-identity
 G-426 main.tex:2164 prose the three perturbation parts hold no Pauli string with both Z and X
 G-445 main.tex:2396 prose true with i indexing an orthonormal eigenbasis of A
 G-446 main.tex:2409 prose the shifted operator and the variance as a sum over eigenvalues are right
 G-448 main.tex:2420 prose the variance is the expectation of the squared shifted operator
 G-449 main.tex:2433 prose true, with the anticommutator on the right: $|\langle A,B \rangle| \leq 2 \|A\| \|B\|$
 G-451 main.tex:2437 prose follows from Cauchy-Schwarz on $A\psi$ and $B\psi$
 G-452 main.tex:2442 prose true; with zero means $\langle A^2 \rangle = (\Delta A)^2$, and the hypothesis is not needed
 G-453 main.tex:2456 prose the standard group axioms, with closure built into the operation
 G-458 main.tex:2479 prose the group of all permutations of n symbols has $n!$ elements
 G-459 main.tex:2479 prose true up to isomorphism, which is Cayley's theorem
 G-461 main.tex:2489 prose the standard external direct product; the subscript i is idle
 G-462 main.tex:2500 prose the six listed pairs are exactly $Z_2 \times Z_3$
 G-463 main.tex:2504 lean $(1,1)$ has order 6, so the group is cyclic of order 6
 G-465 main.tex:2511 prose each row is $k(1,1) = (k \bmod 2, k \bmod 3)$; the map is an isomorphism
 G-468 main.tex:2537 prose the standard meanings of unity and unit
 G-472 main.tex:2553 prose every nonzero class mod p is invertible, by Bezout
 G-473 main.tex:2558 prose the additive group has exponent p and order p^n , so it is $(Z_p)^n$
 G-474 main.tex:2562 prose the multiplicative group of a finite field is cyclic
 G-476 main.tex:2567 prose the standard two-sided ideal, printed as two-side
 G-488 pics/HHL-circuit.png prose the inverse phase estimation is the inverse: QFT, controlled $U^\dagger$, H
 G-496 pics/PCE-encoding.png prose 3 pairs times 3 Paulis give 9 distinct correlators for 9 vertices
 G-509 main.tex:1508 lean under the draft's own ω the collapse holds at every k , not only $k \leq n$

TRUE PROOF INVALID

G-13 main.tex:898 lean the statement holds on the normalised reading; the proof does not
G-226 main.tex:1211 prose membership is shown; hardness is sent to an appendix that does not exist
G-232 main.tex:1222 prose true by sparsification and padding; the printed proof has neither
G-239 main.tex:1254 lean true, and in P by exact diagonalisation; the guiding-state proof fails

GAP

G-15 main.tex:926 lean the overlap lemma gives an existential, not a reduction
G-17 main.tex:955 lean the hardness turns on a reading the draft never chooses
G-36 main.tex:1567 lean the running time omits the parameter the problem is sized by
G-38 main.tex:1468 lean complete is undefined and the display need not be unitary
G-48 main.tex:79 prose the formal definition of a gadget is marked (under editing)
G-49 main.tex:79 prose the construction stops at steps the draft marks incorrect and unfinished
G-71 main.tex:115 prose the section's two paragraphs of content are marked (To be updated)
G-72 main.tex:117 prose a stated requirement on a definition that is itself unwritten
G-74 main.tex:131 prose all 15 data cells are empty and the caption is marked (To be updated)
G-86 main.tex:204 prose the reduction the section is named for is never given, in either copy
G-103 main.tex:296 prose the reduction that would make positive semi-definiteness free is absent
G-109 main.tex:308 prose the steps are listed and no gate count is derived from them
G-136 main.tex:458 prose deferred, and the section it defers to never mentions a witness matrix
G-149 main.tex:538 prose the verifier's gap is $(b - a)/m$, and the remark omits the division
G-152 main.tex:577 prose the step from precision to a spectral gap is omitted, not supplied
G-210 main.tex:985 prose a GLH instance is any k -local Hamiltonian; the reduction to ZX form is not given
G-212 main.tex:999 prose the X_i terms of H_{ZX} are never replaced, so no ZZ Hamiltonian results
G-214 main.tex:1001 prose the remark asserts a step whose precision argument it omits
G-230 main.tex:1218 prose the speed-up is asserted with no algorithm and no running time
G-276 main.tex:1630 prose the reduction is deferred, and both sections named for it are under editing
G-329 main.tex:2095 prose the proof idea for the cited perturbation theorem is deferred
G-344 main.tex:2189 prose the unwanted terms are never cancelled; the parameters are deferred
G-346 main.tex:2224 prose the subsection is empty and marked so
G-347 main.tex:2227 prose the subsection is empty and marked so
G-349 main.tex:2230 prose a quantum reduction transfers quantum hardness only; classical half unargued
G-362 main.tex:1454 prose the log-depth construction is deferred in due course and never given
G-373 main.tex:1519 prose the plan's second step, from cyclic groups to a general group, is omitted
G-379 main.tex:1534 prose the erasure of x is asserted; the phase-estimation step is omitted
G-385 main.tex:1572 prose the lemma's proof is deferred in due course, and a qed closes nothing
G-411 main.tex:1047 prose the details are deferred, and the draft marks the place (to be updated)
G-429 main.tex:2194 prose the listed differences are deferred: one heading, then (to be updated)
G-430 main.tex:2195 prose asserted as a heading, with no body and the marker (to be updated)
G-436 main.tex:2336 prose the appendix is marked (to be updated); no hardness result appears

OPEN

G-93 main.tex:243 prose the draft's thesis question, never applied to a QCMA-complete family
G-236 main.tex:1232 prose an open problem stated as known; Grover's bound concerns black-box queries
G-454 main.tex:2465 prose a question the draft poses and answers only with a link
G-456 main.tex:2471 prose a question the draft poses and answers only with a link
G-460 main.tex:2485 prose a question, answered in the draft by the direct product that follows

ILL POSED

G-1 main.tex:881 lean normalisation present at three lines, absent at two
G-2 main.tex:885 lean fidelity squared in the body, unsquared in the appendix
G-3 main.tex:937 lean the guiding state is promised over two inequivalent classes
G-8 main.tex:1075 lean the symmetric gap again, now in the canonical problem
G-9 main.tex:1111 lean the second definition disagrees with the first
G-18 main.tex:919 lean the promise parameter is never bound
G-19 main.tex:948 lean there is no subset state in GalH to replace
G-33 main.tex:1554 lean one symbol, two meanings, one caption
G-34 main.tex:1433 lean the index is written both up and down in one definition
G-35 main.tex:1451 lean epsilon is unconstrained, and one reading forces an empty circuit
G-37 main.tex:1465 lean the display maps one register to another and is not square
G-41 main.tex:1504 lean the bit order reverses across an asserted equality
G-42 main.tex:1539 lean the phase is fixed only modulo one, so the task has no answer
G-47 main.tex:70 prose the class the title names has no definition in the draft
G-51 main.tex:81 prose the draft calls the same design problem an LP and not an LP
G-69 main.tex:109 prose the graph is never defined and no measure is read off it
G-73 main.tex:128 prose the gloss drops the efficiency the draft's own isomorphism requires
G-83 main.tex:180 prose the two arguments are swapped between announcement and definition
G-91 main.tex:241 prose the LP's variables are weights, not the assignments whose space changes
G-99 main.tex:276 prose set equality written where membership is meant
G-104 main.tex:300 prose the precision epsilon is used in the output and never introduced
G-106 main.tex:305 prose kappa is applied as a function to the quotient by epsilon
G-110 main.tex:330 prose the system register is written $|\psi\rangle$, with no index j
G-112 main.tex:330 prose a basis state labelled by a number that need not be an integer
G-115 main.tex:351 prose a vector scaled by kappa is not a state, and the step is renormalisation
G-121 main.tex:400 prose the assignment space S contains no basis state, the solutions named
G-122 main.tex:400 prose the state space is a continuum; only the finite set S is doubly exponential
G-123 main.tex:409 prose the displayed witness has norm $2^{\{n/2\}}$ and is not a state
G-124 main.tex:415 prose completeness is a property of problems, not of one Hamiltonian

G-127 main.tex:419 prose a theorem environment that defines a problem and asserts nothing

G-128 main.tex:426 prose the formula after i.e. has no inequality

G-133 main.tex:441 prose each term is defined through the other, and neither by given versus promised

G-134 main.tex:444 prose the ground state is not unique, so the promise is not a property of H

G-137 main.tex:474 prose the procedure so named estimates no phase

G-138 main.tex:485 prose an upper-bound symbol where the gap needs a lower bound

G-139 main.tex:486 prose the promise gap is used as the energy threshold

G-143 main.tex:513 prose the answer register of the stated space is absent from the state

G-144 main.tex:516 prose r is never defined, and the state is normalised only if $r = m$

G-145 main.tex:518 prose a witness need not be an eigenvector, and the proof never uses one

G-146 main.tex:527 prose a unitary W_j exists only if every eigenvalue of H_j lies in $[0, 1]$

G-148 main.tex:538 prose d ranges over Z , where $d \leq 0$ gives no inverse-polynomial gap

G-150 main.tex:541 prose summed over i with H_j inside, so the right side has j free

G-151 main.tex:547 prose η has no expansion in W_j 's eigenbasis; the display uses H_j 's

G-156 main.tex:599 prose picking one term gives $\langle \psi | H_1 | \psi \rangle$, not $\langle \psi | H | \psi \rangle$

G-159 main.tex:613 prose the parenthesis neither restricts H nor holds of every LH instance

G-160 main.tex:618 prose the conclusion names no threshold, no error and no range of n

G-161 main.tex:626 prose a ket is called a subspace

G-162 main.tex:630 prose coefficients α_0, α_1 become α_1, α_2 in the next line

G-169 main.tex:653 prose a growth rate is announced and a pointwise inequality is proved

G-172 main.tex:678 prose $b - a = 1/p_3$ is written where the hypothesis is $b - a \geq 1/p_3$

G-173 main.tex:684 prose a lower bound on the overlap is read as an upper bound

G-176 main.tex:695 prose contradicts the lemma it summarises, and most cases is undefined

G-178 main.tex:717 prose the identity factor is on qubit 3, which the projector already uses

G-179 main.tex:717 prose the identity is on qubits 3 and 4, which the projector already uses

G-183 main.tex:736 prose the weight list stops at w_2 ; the parity-check gadget has twelve constraints

G-185 main.tex:814 prose "the correct set" presumes one parameter set, and gadgets are not unique

G-186 main.tex:817 prose no P-Q gadget is in the table, whose caption calls HDQI a C-Q gadget

G-188 main.tex:817 prose the pilot state and its parameters occur nowhere else in the draft

G-192 main.tex:832 prose module I outputs parameters, but P, BPP, BQP and NP are decision classes

G-195 main.tex:853 prose the quantum LP is never written, so its inequality signs are not determined

G-205 main.tex:874 prose a Hamiltonian is not complete for a class; the guide is a promise, not a theorem

G-207 main.tex:968 prose ZZ-Ham_yes and ZZ-Ham_no are named here only; two ZZ problems disagree

G-211 main.tex:995 prose h_i and J_{ij} are reused from H_{ZX} , where they are arbitrary reals

G-219 main.tex:1161 prose the thresholds a and b are neither part of the input nor parameters

G-220 main.tex:1180 prose the thresholds a and b are neither part of the input nor parameters

G-224 main.tex:1197 prose the clause defining H_v lacks a noun and a negation before nor

G-225 main.tex:1205 prose the thresholds a and b are neither part of the input nor parameters

G-231 main.tex:1220 prose the lead-in requires $\epsilon > 1$, the theorem it introduces $\epsilon > 0$

G-233 main.tex:1225 prose ETH gives some $\alpha > 1$; the parenthesis does not say some or every

G-237 main.tex:1245 prose a and b are unbound thresholds and also the letters of two coefficient sets

G-242 main.tex:1262 prose a mixed state is not a guiding vector; the proof then uses a pure state

G-247 main.tex:1281 prose a big-O bound used as a lower bound is met by $\gamma \geq 0$

G-254 main.tex:1287 prose δ is the promise gap $b - a$ elsewhere, here an estimate's error

G-255 main.tex:1294 prose with $\text{poly}(n)$ non-zero entries in all, λ_0 is computable outright

G-259 main.tex:1336 prose the words ask for γ above $1/\text{poly}(n)$; the interval admits 2^{-n}

G-260 main.tex:1347 prose γ is a complex number, set equal to the overlap, and never promised

G-261 main.tex:1347 prose $|b - a| \geq \delta$ orders nothing, so $a = 1$, $b = 0$ makes both cases hold

G-262 main.tex:1347 prose the standing assumption fixes norm at most 1; $\text{GLH}(\gamma, \delta)$ allows $\text{poly}(n)$

G-271 main.tex:1590 prose k variables are named, and f is then applied to n of them

G-298 main.tex:1814 prose b is bound by no quantifier; the witness $b(a)$ from M is meant

G-301 main.tex:1844 prose the auxiliary variable count fixes the LP; it cannot be one of its variables

G-307 main.tex:1929 prose one symbol, L_- , names both the low- and the high-energy eigenspace

G-309 main.tex:1938 prose the relation to the spectral radius is announced but not displayed

G-312 main.tex:1952 prose the lemma's J only bounds H_2 's gap from below

G-313 main.tex:1960 prose the perturbed Hamiltonian is $H^- = H + V$; V is the perturbation

G-315 main.tex:1981 prose with both inequalities strict, the sum misses H 's eigenvectors at λ_*

G-318 main.tex:2041 prose λ_- is used in the gap hypothesis and never defined

G-323 main.tex:2058 prose no invertibility hypothesis; $[[0,1],[1,0]]$ is invertible with $A = D = 0$

G-332 main.tex:2105 prose the second sum runs over j and its summand $\Delta_i X_i$ does not use j

G-335 main.tex:2121 prose the effective interaction targeted two lines earlier is ZX , not ZZ

G-336 main.tex:2137 prose the H named as target is the ancilla penalty; the target is H_{eff}

G-354 main.tex:2237 prose the field F_q exists only for prime powers q ; any positive q is allowed

G-355 main.tex:2244 prose p is declared a positive integer and then used as a noise distribution

G-359 main.tex:1438 prose computed exactly is not what the theorem that follows shows

G-365 main.tex:1489 prose the operator uses a bare ω ; only the subscripted ω_m is defined

G-374 main.tex:1524 prose $x^\sim$ is defined as a vector and then used as a basis label inside a ket

G-376 main.tex:1526 prose inverse QFT over Z_N makes the recipe circular; another is not named

G-382 main.tex:1563 prose a real phase is used as a basis label of an m -qubit register

G-383 main.tex:1565 prose m is the register width in the procedure and the term count in its analysis

G-388 main.tex:2275 prose by position it names the theorem; by content the definition above it

G-394 main.tex:2650 prose the input threshold is s ; the question compares $P(y)$ with an unbound a

G-396 main.tex:622 prose $|\phi\rangle$ is bound nowhere in §8.3; the display states nothing about any state

G-397 main.tex:624 prose no relation between p_1 and p_2 is stated, so every pair qualifies

G-398 main.tex:699 prose an upper-bound symbol where the gap needs a lower bound

G-399 main.tex:700 prose the promise gap is used as the energy threshold

G-401 main.tex:1018 prose the guided state is labelled a witness and described as an input

G-402 main.tex:1018 prose the block never defines U , and j indexes both powers and terms

G-406 main.tex:1041 prose boson sampling is a sampling problem; it computes no function

G-408 main.tex:1041 prose true for the approximate QFT; the sentence is about computing it exactly

G-409 main.tex:1044 prose ϵ is unconstrained, and one reading forces an empty circuit

G-410 main.tex:1047 prose the theorem approximates the QFT; QNC holds operators computed exactly

G-414 main.tex:1375 prose the prover is made the decider; in a Merlin-Arthur protocol Arthur decides

G-416 main.tex:1378 prose $p = \text{poly}(n)$ allows $p > n$, and then no $(n - p)$ -qubit state exists

G-417 main.tex:1378 prose how an arbitrary state vector is given as input is not said

G-421 main.tex:1407 prose the absolute value allows $b < a$, and δ is not quantified in n

G-434 main.tex:2206 prose no thresholds, gap or coefficient bounds, so no decision problem is defined

G-435 main.tex:2208 prose the second sum runs over j and its summand $\Delta_i X_i$ does not use j

G-438 main.tex:2339 prose the vector is not normalised, so it is a state only for special P and f

G-439 main.tex:2341 prose a polynomial has one degree; for all l asks it to have every degree

G-442 main.tex:2351 prose the theorem it summarises samples approximately; the box does not say so

G-457 main.tex:2476 prose finiteness is not part of the notion: $\text{Sym}(N)$ is an infinite permutation group

G-464 main.tex:2510 prose the first and third column headings are swapped

G-466 main.tex:2519 prose the kernel is the subset $\{0\}$ of Z_6 ; $(0,0)$ is the codomain's identity

G-467 main.tex:2525 prose addition is denoted a_b , and two axioms quantify over G , not R

G-469 main.tex:2542 prose the quantifiers are garbled; read as intended it is left cancellation only

G-471 main.tex:2549 prose with the block's own integral domain, the quaternions are a field

G-475 main.tex:2558 prose a definition that asserts two properties of $\text{GF}(p^n)$ and never says what it is

G-477 pics/local-qpe.png prose the index register's input is not a unit vector

G-478 pics/local-qpe.png prose the box reads as the j -th power W^j , the text defines a family W_j

G-480 pics/qpe-generic01.png prose the guided state is labelled a witness and captioned as an input

G-481 pics/qpe-generic01.png prose the caption names the input $|\eta\rangle$, the image draws $|u\rangle$

G-482 pics/qpe-generic01.png prose the first include leaves U undefined, and j indexes both powers and terms

G-495 pics/NDTM_with_GLH-Oracle.png prose outside GLH 's promise the oracle's answer, and so acceptance, is unfixed

G-497 pics/PCE-encoding.png prose a 3-qubit encoding feeds a circuit drawn on 4 wires

G-498 pics/QCMA_in_NP^BQP.png prose an n -qubit state is placed on a 2-sphere

G-499 pics/UQA_circuit.png prose an open-circle control means control on $|0\rangle$, not a register-valued power

G-500 pics/VQLS.png prose a cost below γ is labelled as the exact normalised solution

G-501 pics/hdqi_summary_screenshot.png prose the asterisk on Ground State has no referent

G-506 pics/qpe_tex_qz.png prose θ is unbound; it exists only if $|\psi\rangle$ is an eigenvector, which is unstated

G-507 pics/qpe-generic.png prose $|\psi\rangle$ returns unchanged only for an eigenvector, which the image does not state

G-508 main.tex:489 prose the caption names a dictionary and fixes no relation its entries assert

FALSE

G-7 main.tex:1103 lean the state space is written over the reals

G-11 main.tex:1133 lean the promise constrains a variable it did not introduce

G-20 main.tex:2321 lean the BQ-operator is never defined

G-21 main.tex:2312 lean both bullets are labelled Completeness

G-24 main.tex:2272 lean the conclusion is about a symbol the statement never introduces

G-27 main.tex:2286 lean the first condition is vacuous under its own typing

G-29 main.tex:2305 lean the display contradicts its own prose, three ways

G-31 main.tex:1481 lean the matrix as printed is not unitary

G-32 main.tex:1517 lean the definition requires no such thing

G-43 main.tex:72 measured the link and its label name different addresses

G-45 main.tex:1072 measured every appendix numeral disagrees with the number printed for it

G-46 manish_assist/commands.tex:67 measured rejected by LaTeX, so \log keeps its kernel meaning

G-50 main.tex:79 prose the first author of the cited work is Trevisan

G-52 main.tex:81 prose no quantum LP, assumption or reduction to QLSP appears in the draft

G-56 main.tex:101 prose the cited gadget reduction starts from 3-local, not 5-local

G-62 main.tex:103 prose the cited paper has two authors

G-66 main.tex:107 prose the referenced subsection reviews perturbative gadgets, not the LP method

G-67 main.tex:109 prose the label names the classical gadget section, in either copy

G-68 main.tex:109 prose every construction in the draft targets the commuting ZZ family

G-70 main.tex:109 prose the section treats GLH, which the draft cites as BQP-complete

G-76 main.tex:162 prose the definition is in Appendix XI, not the last section

G-77 main.tex:162 prose the definition makes every gadget finite

G-81 main.tex:173 prose no quantum gadget section discusses the hereditary property

G-89 main.tex:239 prose no section frames the known perturbative gadgets as a linear program

G-94 main.tex:249 lean the source requires $f^2 = I$; as written every unitary qualifies

G-95 main.tex:252 lean not unitary unless $\sin 2\theta = 0$, so not a quantum boolean function

G-105 main.tex:303 prose no bound on any parameter is derived in the section

G-111 main.tex:330 prose phase estimation of e^{iA} reads $\lambda_j/(2\pi)$, not λ_j

G-114 main.tex:341 lean the inverse qft alone does not return the phase register to $|0^n\rangle$

G-118 main.tex:389 lean the expansion drops the factor $1/8$

G-119 main.tex:389 lean -III is written where -IIZ belongs, so the sum is traceless

G-120 main.tex:396 lean 256 vectors of norm $\sqrt{8}$ support no universal computation

G-125 main.tex:417 prose the quoted theorem promises a low-energy state, not the ground state

G-129 main.tex:430 prose a subset state is the uniform superposition, with no free amplitudes

G-130 main.tex:430 prose a subset state can be a product state

G-131 main.tex:435 prose the full subset $S = \{0,1\}^n$ gives $|+\rangle^n$, which is easy to represent

G-132 main.tex:437 prose the overlap is a promise restricting instances, not an assertion

G-135 main.tex:444 prose the cited paper's GalH has other guiding classes and a fidelity parameter

G-140 main.tex:495 prose the appendix has one subsection; QPE is its fourth subsubsection

G-154 main.tex:592 prose the problem decides $\lambda_0(H)$, not the energy of a given state

G-155 main.tex:599 prose a valid LH witness needs low energy, not overlap with the ground state

G-158 main.tex:611 prose the lemma assumes Case III's overlap, and Case III's paragraph treats Case I

G-163 main.tex:632 lean the cross term is dropped; it vanishes only when ψ_0 is an eigenvector

G-164 main.tex:636 lean the orthogonal amplitude is bounded by $\sqrt{2\epsilon - \epsilon^2}$, not ϵ

G-165 main.tex:638 prose the draft's own k-LH bounds each term, so $\|H\|$ can reach m

G-166 main.tex:644 lean fails even with ψ_0 the exact ground state: $3/4$ against $1/2$ at $n = 1$

G-167 main.tex:648 lean an upper bound on the orthogonal energy is used as a lower bound

G-168 main.tex:650 prose both thresholds become $(a + b)/2$, so the new promise gap is zero

G-170 main.tex:661 lean $p_2 = p_3 + 1$ meets the condition and the energies cross

G-171 main.tex:678 prose the bound $\Delta \leq b - a$ holds for every $p_2 \geq 1$ and says nothing of p_2

G-174 main.tex:688 prose the per-run probability is the squared overlap $1/p_1^2$, not $1/p_2$

G-175 main.tex:695 prose the condition admits an energy deviation larger than $b - a$

G-182 main.tex:724 lean a sublevel set of a quadratic form is not closed under scaling

G-187 main.tex:817 prose the reference points to no section: HDQI is not in this draft

G-189 main.tex:817 prose the section formulates no reduction and defines no weakly commuting family

G-190 main.tex:819 prose the reference points to no section: the QCPCP reduction is not in this draft

G-191 main.tex:819 prose the section formulates no QCPCP reduction

G-193 main.tex:841 prose the table has four columns; the gadget types are its rows

G-194 main.tex:851 lean an unsatisfiable formula can yield $7m - 1$ satisfied 2-clauses, not $6m$

G-196 main.tex:855 prose no quantum-LP framework is formulated in §9 or later

G-197 main.tex:861 prose only the ZX family is described; the other two are never defined

G-198 main.tex:869 prose the cited gadget simulates ZX with ZZ and XX, not ZZ with ZX

G-201 main.tex:871 prose the section automates no gadget and generalises no LP

G-202 main.tex:873 lean on basis states ZX terms read 0 and ZZ reads ± 1 ; ZX cannot reproduce ZZ

G-203 main.tex:873 prose the set of basis vectors is not a subspace; its span is the whole space

G-204 main.tex:873 prose none of the three promised analyses follows

G-208 main.tex:977 prose the drawing carries an X_{8X_4} term, which no ZX-Hamiltonian has

G-209 main.tex:983 prose the proof gives no correctness argument, and no marker covers its absence

G-213 main.tex:1001 lean for a subset state α_{ij} is a sum over $|S|^2$ pairs, computable in P

G-215 main.tex:1007 lean the displayed equation has nontrivial solutions, and $p=q=r=s=0$ solves none

G-216 main.tex:1153 prose no composition technique and no newly designed promise problem appear

G-222 main.tex:1193 lean no single U diagonalises $X_1 + Z_2$; a product of per-qubit unitaries does

G-227 main.tex:1213 prose the label is defined nowhere, and no appendix reduces MAX2SAT to $\{ZZ, Z\}$

G-229 main.tex:1218 prose m edges touch up to $2m$ qubits, so the space has up to 4^m assignments

G-234 main.tex:1225 prose 3-CNFs with m clauses on $3m$ distinct variables are always satisfiable

G-235 main.tex:1231 prose the heading asserts hardness for BQP; the body says likely and proves none

G-240 main.tex:1258 lean for $X_1X_2 + Y_1Y_2$ the ground state is the singlet, orthogonal to the guide

G-241 main.tex:1262 lean for $H = \sum X_k$ no poly-size subset state has overlap above $\text{poly } 2^{-(n/2)}$

G-243 main.tex:1262 prose no appendix on QMA_{log} exists, and the label is defined in neither paper

G-244 main.tex:1264 prose that vector has norm $2^{-(m/2)}$, and $\beta_i^2 = 1$ does not normalise complex β

G-245 main.tex:1266 lean $(|0\rangle - |1\rangle)/\sqrt{2}$ is orthogonal to the uniform state at $m = 1$

G-246 main.tex:1277 lean the problem is in P, by exact diagonalisation on at most $2m$ qubits

G-249 main.tex:1282 prose the cited NP-completeness is for a guidable problem, not an unguided one

G-256 main.tex:1315 lean for $u = (1, 1)$ the stated probabilities sum to $\sqrt{2}$

G-257 main.tex:1320 prose the Born rule gives $|u_i|^2 / \|u\|^2$; for $|+\rangle$ the stated values sum to $\sqrt{2}$

G-264 main.tex:1360 prose the cited sources give BQP containment only for γ at least $1/\text{poly}$

G-266 main.tex:1369 prose the appendix's own 1-local and log-edge cases need no eigenvector

G-268 main.tex:1425 prose of the five topics, only Fourier analysis has a following section

G-269 main.tex:1425 prose coding theory, lattices, symplectic spaces and symmetric polynomials go unused

G-279 main.tex:1648 lean the objective never reads b_l , so it does not count satisfied equations

G-281 main.tex:1670 cited TSSW's own strict-gadget clause is existential; the draft prints it universal

G-286 main.tex:1726 lean the 7-gadget is S_2 -partial and fails the appendix's strictness

G-289 main.tex:1760 lean the table picks $y = 1$ at 001, 010, 100, where W_b has $y = 0$ and 1 fails

G-290 main.tex:1764 lean 12 of the 48 printed clause values are wrong

G-291 main.tex:1784 prose every gadget is finite by definition; hereditary bounds the search

G-292 main.tex:1787 lean up to complementation there are $2^{(|S|-1)}$ columns, not $2^{|S|}$

G-293 main.tex:1792 lean the best equal weight gives $2(k-1)/(k-2)$, which is above 2

G-294 main.tex:1792 lean the equal-weight gadget with $k = 4$ has performance $3 > 15/8$

G-297 main.tex:1796 lean a substituted constraint can become the constant 1, which is not from F

G-299 main.tex:1832 cited the example reduces PC to 2CSP; TSSW's own 2SAT bound is $\alpha \geq 11$

G-300 main.tex:1844 lean 14 of the 16 are nonconstant, and 10 depend on both variables

G-303 main.tex:1852 lean a conjunction of two literals is not a 2SAT clause

G-304 main.tex:1860 measured no non-negative weights make the twelve constraints a 5-gadget for PC_0

G-305 main.tex:1862 prose no section poses an LP or gadget-search problem for quantum constraints

G-319 main.tex:2045 prose the restriction must be to H^- 's own low space L_-^- , not to H 's L_-

G-322 main.tex:2066 lean the top-right block needs $A^{-\{-1\}}$, not A : $[[2,1],[0,1]]$ gives -2 for $-1/2$

G-325 main.tex:2081 prose a trailing inverse turns $zI - G_-^{-\{-1\}}$ into $zI - G_-^-$

G-326 main.tex:2085 prose the left side carries the extra inverse; the right side is $\text{Sigma}_-(z)$

G-328 main.tex:2093 lean the fourth term lacks its trailing V_{+-} and maps L_+ into L_-

G-338 main.tex:2144 lean the sign of A is wrong: on L_- the A -term of V_1 is $-A Z_i$

G-340 main.tex:2179 lean the resolvent gives $B^2/(z - \delta)$; the draft puts 1 in place of z

G-341 main.tex:2183 lean the Z_i summand must carry the factor C from V_{++}

G-342 main.tex:2183 lean $(X_j + 1)^3$ is already $4(X_j + 1)$; the printed $4D(X_j + 1)^3$ is $16D(X_j + 1)$

G-343 main.tex:2186 prose false for three of the four terms under the section's own target $Z_i X_j$

G-356 main.tex:2251 lean $f^\perp$ is supported on $H^\perp$, not constant on it: $Z/4$, $H = \{0,2\}$ gives 1, -1

G-357 main.tex:1432 prose the cited paper is by Moore and Nilsson, not Nielsen

G-381 main.tex:1563 lean exact only when $2^m \phi$ is an integer; at $m = 1$, $\phi = 1/4$ each outcome is $1/2$

G-387 main.tex:2260 prose completeness is defined for languages; the draft's uses are promise problems

G-392 main.tex:2643 prose the heading states as shown a completeness the appendix never treats

G-395 main.tex:2654 measured math-mode quantifiers in text mode: four Missing \$ inserted errors

G-400 main.tex:1017 lean the outputs are λ_0 and $|\psi_0\rangle$ only with probability about $|\langle \psi_0 | u \rangle|^2$

G-404 main.tex:1040 prose the heading names finite quantum gadgets; the subsection has none

G-412 main.tex:1172 prose no bound on the frequency of any Pauli clause is derived in the draft

G-415 main.tex:1375 prose the key NHE+20 has no entry in references.bib

G-419 main.tex:1398 prose no analysis of pinned problems with $O(\log^k n)$ free qubits exists in the draft

G-424 main.tex:2164 lean no X factor in V_{3++} , and no Z factor in V_{2-+} or V_{2+-}

G-425 main.tex:2164 lean the off-diagonal blocks V_{2-+} and V_{2+-} are not Hermitian

G-427 main.tex:2167 lean the sign of A is wrong: on L_- the A -term of V_1 is $-A Z_i$

G-428 main.tex:2170 lean the last term drops the coefficient D of the identity

G-431 main.tex:2203 prose no gadget follows: the subsection ends with a definition

G-441 main.tex:2344 prose the reference points to no section: the DQI algorithm is not in this draft

G-444 main.tex:2380 lean the phase has no t , so the printed state does not evolve

G-447 main.tex:2416 lean the bracket is already $(a - \langle A \rangle)^2$; the extra square gives a fourth moment

G-450 main.tex:2437 lean the commutator form is not a restatement of the anticommutator form above

G-470 main.tex:2545 lean $2Z$ has no unity and the quaternions do not commute, yet both meet the definition

G-479 pics/qpe-generic01.png lean the outputs are λ_0 and $|\psi_0\rangle$ only with probability about $|\langle \psi_0 | u \rangle|^2$

G-483 pics/boolean-cubes.png measured the 4-cube prints 0100 twice and omits 0110; four drawn edges are not bit flips

G-486 pics/bqp-npCartesianProduct.png measured $(1, 0, 1)$ is in the product and not on the sphere

G-487 pics/HHL-Algorithm-Schematic-1.ppm lean the uncompute box applies controlled U where the inverse needs $U^\dagger$

G-505 pics/qpe_tex_qz.png lean exact only when $2^t \theta$ is an integer; at $t = 1$, $\theta = 1/4$ each is $1/2$

G-510 pics/HHL-circuit.png lean at $A = \text{diag}(\pi, \pi/2)$ with $m = 1$ the post-selected output is not $|x\rangle$

G-511 pics/HHL-circuit.png lean off the 2^m grid the clock is left mixed, of purity $7/9$, not in $|0\rangle^{\otimes m}$

EXTERNAL

G-4 main.tex:1090 cited not adjudicable here; no row consumes it

G-5 main.tex:893 cited not adjudicable here; transcription defective

G-6 main.tex:942 cited not adjudicable here; two defects, both checked

G-10 main.tex:1124 cited imported; ordered where its own problem is not

G-12 main.tex:1142 cited imported; p typed as a number where the problem uses a polynomial

G-53 main.tex:81 prose hardness holds in a parameter regime the sentence does not state

G-54 main.tex:83 prose a null result about the literature, which no reading of the draft settles

G-57 main.tex:101 prose an attribution to a survey column not frozen here

G-58 main.tex:101 prose a citation to Oliveira and Terhal, not frozen here

G-59 main.tex:101 prose a citation to Biamonte and Love, not frozen here

G-60 main.tex:101 prose a citation to Cubitt and Montanaro, not frozen here

G-61 main.tex:103 prose a priority claim that needs the source and a literature search

G-63 main.tex:103 prose a pinpoint citation into a source not frozen here

G-64 main.tex:103 prose an attribution of an opinion, checked against the source

G-65 main.tex:103 prose a null result about the literature after 2014

G-78 main.tex:165 prose by its title the cited paper concerns complete sets, not gadget reductions

G-79 main.tex:165 prose a verbatim quotation to be checked against the source

G-82 main.tex:180 prose the reviewed route uses Håstad's verifier, which the attribution must match

G-85 main.tex:191 prose the three-query linear PCP of Håstad, cited without a theorem number

G-90 main.tex:239 prose the conditions of the HHL algorithm, named but not stated

G-101 main.tex:286 prose a claim about classical algorithms, from an unnamed source

G-102 main.tex:286 prose the running time omits condition number and precision

G-107 main.tex:305 prose a gate count cited to a lecture note that is a URL, not a source

G-108 main.tex:307 prose a bound attributed to a survey that is not frozen here

G-126 main.tex:417 prose the primary source is unfrozen; a frozen secondary account agrees

G-141 main.tex:498 prose an attribution to a survey theorem that is not frozen here

G-199 main.tex:869 prose holds if 2-local ZX Hamiltonians are QMA-complete, as Biamonte-Love state

G-200 main.tex:869 prose a null claim about the literature, made without a search on record

G-217 main.tex:1154 prose attributed to Cubitt and Montanaro, a source not frozen in 2b

G-218 main.tex:1157 prose a claim of verbatim quotation from a source not frozen in 2b

G-223 main.tex:1195 prose no \cite; Barahona 1982 is absent from references.bib and not frozen

G-238 main.tex:1250 prose cited to Cubitt and Montanaro, a source not frozen in 2b

G-248 main.tex:1282 prose attributed to Le Gall 2024, not frozen; a frozen GL22 theorem covers it

G-252 main.tex:1284 prose uncited; no source for guided LH at exponential precision is frozen

G-253 main.tex:1284 prose uncited; Fefferman and Lin 2016 is neither in the bibliography nor frozen

G-263 main.tex:1356 prose attributed to Le Gall 2024, not frozen; frozen results need more overlap

G-306 main.tex:1922 prose a survey claim; its only citation is commented out and unfrozen

G-330 main.tex:2101 prose the gadget’s role in Biamonte and Love’s proof; uncited here, unfrozen

G-331 main.tex:2103 prose a completeness theorem stated inside a definition, cited nowhere here

G-348 main.tex:2230 prose an attribution to Regev with no citation; no source is frozen

G-350 main.tex:2230 prose the coding analogue of the reduction, stated without a source

G-352 main.tex:2235 prose a code-based reduction attributed to Regev; the source is unnamed

G-371 main.tex:1517 prose a strategy attributed to the literature with no citation

G-384 main.tex:1572 prose the lemma’s proof credited to Nielsen and Chuang §5.2; the book is unfrozen

G-393 main.tex:2643 prose hardness rests on MAX-2SAT completeness, which is unfrozen here

G-405 main.tex:1041 prose an uncited conjecture about shallow circuits; no source is named or frozen

G-407 main.tex:1041 prose attributed to Hales-Hallgren, unfrozen; Cleve-Watrous give depth $2n - 1$

G-418 main.tex:1390 prose attributed to an unresolved key; the pinned-QMA source is not frozen

G-420 main.tex:1402 prose whether CGW’s problem is over the whole space or one magnetisation sector

G-422 main.tex:1927 prose a description cited to an unfrozen web page

G-423 main.tex:1927 prose a description cited to an unfrozen web page

G-432 main.tex:2203 prose the gadget’s role in Biamonte and Love’s proof; uncited here, unfrozen

G-433 main.tex:2206 prose a completeness theorem stated inside a definition, cited nowhere here

G-437 main.tex:2337 prose the form of DQI states is the DQI papers’ content; none is cited or frozen

G-440 main.tex:2344 prose cited to Marwaha et al., unfrozen; plausible for normalised, succinct states

G-443 main.tex:2365 prose the book is known as The Theoretical Minimum; no bibliographic entry is given

G-455 main.tex:2468 prose credits Lagrange with the axioms; history known here only from memory

G-489 pics/hhl_variants_runtime.png prose the conjugate-gradient bound, attributed to an unfrozen source

G-490 pics/hhl_variants_runtime.png prose the HHL bound, attributed to an unfrozen source

G-491 pics/hhl_variants_runtime.png prose the VTAA bound, recalled with a worse epsilon-dependence than printed

G-492 pics/hhl_variants_runtime.png prose the Childs-Kothari-Somma bound, attributed to an unfrozen source

G-493 pics/hhl_variants_runtime.png prose the dense QLSA bound, attributed to an unfrozen source

G-502 pics/hdqi_summary_screenshot.png prose a claim about the HDQI algorithm, whose source is neither named nor frozen

G-503 pics/q-gadgets001.jpeg prose a ground state as HDQI’s output, unqualified; source unnamed, unfrozen

G-504 pics/main_figure.png prose a summary of the HDQI result, whose source is neither named nor frozen